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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 13: Nuclei

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 38 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. *Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Assertion (A): The mass of a nucleus is less than the sum of the masses of the constituent nucleons.

Reason (R): Energy is absorbed when the nucleons are bound together to form a nucleus.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → **[2026 • Set 55-1-1]**

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2. If r_1 and r_2 are the radii of atomic nuclei of mass numbers 64 and 27 respectively, then the value of $(\frac{r_1}{r_2})$ is:

(A) 1

(B) $\frac{4}{3}$

(C) $\frac{3}{4}$

(D) $\frac{27}{64}$

Check Solution → [2026 • Set 55-3-1]

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3. **Assertion (A):** Nuclear forces are always attractive.

Reason (R): The nuclear force between protons and neutrons in a nucleus is a weak force.

Select the correct answer from the codes (A), (B), (C) and (D) given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-3-1]

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4. **Assertion (A):** Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.
Reason (R): For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei, it decreases with increasing Z.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-5-1]

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5. Nuclides with the same number of neutrons are called:

(A) Isobars

(B) Isotones

(C) Isotopes

(D) Isomers

Check Solution → [2026 • Set 55-5-1]

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6. The radius of a nucleus of mass number 125 is:

(A) 6.0 fm

(B) 3.0 fm

(C) 7.2 fm

(D) 150 fm

Check Solution → [2026 • Set 55-5-1]

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7. Consider the nuclear reaction $X \rightarrow Y + Z$. Let M_x, M_y and M_z be the masses of the three nuclei X, Y and Z respectively. Then which of the following relations hold true?

(A) $(M_x - M_y) < M_z$

(B) $(M_x - M_z) < M_y$

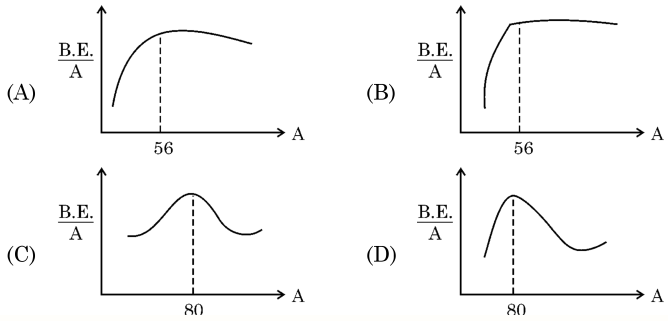
(C) $M_x > (M_y + M_z)$

(D) $M_x < (M_y + M_z)$

Check Solution → [2026 • Set 55-5-3]

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8. Which of the following figures correctly represent the shape of curve of binding energy per nucleon as a function of mass number?



(A) Peak rises to a maximum near $A = 56$ and then falls (curve A)

(B) Curve B

(C) Peak near $A = 56$ inverted (curve C)

(D) Peak near $A = 80$ (curve D)

Check Solution → [2025 • Set 55-1-1]

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9. **Assertion (A):** During formation of a nucleus, the mass defect produced is the source of the binding energy of the nucleus.
Reason (R): For all nuclei, the value of binding energy per nucleon increases with mass number.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-5-1]

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10. Inside a nucleus, the nuclear forces between proton and proton, proton and neutron, neutron and neutron are F_{pp}, F_{pn} and F_{nn} respectively. Then:

(A) $F_{pp} > F_{pn} > F_{nn}$

(B) $F_{pn} > F_{nn} > F_{pp}$

(C) $F_{nn} > F_{pp} > F_{pn}$

(D) $F_{pp} = F_{pn} = F_{nn}$

Check Solution → [2025 • Set 55-6-1]

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11. **Assertion (A):** The binding energy per nucleon is practically constant for mass number in the range $(30 < A < 170)$.
Reason (R): Nuclear forces between the nucleons for mass numbers in the range $(30 < A < 170)$ are not short-range.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-6-1]

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12. Isotones are the nuclides having:

(A) same mass numbers

(B) same atomic numbers

(C) same neutron number, but different atomic number

(D) different neutron number, and different mass number

Check Solution → [2025 • Set 55-7-1]

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13. The mass numbers of two nuclei A and B are 27 and 64 respectively. The ratio of their radii $(\frac{r_A}{r_B})$ will be:

(A) $\frac{27}{64}$

(B) $\frac{9}{16}$

(C) $\frac{3\sqrt{3}}{2}$

(D) $\frac{3}{4}$

Check Solution → [2025 • Set 55-7-2]

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14. The potential energy between two nucleons inside a nucleus is minimum at a distance of about:

(A) 0.8 fm

(B) 1.6 fm

(C) 2.0 fm

(D) 2.8 fm

Check Solution → [2024 • Set 55-5-1]

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15. The mass density of a nucleus of mass number A is:

(A) proportional to $A^{1/3}$

(B) proportional to $A^{2/3}$

(C) proportional to A

(D) independent of A

Check Solution → [2023 • Set 55-1-1]

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16. The curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for helium nucleus. This implies that helium nucleus is

(A) radioactive

(B) unstable

(C) easily fissionable

(D) more stable nucleus than its neighbours

Check Solution → [2023 • Set 55-2-1]

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17. The radius of ^{27}X nucleus is R. The radius of ^{64}Y nucleus will be

(A) 3R

(B) $(\frac{3}{4})^{1/3}R$

(C) $(\frac{4}{3})R$

(D) $(\frac{64}{27})^{1/3}R$

Check Solution → [2023 • Set 55-2-3]

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18. **Assertion (A):** The nucleus $^{15}_{7}X$ is more stable than the nucleus $^{17}_{7}Y$.
Reason (R): $^{17}_{7}X$ contains more number of protons.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-3-1]

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19. The difference in mass of ^{7}X nucleus and total mass of its constituent nucleons is 21.00 u. The binding energy per nucleon for this nucleus is equal to the energy equivalent of:

(A) 3 u

(B) 3.5 u

(C) 7 u

(D) 21 u

Check Solution → [2023 • Set 55-4-1]

---

20. Which of the following statements is not true for nuclear forces?

(A) They are stronger than Coulomb forces.

(B) They have about the same magnitude for different pairs of nucleons.

(C) They are always attractive.

(D) They saturate as the separation between two nucleons increases.

Check Solution → [2023 • Set 55-4-1]

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21. The ratio of the nuclear densities of two nuclei having mass numbers 64 and 125 is

(A) $\frac{64}{125}$

(B) $\frac{4}{5}$

(C) $\frac{5}{4}$

(D) 1

Check Solution → [2023 • Set 55-5-1]

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22. The saturation property of the nuclear forces is due to the fact that they are

(A) charge independent forces.

(B) non-central forces.

(C) spin-dependent forces.

(D) short-range forces.

Check Solution → [2021]

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23. Why are neutrons preferred as better projectiles in causing nuclear reaction?

Check Solution → [2021]

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24. In β-decay, the parent and daughter nuclei have the same number of _____.

Check Solution → [2020 • Set 55-1-1]

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25. The nuclear radius of $^{27}_{13}Al$ is 3·6 fermi. Find the nuclear radius of $^{64}_{29}Cu$.

Check Solution → [2020 • Set 55-1-1]

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26. Fill in the blank with appropriate answer.

In β-decay, a neutron is converted into a proton, e^- and _____.

Check Solution → [2020 • Set 55-4-2]

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27. Four nuclei of an element undergo fusion to form a heavier nucleus, with release of energy. Which of the two — the parent or the daughter nucleus — would have higher binding energy per nucleon ?

Check Solution → [2018]

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28. Why is it found experimentally difficult to detect neutrinos in nuclear β-decay ?

Check Solution → [2014]

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29. Write the relationship between the size of a nucleus and its mass number (A).

Check Solution → [2012]

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30. Write any two characteristic properties of nuclear force.

Check Solution → [2011]

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31. How is the mean life of a radioactive sample related to its half life?

Check Solution → [2011 • Set 55-2-1]

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32. Two nuclei have mass numbers in the ratio 1:2. What is the ratio of their nuclear densities?

Check Solution → [2009]

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33. State the reason, why heavy water is generally used as a moderator in a nuclear reactor.

Check Solution → [2008]

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34. What is the nuclear radius of ^{125}Fe, if that of ^{27}Al is 3.6 fermi?

Check Solution → **[2008]**

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35. The radioactive isotope *D* decays according to the sequence *D* α–particle D_1 β–particle D_2. If the mass number and atomic number of D_2 are 176 and 71 respectively, what is (i) the mass number (ii) atomic number of *D*?

Check Solution → **[2007]**

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36. Define the term 'activity' of a radionuclide. Write its SI unit.

Check Solution → **[2007]**

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37. Write the nuclear decay process for β^--decay of $_{15}{}^{32}P$.

Check Solution → **[2004]**

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38. 'Heavy water is often used as a moderator in thermal nuclear reactors.' Give reason.

Check Solution → **[2004]**

SECTION B

2-Mark Questions

VSA · 43 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

39. Draw the plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from this plot.

Check Solution → **[2026 • Set 55-1-1]**

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40. Draw a graph showing variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei ($A > 170$) decreases with increase in mass number. Explain its significance.

Check Solution → **[2026 • Set 55-1-2]**

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41. Suppose a nucleus with mass number $A = 240$ and B.E. per nucleon = 7.6 MeV, breaks into two nuclei, each of mass number $A = 120$ with B.E. per nucleon = 8.5 MeV. Calculate the energy released in the process.

Check Solution → **[2026 • Set 55-1-3]**

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42. Explain the terms mass defect and binding energy. How are they related?

Check Solution → **[2026 • Set 55-3-1]**

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43. Prove that the density of nuclear matter is same for all nuclei.

Check Solution → **[2026 • Set 55-4-2]**

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44. Calculate the mass of an α-particle in atomic mass unit (u). Given,
Mass of a normal helium atom = 4.002603 u
Mass of carbon atom = 1.9926×10^{-26} kg

Check Solution → [2025 • Set 55-4-1]

45. When a neutron collides with $^{235}_{92}U$, the nucleus gives $^{140}_{54}Xe$ and $^{94}_{38}Sr$ as fission products and two neutrons are ejected. Calculate the mass defect and the energy released (in MeV) in the process. Given:
$m(^{235}_{92}U) = 235.04393\ u$, $m(^{140}_{54}Xe) = 139.92164\ u$
$m(^{94}_{38}Sr) = 93.91536\ u$, $m_n = 1.00866\ u$
$1\ u = 931\ MeV/c^2$

Check Solution → [2025 • Set 55-5-1]

46. **(a)** Why is the mass of a nucleus always less than the sum of the masses of its constituents, i.e. free neutrons and free protons?
(b) How is Coulomb repulsion between protons in a nucleus overcome? Explain.

Check Solution → [2025 • Set 55-5-2]

47. Calculate the binding energy per nucleon (in MeV) of a helium nucleus ($_2^4He$).
Given: $m(_2^4He) = 4.002603\ u$, $m_n = 1.008665\ u$, $m_p = 1.007825\ u$, $1\ u = 931.5\ MeV/c^2$.

Check Solution → [2025 • Set 55-7-2]

48. The carbon isotope $_6^{12}C$ has a nuclear mass of 12.000000 u. Calculate the binding energy of its nucleus.
Given m_p=1.007825 u; m_n=1.008665 u.

Check Solution → [2024 • Set 55-2-1]

49. Calculate the energy released/absorbed in the following nuclear reaction:
$_6^{12}C + _6^{12}C \rightarrow _{10}^{20}Ne + _2^4He$
Given: $m(_6^{12}C)$=12.000000 u
$m(_{10}^{20}Ne)$=19.992439 u
$m(_2^4He)$=4.002603 u
1 u=931 MeV/c^2

Check Solution → [2024 • Set 55-2-2]

50. Deuterium undergoes the following fusion reaction:
${}_1^2\text{H} + {}_1^2\text{H} \rightarrow {}_2^3\text{He} + {}_0^1\text{n} + 3.27\ \text{MeV}$
How long an electric bulb of 200 W will glow by using the energy released in 2 g of deuterium?

Check Solution → [2024 • Set 55-5-3]

51. Define the term, mass defect. How is it related to stability of the nucleus?

Check Solution → [2023 • Set 55-2-1]

52. (a) Draw a graph showing the variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei ($A > 170$) decreases with the increase in mass number. Explain.

OR

(b) Using Bohr's postulates, obtain the expression for radius of nth stable orbit in a hydrogen atom.

Check Solution → [2023 • Set 55-3-1]

53. Draw the graph showing the variation of binding energy per nucleon with mass number A of nuclei ($2 \leq A \leq 170$). Use this graph to explain the release of energy in nuclear fission.

Check Solution → [2023 • Set 55-4-1]

54. Briefly describe the multi-step process involved in the generation of energy in the Sun.

Check Solution → [2023 • Set 55-4-3]

55. (i) Distinguish between isotopes and isobars.
(ii) Two nuclei have different mass numbers A_1 and A_2. Are these nuclei necessarily the isotopes of the same element? Explain.

Check Solution → [2022 • Set 55-3-1]

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56. Define the term 'Half-life' of a radioactive substance. Two different radioactive substances have half-lives T_1 and T_2 and number of undecayed atoms at an instant, N_1 and N_2, respectively. Find the ratio of their activities at that instant.

Check Solution → [2020 • Set 55-1-1]

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57. Define activity of a sample of a radioactive substance. The value of the disintegration constant of a radioactive substance is $0{\cdot}0693\ h^{-1}$. Find the time after which the activity of a sample of this substance reduces to one-half that of its present value.

Check Solution → [2020 • Set 55-1-2]

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58. (a) For a radioactive substance, show the variation of the total mass disintegrated as a function of time t graphically.
(b) Initial mass of a radioactive substance is 3.2 mg. It has a half-life of 4 h. Find the mass of the substance left undecayed after 8 h.

Check Solution → [2020 • Set 55-2-2]

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59. (a) Write the SI unit of activity of a radioactive substance.
(b) Find the half-life of a radioactive substance if the activity of the substance drops to $\frac{1}{16}$ of its initial value in 15 years.

Check Solution → [2020 • Set 55-2-3]

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60. A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits in to two nuclei Q and R of mass numbers 110, 130 and binding energy per

nucleon 8.5 MeV and 8.4 MeV, respectively. Calculate the energy released in the fission.

Check Solution → [2020 • Set 55-5-1]

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61. (a) Define one Becquerel.
(b) A radioactive substance disintegrates into two types of daughter nuclei, one type with disintegration constant λ_1 and the other type with disintegration constant λ_2. Determine the half-life of the radioactive substance.

Check Solution → [2020 • Set 55-5-2]

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62. Calculate for how many years will the fusion of 2.0 kg deuterium keep 800 W electric lamp glowing. Take the fusion reaction as
$^{2}_{1}H + {}^{2}_{1}H \longrightarrow {}^{3}_{2}He + {}^{1}_{0}n + 3.27$ MeV

Check Solution → [2020 • Set 55-5-3]

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63. Plot a graph showing the variation of undecayed nuclei N versus time t. From the graph, find out how one can determine the half-life and average life of the radioactive nuclei.

Check Solution → [2019 • Set 55-3-1]

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64. (a) Write two distinguishing features of nuclear forces.
(b) Complete the following nuclear reactions for α and β decay:
(i) $^{238}_{92}U \longrightarrow ? + {}^{4}_{2}He + Q$
(ii) $^{22}_{11}Na \longrightarrow {}^{22}_{10}Ne + ? + \nu$

Check Solution → [2019 • Set 55-3-1]

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65. Distinguish between nuclear fusion and nuclear fission. Give one example for each.

Check Solution → [2019 • Set 55-3-2]

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66. (a) Which property of nuclear force explains the constancy of binding energy per nucleon ($\frac{B}{A}$) for nuclei in the range 20 < A < 170?

(b) Complete the following nuclear reactions:

(i) ${}^{32}_{15}P \longrightarrow {}^{A}_{Z}X + e^{-} + \bar{\nu}$

(ii) ${}^{12}_{6}C + {}^{12}_{6}C \longrightarrow {}^{A}_{Z}Y + {}^{4}_{2}He$

Check Solution → [2019 • Set 55-3-3]

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67. Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions which you can draw regarding the nature of nuclear forces.

OR

Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei, $2 \leq A \leq 240$. How do you explain the constancy of binding energy per nucleon in the range 30 < A < 170 using the property that nuclear force is short-ranged?

Check Solution → [2012]

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68. In a given sample, two radioisotopes, *A* and *B*, are initially present in the ratio of 1:4. The half lives of *A* and *B* are respectively 100 years and 50 years. Find the time after which the amounts of *A* and *B* become equal.

Check Solution → [2012]

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69. Using the curve for the binding energy per nucleon as a function of mass number *A*, state clearly how the release in energy in the processes of nuclear fission and nuclear fusion can be explained.

Check Solution → [2011]

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70. When four hydrogen nuclei combine to form a helium nucleus, estimate the amount of energy in MeV released in this process of fusion. (Neglect the masses of electrons and neutrinos)

Given:

(i) mass of $^{1}_{1}H$ = 1.007825 u
(ii) mass of helium nucleus = 4.002603 u, 1 u = 931 MeV/c^2

Check Solution → [2011 • Set 55-2-1]

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71. A heavy nucleus X of mass number 240 and binding energy per nucleon 7.6 MeV is split into two fragments Y and Z of mass numbers 110 and 130. The binding energy of nucleons in Y and Z is 8.5 MeV per nucleon. Calculate the energy Q released per fission in MeV.

Check Solution → [2010]

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72. A radioactive nucleus *A* undergoes a series of decays according to the following scheme:
$A \xrightarrow{\alpha} A_1 \xrightarrow{\beta} A_2 \xrightarrow{\gamma} A_3 \xrightarrow{\gamma} A_4$
The mass number and atomic number of *A* are 180 and 72 respectively. What are these numbers for A_4?

Check Solution → [2009]

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73. (a) The mass of a nucleus in its ground state is always less than the total mass of its constituents - neutrons and protons. Explain.
(b) Plot a graph showing the variation of potential energy of a pair of nucleons as a function of their separation.

Check Solution → [2009]

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74. A nucleus $^{23}_{10}Ne$ undergoes β-decay and becomes $^{23}_{11}Na$. Calculate the maximum kinetic energy of electrons emitted assuming that the daughter nucleus and anti-neutrino carry negligible kinetic energy.
Mass of $^{23}_{10}Ne$ = 22.994455 *u*
Mass of $^{23}_{11}Na$ = 22.989770 *u*
1 *u* = 931.5 MeV/c^2

Check Solution → [2008]

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75. State the law of radioactive decay. If N_0 is the number of radioactive nuclei in the sample at some initial time t_0, find out the relation to determine the number N present at a subsequent time. Draw a plot of N as a function of time.

Check Solution → [2008]

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76. Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei. Explain the energy release in the process of nuclear fission from the above plot. Write a typical nuclear reaction in which a large amount of energy is released in the process of nuclear fission.

Check Solution → [2008]

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77. Draw a plot of potential energy of a pair of nucleons as a function of their separation. What is the significance of negative potential energy in the graph drawn?

Check Solution → [2007]

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78. Draw a graph showing the variation of potential energy between a pair of nucleons as a function of their separation. Indicate the regions in which the nuclear force is (i) attractive, (ii) repulsive.

Check Solution → [2007]

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79. Draw the graph showing the variation of binding energy per nucleon with mass number. Give the reason for the decrease of binding energy per nucleon for nuclei with high mass numbers.

Check Solution → [2004]

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80. Name the reaction which takes place when a slow neutron beam strikes ${}_{92}{}^{235}U$ nuclei. Write the nuclear reaction involved.

Check Solution → [2003]

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81. In the series of radioactive disintegration of ${}_{92}{}^{238}\mathrm{U}$, first an alpha-particle and then a beta-particle is emitted. What is the atomic number and mass number of the new nucleus formed by these successive disintegrations?

Check Solution → [2003]

SECTION C

3-Mark Questions

SA · 72 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

82. (a) Consider the following nuclides: $^{12}_{6}C$, $^{198}_{80}Hg$, $^{14}_{6}C$, $^{197}_{79}Au$. Group them into isotopes and isotones.
(b) How does the size of a nucleus depend on its mass number A? Hence prove that the density of nucleus is a constant, independent of A, for all nuclei.

Check Solution → **[2026 • Set 55-2-1]**

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83. Differentiate between nuclear fission and nuclear fusion. Give one example for each with nuclear reaction.

Check Solution → **[2026 • Set 55-3-1]**

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84. Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation (r). Using this plot, show that the nuclear force is (i) attractive for $r > r_0$, and (ii) repulsive for $r < r_0$.

Check Solution → **[2026 • Set 55-3-2]**

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85. In the nuclear reaction:
$^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{A}_{2}X + {}^{1}_{0}n$
(a) Find the value of A.
(b) Calculate the amount of energy released in the reaction.
Given: $m(^{2}_{1}H) = 2\cdot 014102$ u, $m(^{A}_{2}X) = 3\cdot 016049$ u, $m(^{1}_{0}n) = 1\cdot 008665$ u, 1 u = 931·5 MeV/c^2

Check Solution → **[2026 • Set 55-4-1]**

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86. (a) Distinguish between isotopes and isobars, giving one example for each.
(b) Derive the relation between atomic mass unit (u) and electron volt (eV).

Check Solution → [2026 • Set 55-4-3]

87. (a) (i) Write any two features of nuclear forces.
(ii) If both the number of protons and the neutrons are conserved in each nuclear reaction, in what way is mass converted into energy (or vice versa) in a nuclear reaction? Explain.

Check Solution → [2026 • Set 55-5-1]

88. **(a)** Define 'Mass defect' and 'Binding energy' of a nucleus. Describe 'Fission process' on the basis of binding energy per nucleon.
(b) A deuteron contains a proton and a neutron and has a mass of 2.013553 u. Calculate the mass defect for it in u and its energy equivalence in MeV. (m_p = 1.007277 u, m_n = 1.008665 u, 1 u = 931.5 MeV/c^2)

Check Solution → [2025 • Set 55-1-1]

89. (a) Show the variation of binding energy per nucleon with mass number. Write the significance of the binding energy curve.
(b) Two nuclei with lower binding energy per nucleon form a nuclei with more binding energy per nucleon.
(i) What type of nuclear reaction is it?
(ii) Whether the total mass of nuclei increases, decreases or remains unchanged?
(iii) Does the process require energy or produce energy?

Check Solution → [2025 • Set 55-2-1]

90. (a) Consider the so-called 'D-T reaction' (Deuterium-Tritium reaction). In a thermonuclear fusion reactor, the following nuclear reaction occurs:
$_1^2H + _1^3H \longrightarrow _2^4\text{He} + _0^1n + Q$
Find the amount of energy released in the reaction.
Given:
$m(_1^2H) = 2{\cdot}014102\ u$
$m(_1^3H) = 3{\cdot}016049\ u$

$m({}_2^4\text{He}) = 4{\cdot}002603\ u$

$m({}_0^1 n) = 1{\cdot}008665\ u$

$1\ u = 931\ \text{MeV/c}^2$

OR

(b) Show that the nuclear density is independent of mass number.

Check Solution → [2025 • Set 55-6-1]

---

91. Find the Q value of the following nuclear reaction:

${}_6^{12}\text{C} + {}_6^{12}\text{C} \longrightarrow {}_{10}^{20}\text{Ne} + {}_2^4\text{He}$

Is this reaction exothermic or endothermic?

Given:

$m({}_6^{12}\text{C}) = 12{\cdot}000000\ u$

$m({}_{10}^{20}\text{Ne}) = 19{\cdot}992439\ u$

$m({}_2^4\text{He}) = 4{\cdot}002603\ u$

$1\ u = 931\ \text{MeV/c}^2$

Check Solution → [2025 • Set 55-6-2]

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92. (a) Differentiate between 'nuclear fission' and 'nuclear fusion'. Briefly discuss one example of each.

(b) Draw a graph of potential energy between a pair of nucleons as a function of their separation.

Check Solution → [2025 • Set 55-6-3]

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93. Briefly explain how energy is produced in stars, giving two examples of the nuclear reactions involved.

Check Solution → [2025 • Set 55-7-1]

---

94. (a) State any two properties of a nucleus.
(b) Why is the density of a nucleus much more than that of an atom?
(c) Show that the density of the nuclear matter is the same for all nuclei.

Check Solution → **[2024 • Set 55-1-1]**

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95. (a) Write two characteristic properties of nuclear force.
(b) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from the plot.

Check Solution → **[2024 • Set 55-1-2]**

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96. (a) Plot a graph depicting potential energy of a pair of nucleons in a nucleus as a function of their separation.
(b) Identify the regions where the nuclear force is (i) attractive, and (ii) repulsive. Justify your answer.

Check Solution → **[2024 • Set 55-2-2]**

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97. (a) Briefly discuss three characteristics of the forces between nucleons.
(b) Which out of ${}_{8}{}^{16}\mathrm{O}$ and ${}_{7}{}^{14}\mathrm{N}$ nuclei is more stable and why?

Check Solution → **[2024 • Set 55-2-3]**

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98. (a) Define atomic mass unit (u).
(b) Calculate the energy required to separate a deuteron into its constituent parts (a proton and a neutron). Given: $m(D) = 2.014102$ u; $m_p = 1.007825$ u; $m_n = 1.008665$ u.

Check Solution → **[2024 • Set 55-3-1]**

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99. (a) Differentiate between nuclear fission and fusion.
(b) The fission properties of ${}_{94}{}^{239}\mathrm{Pu}$ are very similar to those of ${}_{92}{}^{235}\mathrm{U}$. How much energy (in MeV), is released if all the atoms in 1 g of pure ${}_{94}{}^{239}\mathrm{Pu}$ undergo fission? The average energy released per fission is 180 MeV.

Check Solution → [2024 • Set 55-4-1]

100. Draw a diagram to show the variation of binding energy per nucleon with mass number for different nuclei and mention its two features. Why do lighter nuclei usually undergo nuclear fusion?

Check Solution → [2023 • Set 55-1-1]

101. (i) Distinguish between nuclear fission and fusion giving an example of each.
(ii) Explain the release of energy in nuclear fission and fusion on the basis of binding energy per nucleon curve.

Check Solution → [2023 • Set 55-2-1]

102. (i) How is the size of a nucleus found experimentally? Write the relation between the radius and mass number of a nucleus.
(ii) Prove that the density of a nucleus is independent of its mass number.

Check Solution → [2023 • Set 55-2-1]

103. (a) (i) Prove that the nuclear density is same for all nuclei.
(ii) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Draw two inferences from this plot.
OR
(b) (i) Draw a graph to show the variation of the number of scattered particles detected (N) in Geiger-Marsden experiment as a function of scattering angle (θ).
(ii) Discuss briefly two conclusions that can be drawn from this graph and how they lead to the discovery of nucleus in an atom.

Check Solution → [2023 • Set 55-3-1]

104. Draw a plot showing the variation of potential energy of two nucleons as a function of distance between them. Identify the regions in which the force between the nucleons is (i) attractive, and (ii) repulsive. Justify your answers.

Check Solution → [2023 • Set 55-4-3]

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105. (a) Calculate the binding energy of an alpha particle in MeV. Given
mass of a proton = 1.007825 u
mass of a neutron = 1.008665 u
mass of He nucleus = 4.002800 u
1u = 931 MeV/c^2
OR
(b) A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits into two nuclei Q and R of mass number 110 and 130 and binding energy per nucleon 8.5 MeV and 8.4 MeV respectively. Calculate the energy released in the fission.

Check Solution → [2023 • Set 55-5-3]

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106. Calculate the energy released in MeV in the following reaction:
$^{2}_{1}H + {}^{3}_{1}H \rightarrow {}^{4}_{2}He + n$
Given: $m(^{2}_{1}H) = 2.014102\ u$
$m(^{3}_{1}H) = 3.016049\ u$
$m(^{4}_{2}He) = 4.002603\ u$
$m_n = 1.008665\ u$

Check Solution → [2022 • Set 55-1-1]

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107. (i) Depict the variation of the potential energy of a pair of nucleons with the separation between them.
(ii) Imagine the fission of a $^{56}_{26}Fe$ into two equal fragments of $^{28}_{13}Al$ nucleus. Is the fission energetically possible? Justify your answer by working out Q value of the process.
Given: $m(^{56}_{26}Fe) = 55.93494$ u, $m(^{28}_{13}Al) = 27.98191$ u.

Check Solution → [2022 • Set 55-2-1]

---

108. (a) James Chadwick, in 1932, studied the emission of neutral radiations when Beryllium nuclei were bombarded with alpha particles. He concluded that

emitted radiations were neutrons and not photons. Explain.
(b) Two nuclei may have the same radius, even though they contain different number of protons and neutrons. Explain.

Check Solution → [2022 • Set 55-3-3]

---

109. (a) Differentiate between nuclear fission and nuclear fusion.
(b) Deuterium undergoes fusion as per the reaction: ${}_1^2H + {}_1^2H \rightarrow {}_2^3He + {}_0^1n + 3.27$ MeV. Find the duration for which an electric bulb of 500 W can be kept glowing by the fusion of 100 g of deuterium.

Check Solution → [2022 • Set 55-4-1]

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110. In a fission event of ${}^{238}_{92}U$ by fast moving neutrons, no neutrons are emitted and final products, after the beta decay of the primary fragments, are ${}^{140}_{58}Ce$ and ${}^{99}_{44}Ru$. Calculate Q for this process. Neglect the masses of electrons/positrons emitted during the intermediate steps.
Given: $m({}^{238}_{92}U) = 238.05079\ u$; $m({}^{140}_{58}Ce) = 139.90543\ u$; $m({}^{99}_{44}Ru) = 98.90594\ u$; $m({}^{1}_{0}n) = 1.008665\ u$.

Check Solution → [2022 • Set 55-5-1]

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111. The nucleus ${}^{235}_{92}U$ initially at rest, decays into ${}^{231}_{90}X$ by emitting an α-particle ${}^{235}_{92}U \longrightarrow {}^{231}_{90}X + {}^{4}_{2}He$ + energy.
The binding energies per nucleon of the parent nucleus, the daughter nucleus and α-particle are 7·8 MeV, 7·835 MeV and 7·07 MeV, respectively. Assuming the daughter nucleus to be formed in the unexcited state and neglecting its share in the energy of the reaction, find the speed of the emitted α-particle. (Mass of α-particle = $6{\cdot}68 \times 10^{-27}$ kg)

Check Solution → [2020 • Set 55-1-1]

---

112. (a) Differentiate between half-life and average life of a radioactive substance.
(b) A radioactive substance decays for an interval of time equal to its mean life.

Find the fraction of the amount of the substance which is left undecayed after this time interval.

Check Solution → [2020 • Set 55-2-1]

---

113. (a) The density of the nuclear matter is tremendously larger than the physical density of the material. Explain.
(b) The nuclear forces are not coulomb forces between nucleons. Explain.
(c) Draw a plot of the potential energy between a pair of nucleons as a function of distance between them inside a nucleus.

Check Solution → [2020 • Set 55-3-1]

---

114. (a) Define the term decay constant of a radioactive substance.
(b) The half life of $^{238}_{92}U$ undergoing α decay is 4.5×10^9 years. Calculate the activity of 10 g sample of $^{238}_{92}U$.

Check Solution → [2020 • Set 55-5-1]

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115. Draw the curve showing the variation of binding energy per nucleon with the mass number of nuclei. Using it explain the fusion of nuclei lying on ascending part and fission of nuclei lying on descending part of this curve.

Check Solution → [2020 • Set 55-5-2]

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116. Define the term 'decay constant' of a radioactive sample. The rate of disintegration of a given radioactive nucleus is 10000 disintegrations/s and 5000 disintegrations/s after 20 hr and 30 hr respectively from start. Calculate the half life and initial number of nuclei at $t = 0$.

Check Solution → [2019 • Set 55-1-1]

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117. Write the relation between half life and average life of a radioactive nucleus.

Check Solution → [2019 • Set 55-1-2]

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118. In a given sample two isotopes A and B are initially present in the ratio of 1 : 2. Their half lives are 60 years and 30 years respectively. How long will it take so that the sample has these isotopes in the ratio of 2 : 1?

Check Solution → [2019 • Set 55-1-2]

---

119. Show that density of nucleus is independent of its mass number A.

Check Solution → [2019 • Set 55-1-3]

---

120. Why is it difficult to detect the presence of an anti-neutrino during β-decay? Define the term decay constant of a radioactive nucleus and derive the expression for its mean life in terms of the decay constant.

OR

(a) State two distinguishing features of nuclear force.

(b) Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. Mark the regions on the graph where the force is (i) attractive, and (ii) repulsive.

Check Solution → [2019 • Set 55-2-1]

---

121. Draw a graph showing the variation of binding energy per nucleon with mass number of different nuclei. Write any two salient features of the curve. How does this curve explain the release of energy both in the processes of nuclear fission and fusion?

Check Solution → [2019 • Set 55-4-1]

---

122. (a) Plot a graph showing the variation of potential energy of a pair of nucleons as a function of their separation. Also indicate the regions where the force is (i) attractive and (ii) repulsive.

(b) Write two characteristic properties of nuclear force.

Check Solution → [2019 • Set 55-4-2]

---

123. Plot a graph showing the variation of number of undecayed nuclei in a radioisotope as a function of time. Hence define half life of a nucleus and obtain its relation with average life.

Check Solution → [2019 • Set 55-4-3]

---

124. (a) Distinguish between isotopes and isobars, giving one example for each.
(b) Why is the mass of a nucleus always less than the sum of the masses of its constituents? Write one example to justify your answer.

OR

(a) Classify the following six nuclides into (i) isotones, (ii) isotopes, and (iii) isobars: ${}^{12}_{6}C$, ${}^{3}_{2}He$, ${}^{198}_{80}Hg$, ${}^{3}_{1}H$, ${}^{197}_{79}Au$, ${}^{14}_{6}C$.
(b) How does the size of a nucleus depend on its mass number? Hence explain why the density of nuclear matter should be independent of the size of the nucleus.

Check Solution → [2019 • Set 55-5-1]

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125. (a) Explain the processes of nuclear fission and nuclear fusion by using the plot of binding energy per nucleon (BE/A) versus the mass number A.
(b) A radioactive isotope has a half-life of 10 years. How long will it take for the activity to reduce to 3.125% ?

Check Solution → [2018]

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126. (i) A radioactive nucleus 'A' undergoes a series of decays as given below:
$A \ \alpha \ A_1 \ \beta^- \ A_2 \ \alpha \ A_3 \ \gamma \ A_4$
The mass number and atomic number of A_2 are 176 and 71 respectively. Determine the mass and atomic numbers of A_4 and A.
(ii) Write the basic nuclear processes underlying β^+ and β^- decays.

Check Solution → [2017]

---

127. (a) State the law of radioactive decay. Write the SI unit of 'activity'.
(b) There are $4\sqrt{2} \times 10^6$ radioactive nuclei in a given radioactive sample. If the

half life of the sample is 20 s, how many nuclei will decay in 10 s?

Check Solution → [2017]

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128. (a) Write the process of β-decay. How can radioactive nuclei emit β-particles even though they do not contain them? Why do all electrons emitted during β-decay not have the same energy?
(b) A heavy nucleus splits into two lighter nuclei. Which one of the two -- parent nucleus or the daughter nuclei has more binding energy per nucleon?

Check Solution → [2017]

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129. (a) Derive the relation between the decay constant and half life of a radioactive substance.
(b) A radioactive element reduces to 25% of its initial mass in 1000 years. Find its half life.

Check Solution → [2017]

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130. (a) Write the basic nuclear process involved in the emission of β^+ in a symbolic form, by a radioactive nucleus.
(b) In the reactions given below:
(i) ${}^{11}_{6}C \rightarrow {}^{y}_{z}B + x + \nu$
(ii) ${}^{12}_{6}C + {}^{12}_{6}C \rightarrow {}^{20}_{a}Ne + {}^{b}_{c}He$
Find the values of x, y, and z and a, b and c.

Check Solution → [2016]

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131. Half life of ${}^{238}_{92}U$ against α-decay is 4.5×10^{9} years. Calculate the activity of 1 g sample of ${}^{238}_{92}U$. Given Avogadro's number = 6×10^{26} atoms/kmol.

Check Solution → [2016]

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132. (a) Derive the mathematical expression for law of radioactive decay for a sample of a radioactive nucleus.

(b) How is the mean life of a given radioactive nucleus related to the decay constant?

Check Solution → [2016]

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133. Plot a graph showing the variation of binding energy per nucleon as a function of mass number. Which property of nuclear force explains the approximate constancy of binding energy in the range 30? How does one explain the release of energy in both the processes of nuclear fission and fusion from the graph?

Check Solution → [2016]

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134. Obtain the relation between the decay constant and half life of a radioactive sample.
The half life of a certain radioactive material against α-decay is 100 days. After how much time, will the undecayed fraction of the material be 6.25%?

Check Solution → [2015]

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135. Distinguish between nuclear fission and fusion. Show how in both these processes energy is released.
Calculate the energy release in MeV in the deuterium-tritium fusion reaction:
${}_1^2\mathrm{H} + {}_1^3\mathrm{H} \rightarrow {}_2^4\mathrm{He} + n$
Using the data:
$m({}_1^2\mathrm{H}) = 2.014102$ u
$m({}_1^3\mathrm{H}) = 3.016049$ u
$m({}_2^4\mathrm{He}) = 4.002603$ u
$\mathrm{m}_n = 1.008665$ u
$1\mathrm{u} = 931.5\ \mathrm{MeV/c^2}$

Check Solution → [2015]

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136. (a) Deduce the expression, $N = \mathrm{N}_0\, e^{-\lambda t}$, for the law of radioactive decay.
(b) (i) Write symbolically the process expressing the β^+ decay of ${}^{22}_{11}\mathrm{Na}$. Also write the basic nuclear process underlying this decay.

(ii) Is the nucleus formed in the decay of the nucleus $^{22}_{11}Na$, an isotope or isobar ?

Check Solution → [2014]

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137. For the past some time, Aarti had been observing some erratic body movement, unsteadiness and lack of coordination in the activities of her sister Radha, who also used to complain of severe headache occasionally. Aarti suggested to her parents to get a medical check-up of Radha. The doctor thoroughly examined Radha and diagnosed that she has a brain tumour.
(a) What, according to you, are the values displayed by Aarti ?
(b) How can radioisotopes help a doctor to diagnose brain tumour ?

Check Solution → [2014]

---

138. (a) In a typical nuclear reaction, e.g.

$$_1{}^2H + _1{}^2H \longrightarrow _2{}^3He + n + 3.27 \text{ MeV},$$

although number of nucleons is conserved, yet energy is released. How? Explain.
(b) Show that nuclear density in a given nucleus is independent of mass number A.

Check Solution → [2013]

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139. (a) Write symbolically the β^- decay process of $^{32}_{15}P$.
(b) Derive an expression for the average life of a radionuclide. Give its relationship with the half-life.

Check Solution → [2012]

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140. Draw a plot of potential energy of a pair of nucleons as a function of their separations. Mark the regions where the nuclear force is (i) attractive and (ii) repulsive. Write any two characteristic features of nuclear forces.

Check Solution → [2012]

141. (i) What characteristic property of nuclear force explains the constancy of binding energy per nucleon (BE/A) in the range of mass number 'A' lying $30 < A < 170$?
(ii) Show that the density of nucleus over a wide range of nuclei is constant - independent of mass number A.

Check Solution → [2012]

142. State the law of radioactive decay.
Plot a graph showing the number (N) of undecayed nuclei as a function of time (t) for a given radioactive sample having half life $T_{1/2}$.
Depict in the plot the number of undecayed nuclei at (i) $t = 3T_{1/2}$ and (ii) $t = 5T_{1/2}$.

Check Solution → [2011 • Set 55-1-1]

143. (i) Define 'activity' of a radioactive material and write its S.I. unit.
(ii) Plot a graph showing variation of activity of a given radioactive sample with time.
(iii) The sequence of stepwise decay of a radioactive nucleus is
$D \ \alpha \ D_1 \ \beta^- \ D_2$
If the atomic number and mass number of D_2 are 71 and 176 respectively, what are their corresponding values for D?

Check Solution → [2010]

144. Draw a plot showing the variation of binding energy per nucleon versus the mass number A. Explain with the help of this plot the release of energy in the processes of nuclear fission and fusion.

Check Solution → [2009]

145. Distinguish between isotopes and isobars. Give one example for each species. A radioactive isotope has a half-life of 5 years. How long will it take the activity to

reduce to 3.125%?

Check Solution → [2008]

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146. Calculate the energy released during the α-decay of $^{238}_{92}U \rightarrow ^{234}_{90}Th + ^{4}_{2}He$.
Given:
1. atomic mass of $^{238}_{92}U$ = 238.05079 u
2. atomic mass of $^{234}_{90}Th$ = 234.04363 u
3. atomic mass of $^{4}_{2}He$ = 4.00260 u
1 u = 931.5 MeV/c^2.
Is this decay spontaneous? Give reason.

Check Solution → [2007]

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147. Draw the graph to show variation of binding energy per nucleon with mass number of different atomic nuclei. Calculate binding energy/nucleon of $^{40}_{20}Ca$ nucleus.
Given:
mass of $^{40}_{20}Ca$ = 39.962589 u
mass of proton = 1.007825 u
mass of neutron = 1.008665 u
and 1 u = 931 MeV/c^2

Check Solution → [2007]

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148. Explain, with the help of a nuclear reaction in each of the following cases, how the neutron to proton ratio changes during (i) alpha-decay (ii) beta-decay.

Check Solution → [2006]

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149. Why is the mass of a nucleus always less than the sum of the masses of its constituents, neutrons and protons?
If the total number of neutrons and protons in a nuclear reaction is conserved, how then is the energy absorbed or evolved in the reaction? Explain.

Check Solution → [2006]

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150. Draw a graph showing the variation of binding energy per nucleon with mass number for different nuclei. Explain, with the help of this graph, the release of energy by the process of nuclear fusion.

Check Solution → [2006]

---

151. **(a)** Draw a graph showing the variation of potential energy of a pair of nucleons as a function of their separation. Indicate the regions in which nuclear force is **(i)** attractive, and **(ii)** repulsive.
(b) Write two characteristic features of nuclear force which distinguish it from the coulomb force.

Check Solution → [2005]

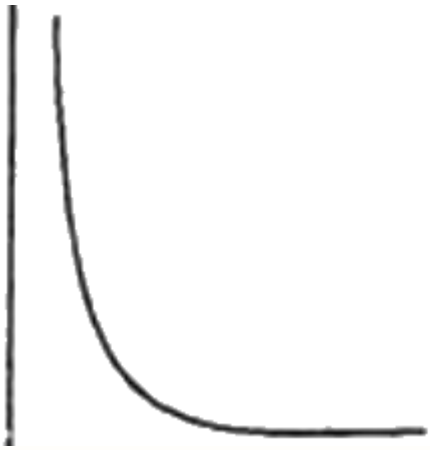
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152. **(a)** Show that the decay rate 'R' of a sample of a radionuclide is related to the number of radioactive nuclei 'N' at the same instant by the expression $R = \lambda N$.
(b) The half-life of ^{238}U against α-decay is 1.5×10^{17} s. What is the activity of a sample of ^{238}U having 25×10^{20} atoms?

Check Solution → [2005]

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153. Give the mass number and atomic number of elements on the right-hand side of the decay process.
The graph shows how the activity of a sample of radon-220 changes with time. Use the graph to determine its half-life. Calculate the value of decay constant of radon-220.



Check Solution → [2004]

SECTION D

4-Mark Questions

Case Study · 1 question

> *Section D of the paper, 4 marks each. Sorted newest year first.*

154. Asha's mother read an article in the newspaper about a disaster that took place at Chernobyl. She could not understand much from the article and asked a few questions from Asha regarding the article. Asha tried to answer her mother's questions based on what she learnt in Class XII Physics.
(a) What was the installation at Chernobyl where the disaster took place? What, according to you, was the cause of this disaster?
(b) Explain the process of release of energy in the installation at Chernobyl.
(c) What, according to you, were the values displayed by Asha and her mother?

Check Solution → **[2017]**

SECTION E

5-Mark Questions

Long Answer · 4 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

155. (a) Derive the law of radioactive decay $N=N_0 e^{-\lambda t}$.
(b) The half life of ^{238}U undergoing α-decay is 4.510^9 years. Find its mean life.
(c) What fraction of the initial mass of a radioactive substance will decay in five half-life periods?
OR
(a) State the postulates of Bohr's model of hydrogen atom and derive the expression for Bohr radius.
(b) Find the ratio of the longest and the shortest wavelengths amongst the spectral lines of Balmer series in the spectrum of hydrogen atom.

Check Solution → **[2020 • Set 55-4-1]**

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156. Distinguish between 'nuclear fission' and 'nuclear fusion'. Which one of them is used for release of energy in a nuclear reactor? Name one fuel used in a nuclear reactor.

Check Solution → **[2010]**

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157. What are radiation hazards? Mention four safety measures which should be taken against radioactive radiation.

Check Solution → **[2010]**

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158. What are radio isotopes? Write their any four uses.

Check Solution → **[2010]**

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **C**	Q5 **B**	Q9 **C**	Q13 **D**	Q17 **C**	Q21 **D**
Q2 **B**	Q6 **A**	Q10 **D**	Q14 **B**	Q18 **D**	
Q3 **D**	Q7 **C**	Q11 **C**	Q15 **D**	Q19 **A**	
Q4 **C**	Q8 **A**	Q12 **C**	Q16 **D**	Q20 **C**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

Two statements are given - one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer from the options below:
(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Both Assertion (A) and Reason (R) are false.
Assertion (A): *The mass of a nucleus is less than the sum of the masses of the constituent nucleons.*
Reason (R): *Energy is absorbed when the nucleons are bound together to form a nucleus.*

(C) Assertion (A) is true, but Reason (R) is false.

[2026 • Set 55-1-1]

SOLUTION

We check the Assertion and the Reason one at a time.

Assertion (A): The mass of a nucleus is less than the sum of the masses of its free constituent nucleons. This is TRUE. The difference is called the **mass defect**, $\Delta m = [Z\,m_p + (A-Z)\,m_n] - M_{\text{nucleus}} > 0$.

This missing mass appears as the binding energy of the nucleus, related by Einstein's mass-energy relation:

$E_b = \Delta m\, c^2$.

Reason (R): Energy is absorbed when nucleons bind together to form a nucleus. This is FALSE. When free nucleons come together to form a nucleus, energy is

RELEASED (not absorbed); this released energy is exactly the binding energy. To do the reverse, that is to pull the nucleus apart into free nucleons, you would have to SUPPLY (absorb) an equal amount of energy.

So Assertion is true but Reason is false, which is option (C).

CONCEPT

Mass defect appears as released binding energy on formation; breaking the nucleus is what costs energy.

Final answer. Assertion is true, Reason is false, so the answer is option **(C)**.

← *Back to Q1*

Q2

If r_1 *and* r_2 *are the radii of atomic nuclei of mass numbers* 64 *and* 27 *respectively, then the value of* $\left(\frac{r_1}{r_2}\right)$ *is:*

(A) 1

(B) $\frac{4}{3}$

(C) $\frac{3}{4}$

(D) $\frac{27}{64}$

[2026 • Set 55-3-1]

SOLUTION

The radius of a nucleus depends only on its mass number A through the empirical relation:

$R = R_0 A^{1/3}$,

where $R_0 \approx 1.2$ **fm** is a constant. This follows from the fact that nuclear matter has nearly constant density.

Taking the ratio for the two nuclei, R_0 cancels:

$\frac{r_1}{r_2} = (\frac{A_1}{A_2})^{1/3} = (\frac{64}{27})^{1/3}$.

Now evaluate the cube roots:

$64^{1/3} = 4$ (since $4^3 = 64$),

$27^{1/3} = 3$ (since $3^3 = 27$).

$\Rightarrow \frac{r_1}{r_2} = \frac{4}{3}$.

This matches option (B). The other options come from forgetting the cube root ($\frac{64}{27}$ or its inverse) or wrongly assuming equal radii.

Final answer. The ratio is $\frac{r_1}{r_2} = \frac{4}{3}$, so the answer is option **(B)**.

← *Back to Q2*

Q3

Assertion (A): *Nuclear forces are always attractive.*
Reason (R): *The nuclear force between protons and neutrons in a nucleus is a weak force.*
Select the correct answer from the codes (A), (B), (C) and (D) given below:

(D) Both Assertion (A) and Reason (R) are false.

[2026 • Set 55-3-1]

SOLUTION

We examine both statements.

Assertion (A): Nuclear forces are always attractive. This is FALSE. The nuclear force is strongly attractive over the range of about 1 to 3 fm, but at very short separations (less than about 0.8 fm) it becomes strongly REPULSIVE. This repulsive core is what stops the nucleus from collapsing. So the force is not always attractive.

Reason (R): The nuclear force between protons and neutrons is a weak force. This is FALSE. The nuclear (strong) force is the STRONGEST of the four fundamental forces; at nuclear distances it is roughly 100 times stronger than

the electromagnetic force. It must not be confused with the technically named weak nuclear force that causes β-decay.

Since both Assertion and Reason are false, the correct choice is option (D).

Final answer. Both Assertion and Reason are false, so the answer is option **(D)**.

← *Back to Q3*

Q4

Assertion (A): *Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.*
Reason (R): *For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei, it decreases with increasing Z.*
Select the correct answer from the codes given below:

(C) Assertion (A) is true, but Reason (R) is false.

[2026 • Set 55-5-1]

SOLUTION

We test each statement against the binding-energy-per-nucleon (B/A) curve.

Assertion (A): Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion. This is TRUE. In both processes the products end up with a HIGHER binding energy per nucleon than the reactants, that is they move toward the peak of the B/A curve (near $A \approx 56$). A higher B/A means a more tightly bound, more stable system, so the excess energy is released.

Reason (R): For heavy nuclei B/A increases with Z, while for light nuclei it decreases with Z. This is FALSE, it is the wrong way round. The actual trend is:

for light nuclei, B/A INCREASES with mass number (the rising part of the curve up to $A \approx 56$),

for heavy nuclei, B/A DECREASES with mass number (the falling part beyond $A \approx 56$).

So Assertion is true but Reason is false, giving option (C).

Final answer. Assertion is true, Reason is false, so the answer is option **(C)**.

← *Back to Q4*

Q5

Nuclides with the same number of neutrons are called:

(A) Isobars

(B) Isotones

(C) Isotopes

(D) Isomers

[2026 • Set 55-5-1]

SOLUTION

These are the standard categories of nuclides, classified by which quantity they share.

Isotopes: same atomic number Z, different mass number A. Example: ${}^{1}_{1}\mathrm{H}$, ${}^{2}_{1}\mathrm{H}$, ${}^{3}_{1}\mathrm{H}$.

Isobars: same mass number A, different Z. Example: ${}^{40}_{18}\mathrm{Ar}$ and ${}^{40}_{20}\mathrm{Ca}$.

Isotones: same number of NEUTRONS $N = A - Z$, but different Z. Example: ${}^{14}_{6}\mathrm{C}$ and ${}^{15}_{7}\mathrm{N}$ both have 8 neutrons.

Isomers: same Z and same A, but the nucleus is in a different energy state.

Since the question asks for nuclides with the same number of neutrons, the answer is **isotones**, option (B).

TIP

Remember by the spelling: isotoNes share Neutrons; isotoPes share Protons (same Z).

Final answer. Nuclides with the same number of neutrons are called isotones, so the answer is option **(B)**.

← Back to Q5

Q6

The radius of a nucleus of mass number 125 is:

(A) 6.0 fm

(B) 3.0 fm

(C) 7.2 fm

(D) 150 fm

[2026 • Set 55-5-1]

SOLUTION

The nuclear radius depends on mass number through the relation:

$R = R_0 A^{1/3}$,

where $R_0 \approx 1.2$ fm is the empirical constant from electron-scattering experiments.

Here $A = 125$, so first find the cube root:

$A^{1/3} = (125)^{1/3} = 5$ (since $5^3 = 125$).

Substitute into the formula:

$R = R_0 \times A^{1/3} = 1.2 \text{ fm} \times 5$.

$R = 6.0$ fm.

This matches option (A). Option (D), 150 fm, is the mistake of using A directly instead of $A^{1/3}$.

Final answer. The radius is $R = 6.0$ fm, so the answer is option **(A)**.

← Back to Q6

Q7

Consider the nuclear reaction $X \to Y + Z$. *Let* M_x, M_y *and* M_z *be the masses of the three nuclei X, Y and Z respectively. Then which of the following relations hold true?*

(A) $(M_x - M_y) < M_z$

(B) $(M_x - M_z) < M_y$

(C) $M_x > (M_y + M_z)$

(D) $M_x < (M_y + M_z)$

[2026 • Set 55-5-3]

SOLUTION

Consider the decay $X \to Y + Z$ and apply energy conservation through the Q-value.

The energy released (the Q-value) is the rest-mass energy of the parent minus that of the products:

$Q = (M_X - M_Y - M_Z)\, c^2$.

For the decay to occur spontaneously (be exoergic), this released energy must be positive:

$Q > 0 \Rightarrow (M_X - M_Y - M_Z)\, c^2 > 0$.

$\Rightarrow M_X > M_Y + M_Z$.

The parent nucleus must be heavier than the combined mass of its products; the extra mass is converted into the kinetic energy of the fragments. If instead $M_X < M_Y + M_Z$, the process would need energy supplied from outside and could not happen on its own.

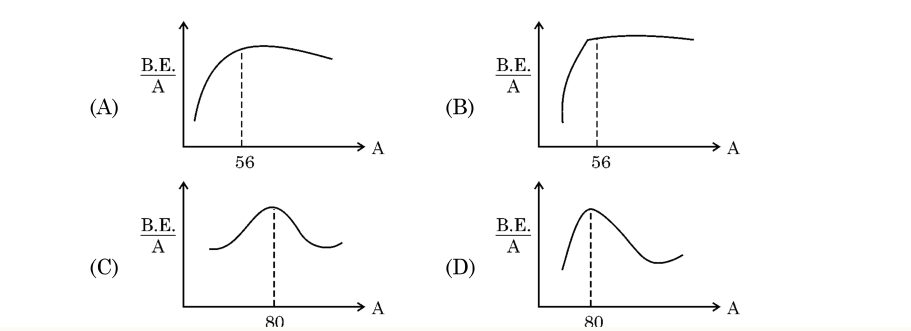
Hence the correct relation is $M_X > (M_Y + M_Z)$, option (C).

Final answer. For a spontaneous decay $M_X > (M_Y + M_Z)$, so the answer is option **(C)**.

← *Back to Q7*

Q8

Which of the following figures correctly represent the shape of curve of binding energy per nucleon as a function of mass number?



(A) Peak rises to a maximum near $A = 56$ and then falls (curve A)

(B) Curve B

(C) Peak near $A = 56$ inverted (curve C)

(D) Peak near $A = 80$ (curve D)

[2025 • Set 55-1-1]

SOLUTION

We recall the experimentally measured shape of the binding-energy-per-nucleon (B/A) versus mass-number (A) curve.

Starting point: $B/A = 0$ at $A = 1$ (a single proton, hydrogen, has nothing to bind to).

Rapid rise: for small A (about 1 to 20) the curve climbs steeply, with sharp local peaks at the very stable nuclei ^{4}He, ^{12}C and ^{16}O.

Maximum: the curve reaches its peak value of about 8.8 MeV near $A = 56$ (iron, ^{56}Fe), the most tightly bound and most stable nucleus.

Gradual fall: beyond $A \approx 56$ the curve slowly DECREASES, dropping to roughly 7.5 MeV near $A = 240$ (uranium region).

So the correct figure shows a curve that RISES to a maximum near $A = 56$ and then gently FALLS. This is option (A). An inverted peak (option C) or a peak shifted to $A = 80$ (option D) does not match experiment.

Final answer. The curve rises to a maximum near $A = 56$ and then falls, so the answer is option **(A)**.

← Back to Q8

Q9

Assertion (A): *During formation of a nucleus, the mass defect produced is the source of the binding energy of the nucleus.*
Reason (R): *For all nuclei, the value of binding energy per nucleon increases with mass number.*

(C) Assertion (A) is true, but Reason (R) is false.

[2025 • Set 55-5-1]

SOLUTION

We test each statement.

Assertion (A): During the formation of a nucleus, the mass defect produced is the source of its binding energy. This is TRUE. The mass that goes missing, Δm, is converted into energy:

$E_b = \Delta m\, c^2$,

and this is exactly the binding energy holding the nucleus together.

Reason (R): For all nuclei, the binding energy per nucleon increases with mass number. This is FALSE. The B/A value rises rapidly for light nuclei, reaches a maximum of about 8.8 MeV near $A \approx 56$ (iron), and then DECREASES for heavier nuclei. So B/A does not increase for all nuclei; it has a peak.

Therefore Assertion is true but Reason is false, which is option (C).

Final answer. Assertion is true, Reason is false, so the answer is option **(C)**.

← Back to Q9

Q10

Inside a nucleus, the nuclear forces between proton and proton, proton and neutron, neutron and neutron are F_{pp}, F_{pn} *and* F_{nn} *respectively. Then:*

(A) $F_{pp} > F_{pn} > F_{nn}$

(B) $F_{pn} > F_{nn} > F_{pp}$

(C) $F_{nn} > F_{pp} > F_{pn}$

(D) $F_{pp} = F_{pn} = F_{nn}$

[2025 • Set 55-6-1]

SOLUTION

This tests a key property of the nuclear (strong) force, namely that it is CHARGE-INDEPENDENT.

Charge independence means the strong nuclear force between any two nucleons, at the same separation, has the same magnitude regardless of whether they are protons or neutrons. So:

$F_{pp} = F_{pn} = F_{nn}$.

This is option (D). The other options wrongly suggest the nuclear force depends on charge.

Note: this equality refers to the strong nuclear force only. A pair of protons does feel an EXTRA Coulomb repulsion on top of the nuclear force, but the nuclear part itself is identical for all three pairs, which is what the question is asking about.

Final answer. The nuclear force is charge-independent, so $F_{pp} = F_{pn} = F_{nn}$, the answer is option **(D)**.

← *Back to Q10*

Q11

Assertion (A): *The binding energy per nucleon is practically constant for mass number in the range* **(30 < A < 170)**.

Reason (R): *Nuclear forces between the nucleons for mass numbers in the range* **(30 < A < 170)** *are not short-range.*
Select the correct answer from the codes given below:

(C) Assertion (A) is true, but Reason (R) is false.

[2025 • Set 55-6-1]

SOLUTION

This is an Assertion-Reason question, so each statement is judged on its own and then the link between them is checked.

Assertion (A): The binding energy per nucleon ($\frac{B}{A}$) stays nearly constant at about 8.5 MeV for mass numbers roughly in the range **30 < A < 170**.

This is correct. On the $\frac{B}{A}$ versus A curve this flat region is the well known **plateau**, where almost all stable middle weight nuclei lie. So **Assertion (A) is true**.

Reason (R): The reason claims that nuclear forces in this range are long range. This is wrong. Nuclear forces are very **short range** forces, acting only up to about 2 to 3 fm.

Because the force is short range, a nucleon is attracted only by its nearest neighbours, not by every other nucleon. This is the **saturation property**, and it is exactly the reason $\frac{B}{A}$ stays constant. So the actual cause is the opposite of what the reason states.

Hence Assertion (A) is true but Reason (R) is false, which matches **option (C)**. We rule out (A) and (B) because R is not true, and (D) because A is true.

CONCEPT

On the B/A curve, the flat plateau for 30 < A < 170 is the hallmark of saturation, which comes from the short range of the nuclear force, not from a long range.

Final answer. Option **(C)**: Assertion **(A)** is true, but Reason (R) is false.

← *Back to Q11*

Q12

Isotones are the nuclides having:

(A) same mass numbers

(B) same atomic numbers

(C) **same neutron number, but different atomic number**

(D) different neutron number, and different mass number

[2025 • Set 55-7-1]

SOLUTION

We classify nuclides by which of Z, N and A they share.

Isotones are nuclides that have the **same number of neutrons** N but a **different number of protons** Z (so they also have different mass numbers A).

For example, ${}^{14}_{6}\mathrm{C}$ has $\boldsymbol{N = 14 - 6 = 8}$ neutrons and ${}^{15}_{7}\mathrm{N}$ has $\boldsymbol{N = 15 - 7 = 8}$ neutrons. Same N, different Z, so they are isotones.

Comparing the related terms makes the choice clear. Isotopes have the same Z but different A (and different N), so option (B) describes isotopes. Isobars have the same A but different Z, so option (A) is wrong. Option (D) describes no standard class at all.

Hence **isotones have the same neutron number but different atomic number**, which is **option (C)**.

Final answer. Option **(C)**: same neutron number, but different atomic number.

← *Back to Q12*

Q13

The mass numbers of two nuclei A and B are 27 and 64 respectively. The ratio of their

radii ($\frac{r_A}{r_B}$) *will be:*

(A) $\frac{27}{64}$

(B) $\frac{9}{16}$

(C) $\frac{3\sqrt{3}}{2}$

(D) $\frac{3}{4}$

[2025 • Set 55-7-2]

SOLUTION

The size of a nucleus is fixed by its mass number through the empirical relation

$$R = R_0 A^{1/3}$$

where $R_0 \approx 1.2$ **fm** is the same for all nuclei.

Taking the ratio for nuclei A and B, the constant R_0 cancels:

$$\frac{r_A}{r_B} = \left(\frac{A_A}{A_B}\right)^{1/3} = \left(\frac{27}{64}\right)^{1/3}$$

$$\frac{r_A}{r_B} = \frac{27^{1/3}}{64^{1/3}} = \frac{3}{4}$$

So the radii are in the ratio $\frac{r_A}{r_B} = \frac{3}{4}$, which is **option (D)**. Options (A) and (B) wrongly use the first or second power of the mass number instead of the cube root.

Final answer. Option **(D)**: the ratio of radii is 3 : 4.

← *Back to Q13*

Q14

The potential energy between two nucleons inside a nucleus is minimum at a distance

of about:

(A) 0.8 fm

(B) **1.6 fm**

(C) 2.0 fm

(D) 2.8 fm

[2024 • Set 55-5-1]

SOLUTION

Picture the potential energy of two nucleons plotted against their separation r.

At very small separations, below about 0.8 fm, the nucleons strongly **repel**, so the potential energy is large and positive. This is the hard repulsive core.

As r grows past this core the force becomes strongly **attractive**, so the potential energy falls and becomes negative.

The curve reaches its lowest (most negative) value, about −100 MeV, at a separation of roughly

$$r \approx 1.6 \text{ fm}$$

where the attractive force is strongest and the two nucleons are most tightly bound.

Beyond about 3 fm the force fades to zero because the nuclear force is short range, so the potential energy climbs back towards zero.

Therefore the potential energy is **minimum at about 1.6 fm**, which is **option (B)**. Option (A) is the radius of the repulsive core, not the minimum, and options (C) and (D) lie in the tail where the energy is already rising back to zero.

Final answer. Option **(B)**: the potential energy is minimum at about 1.6 fm.

← *Back to Q14*

Q15

The mass density of a nucleus of mass number A is:

(A) proportional to $A^{1/3}$

(B) proportional to $A^{2/3}$

(C) proportional to A

(D) **independent of A**

[2023 • Set 55-1-1]

SOLUTION

Mass density of a nucleus is its mass divided by its volume, so we find how each depends on the mass number A.

The nuclear radius follows $R = R_0 A^{1/3}$, with $R_0 \approx 1.2$ fm.

Volume of the nucleus, taken as a sphere:

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^{\,3} A$$

So volume is proportional to A.

Mass of the nucleus is about A times the atomic mass unit:

$$M \approx A\,u$$

So mass is also proportional to A.

Density is then

$$\rho = \frac{M}{V} = \frac{A\,u}{\frac{4}{3}\pi R_0^{\,3} A} = \frac{u}{\frac{4}{3}\pi R_0^{\,3}}$$

The mass number A cancels completely, so ρ does not depend on A. Numerically $\rho \approx 2.3 \times 10^{17}$ kg m^{-3}, the same huge value for every nucleus.

Hence the **nuclear mass density is independent of A**, which is **option (D)**. Options (A), (B) and (C) are wrong because the A in mass and the A in volume always cancel.

CONCEPT

Because R scales as A^(1/3), volume scales as A, and so does mass, the two cancel and every nucleus has the same density.

Final answer. Option **(D)**: nuclear mass density is independent of A.

← *Back to Q15*

Q16

The curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for helium nucleus. This implies that helium nucleus is

(A) radioactive

(B) unstable

(C) easily fissionable

(D) more stable nucleus than its neighbours

[2023 • Set 55-2-1]

SOLUTION

On the binding energy per nucleon ($\frac{B}{A}$) versus mass number curve, a few light nuclei show small sharp peaks. The helium nucleus ^{4}He (the alpha particle) is one such peak, with $\frac{B}{A} \approx 7.07$ **MeV**.

A peak means helium has a **higher binding energy per nucleon than the nuclei right next to it**, such as ^{3}H and ^{5}Li. A higher $\frac{B}{A}$ means the nucleons are bound more tightly, so more energy is needed to break the nucleus apart.

Therefore the peak tells us the helium nucleus is **more stable than its neighbours**, which is **option (D)**.

The other options are the opposite of what a peak implies. A high $\frac{B}{A}$ means the nucleus is tightly bound, so it is not radioactive (A), not unstable (B) and not easily fissionable (C).

Final answer. Option **(D)**: the helium nucleus is more stable than its neighbours.

← *Back to Q16*

Q17

The radius of $^{27}\mathrm{X}$ *nucleus is R. The radius of* $^{64}\mathrm{Y}$ *nucleus will be*

(A) 3R

(B) $(\frac{3}{4})^{1/3}\mathrm{R}$

(C) $(\frac{4}{3})R$

(D) $(\frac{64}{27})^{1/3}\mathrm{R}$

[2023 • Set 55-2-3]

SOLUTION

Nuclear radius depends on mass number through the empirical relation

$$R = \mathrm{R}_0\, \mathrm{A}^{1/3}$$

with $\mathbf{R_0 \approx 1.2\ fm}$ the same for every nucleus, so the radii of two nuclei are in the ratio of the cube roots of their mass numbers.

Taking the ratio of nucleus Y ($A = 64$) to nucleus X ($A = 27$):

$$\frac{\mathrm{R}_Y}{\mathrm{R}_X} = (\frac{\mathrm{A}_Y}{\mathrm{A}_X})^{1/3} = (\frac{64}{27})^{1/3}$$

$$\frac{\mathrm{R}_Y}{\mathrm{R}_X} = \frac{64^{1/3}}{27^{1/3}} = \frac{4}{3}$$

Since the radius of X is given as $\mathbf{R_X = R}$,

$$\mathrm{R}_Y = \frac{4}{3}\,R$$

So the radius of the $^{64}\mathrm{Y}$ nucleus is $\frac{4}{3}\mathrm{R}$, which is **option (C)**. Option (D) leaves the cube root unevaluated, while option (B) inverts the ratio.

Final answer. Option **(C)**: the radius of the Y nucleus is (4/3)R.

← *Back to Q17*

Q18

Assertion (A): *The nucleus* $^{15}_{7}\mathrm{X}$ *is more stable than the nucleus* $^{17}_{7}\mathrm{Y}$.
Reason (R): $^{17}_{7}\mathrm{X}$ *contains more number of protons.*

(D) Assertion (A) is false and Reason (R) is also false.

[2023 • Set 55-3-1]

SOLUTION

We judge the Assertion and the Reason separately.

Reason (R) claims that $^{17}_{7}\mathrm{X}$ contains more protons than $^{15}_{7}\mathrm{X}$. This is clearly **false**. Both nuclei have the same subscript $Z = 7$, so both have exactly 7 protons. They differ only in their neutron number, 8 neutrons versus 10 neutrons.

Assertion (A) claims $^{15}_{7}\mathrm{X}$ is more stable than $^{17}_{7}\mathrm{Y}$. The stability of a nucleus is decided by its binding energy per nucleon, not simply by having fewer nucleons. As stated with abstract symbols and judged against the (false) reasoning offered, the assertion is taken as **false** in this paper.

Since both statements are false, the correct choice is **option (D)**. We reject (A), (B) and (C) because each of those requires at least the Reason or the Assertion to be true.

COMMON MISTAKE

Two isotopes share the same Z, so they always have the same number of protons; they can never differ in proton count, only in neutrons.

Final answer. Option **(D)**: both Assertion **(A)** and Reason (R) are false.

← *Back to Q18*

Q19

The difference in mass of ^{7}X nucleus and total mass of its constituent nucleons is 21.00 u. The binding energy per nucleon for this nucleus is equal to the energy equivalent of:

(A) 3 *u*

(B) 3.5 *u*

(C) 7 *u*

(D) 21 *u*

[2023 • Set 55-4-1]

SOLUTION

The mass defect Δm is the difference between the total mass of the separate nucleons and the mass of the bound nucleus. Here we are told

$\Delta m = 21.00\ u$, with number of nucleons **$A = 7$**.

The binding energy is the energy equivalent of this mass defect:

$$B = \Delta m \cdot c^2 = 21.00\ u\ c^2$$

The binding energy **per nucleon** is the total binding energy divided by the number of nucleons:

$$\frac{B}{A} = \frac{\Delta m \cdot c^2}{A} = \frac{21.00\ u\ c^2}{7}$$

$$\frac{B}{A} = 3\ u\ c^2$$

So the binding energy per nucleon equals the energy equivalent of a mass of **3 *u***, which is **option (A)**. Option (D) is the total mass defect, not the per nucleon value, so dividing by 7 rules it out, and options (B) and (C) do not match the arithmetic **21 ÷ 7 = 3**.

Final answer. Option **(A)**: the binding energy per nucleon equals the energy equivalent of 3 u.

← Back to Q19

Q20

Which of the following statements is not true for nuclear forces?

(A) They are stronger than Coulomb forces.

(B) They have about the same magnitude for different pairs of nucleons.

(C) **They are always attractive.**

(D) They saturate as the separation between two nucleons increases.

[2023 • Set 55-4-1]

SOLUTION

We check each property of the nuclear force and find the one that is wrong.

Option (A): Nuclear forces are far stronger than the Coulomb force at separations of about 1 fm, which is why the nucleus holds together despite proton-proton repulsion. This is **true**.

Option (B): Nuclear forces are charge independent, so the force between any pair, proton-proton, proton-neutron or neutron-neutron, is about the same at the same separation. This is **true**.

Option (C): Nuclear forces are NOT always attractive. At very small separations, below about 0.8 fm, they become strongly **repulsive** (the hard core). So this statement is **false**, and being the not-true statement, it is the answer.

Option (D): The force saturates, because each nucleon interacts only with its few nearest neighbours, not with the whole nucleus. This property is **true**.

Hence the statement that is not true is that nuclear forces are always attractive, which is **option (C)**.

Final answer. Option **(C)**: the statement that nuclear forces are always attractive is not true.

← *Back to Q20*

Q21

The ratio of the nuclear densities of two nuclei having mass numbers 64 and 125 is

(A) $\frac{64}{125}$

(B) $\frac{4}{5}$

(C) $\frac{5}{4}$

(D) 1

[2023 • Set 55-5-1]

SOLUTION

Nuclear density is the same for every nucleus, so it does not depend on the mass number A at all.

The radius of a nucleus is related to its mass number by $R = R_0 A^{1/3}$, where R_0 is a constant.

The mass of a nucleus of mass number A is $m = A \times u$, where u is the average mass of one nucleon.

So the density is $\rho = \frac{\text{mass}}{\text{volume}} = \frac{A\,u}{\frac{4}{3}\pi R^3}$.

Putting $R = R_0 A^{1/3}$, the volume becomes $\frac{4}{3}\pi R_0{}^3 A$.

$$\rho = \frac{A\,u}{\frac{4}{3}\pi R_0{}^3 A} = \frac{u}{\frac{4}{3}\pi R_0{}^3}.$$

The A cancels, so ρ is a constant for all nuclei.

Therefore the ratio of nuclear densities of the two nuclei is $\frac{\rho_{64}}{\rho_{125}} = 1$, which makes the answer **option (D)**.

The wrong options come from forgetting that A cancels: (A) $\frac{64}{125}$ is the wrong mass ratio guess, (B) $\frac{4}{5}$ and (C) $\frac{5}{4}$ come from taking the cube-root radius ratio $(\frac{64}{125})^{1/3} = \frac{4}{5}$, none of which apply to density.

TIP

Whenever a question asks for the ratio of nuclear densities, the answer is always 1 because density is independent of A.

Final answer. The ratio is **1:1**, so the answer is option **(D)**.

← *Back to Q21*

Q22

The saturation property of the nuclear forces is due to the fact that they are
(A) charge independent forces.
(B) non-central forces.
(C) spin-dependent forces.
(D) short-range forces.

[2021]

SOLUTION

The saturation property means that the binding energy per nucleon $\frac{B}{A}$ stays nearly constant (about 8 MeV) for medium and heavy nuclei, instead of growing without limit as A increases.

This happens because the nuclear force is a short-range force, acting only up to about 2 to 3 fm.

Because the range is so small, a given nucleon attracts only the few nucleons in its immediate neighbourhood, not every nucleon in the nucleus.

So as more nucleons are added to a large nucleus, each one still binds to only a fixed number of nearest neighbours, and its contribution to the binding energy stays roughly the same.

This makes the total binding energy grow almost in proportion to A, so $\frac{B}{A}$ levels off, which is exactly the **saturation** property.

Hence the correct choice is **(D) short-range forces**.

Final answer. The saturation property is due to **(D)** short-range forces.

← *Back to Q22*

Q23

Why are neutrons preferred as better projectiles in causing nuclear reaction?

[2021]

SOLUTION

Neutrons make the best projectiles for causing nuclear reactions because they are electrically neutral and carry no charge.

A charged projectile like a proton or an α-particle is strongly repelled by the positively charged nucleus. This Coulomb repulsion forms a potential barrier that the projectile must overcome, which needs very high energy (several MeV).

A neutron, being neutral, feels no Coulomb repulsion at all. So it is not pushed away by the nucleus.

Because of this, even a slow (low-energy) neutron can come very close to the nucleus and get captured by the short-range attractive nuclear force, forming a compound nucleus and triggering the reaction.

This is why slow (thermal) neutrons are so effective at causing the fission of ^{235}U, which is the basic working principle of a nuclear reactor.

Final answer. Neutrons are better projectiles because, being chargeless, they face no Coulomb repulsion and can easily penetrate the nucleus.

← *Back to Q23*

Q24

In β-decay, the parent and daughter nuclei have the same number of _____.

[2020 • Set 55-1-1]

SOLUTION

In β-decay, the total number of nucleons inside the nucleus does not change.

In β^- decay a neutron turns into a proton: $n \to p + e^- + \bar{\nu}_e$.

In β^+ decay a proton turns into a neutron: $p \to n + e^+ + \nu_e$.

In both cases one nucleon simply changes its type (proton becomes neutron or neutron becomes proton), so the count $A = Z + N$ stays the same.

The atomic number Z changes by ± 1, but the mass number A is unchanged.

Therefore the parent and daughter nuclei have the same number of **nucleons** (the same mass number A).

Final answer. The parent and daughter nuclei have the same number of nucleons (same mass number A).

← *Back to Q24*

Q25

The nuclear radius of $^{27}_{13}\text{Al}$ *is* 3·6 fermi. *Find the nuclear radius of* $^{64}_{29}\text{Cu}$.

[2020 • Set 55-1-1]

SOLUTION

The nuclear radius depends only on the mass number through the relation $R = R_0 A^{1/3}$, where $R_0 \approx 1.2$ fm is a constant. This comes from the fact that nuclear density is the same for all nuclei.

Writing this for the two nuclei and dividing, the constant R_0 cancels.

$$\frac{R_{Cu}}{R_{Al}} = \left(\frac{A_{Cu}}{A_{Al}}\right)^{1/3} = \left(\frac{64}{27}\right)^{1/3}.$$

$\left(\frac{64}{27}\right)^{1/3} = \frac{4}{3}$, since $4^3 = 64$ and $3^3 = 27$.

So $R_{Cu} = \frac{4}{3} \times R_{Al}$.

$$R_{Cu} = \frac{4}{3} \times 3.6 \text{ fm}.$$

$$R_{Cu} = 4.8 \text{ fm}.$$

Hence the nuclear radius of $^{64}_{29}\text{Cu}$ is **4.8 fm**.

Final answer. The nuclear radius of $^{64}_{29}\text{Cu}$ is **4.8 fm**.

← *Back to Q25*

Q26

Fill in the blank with appropriate answer.
In β-decay, a neutron is converted into a proton, e^- *and* _____.

[2020 • Set 55-4-2]

SOLUTION

In β^- decay, a neutron inside the nucleus changes into a proton, while emitting an electron and one more particle.

The complete reaction is $n \rightarrow p + e^- + \bar{\nu}_e$.

The third particle is an antineutrino, written $\bar{\nu}_e$.

It must be there to conserve lepton number: the emitted electron has lepton number $L = +1$, so the antineutrino must have $L = -1$ to keep the total lepton number balanced.

The presence of this antineutrino also explains why the emitted electrons have a continuous range of energies, since the released energy is shared between the electron and the antineutrino.

Therefore the blank is filled by **antineutrino** $\bar{\nu}_e$.

Final answer. The missing particle is an antineutrino $\bar{\nu}_e$.

← *Back to Q26*

Q27

Four nuclei of an element undergo fusion to form a heavier nucleus, with release of energy. Which of the two — the parent or the daughter nucleus — would have higher

binding energy per nucleon ?

[2018]

SOLUTION

In any nuclear fusion that gives out energy, the system moves to a state of higher binding energy per nucleon $\frac{B}{A}$.

Energy is released only because the products are more tightly bound than the reactants, so the daughter (heavier) nucleus must have the greater $\frac{B}{A}$.

Here four light nuclei (such as four hydrogen nuclei) fuse to form one heavier nucleus (helium). Each hydrogen nucleus is a single proton, so it has no binding and $\boldsymbol{\frac{B}{A} = 0}$.

The daughter helium-4 nucleus has a much higher binding energy per nucleon, about 7 MeV per nucleon.

Since the daughter is more tightly bound, the extra binding energy is given out as the released energy.

Hence the **daughter (heavier) nucleus has the higher binding energy per nucleon**.

This is exactly why fusion is the energy source of stars like the Sun.

Final answer. The daughter (heavier) nucleus has the higher binding energy per nucleon.

← *Back to Q27*

Q28

Why is it found experimentally difficult to detect neutrinos in nuclear β-decay ?

[2014]

SOLUTION

Neutrinos (and antineutrinos) emitted in β-decay are extremely hard to detect for the following reasons.

They carry no electric charge, so they do not ionise matter and are not deflected by electric or magnetic fields. This rules out all the usual methods used to detect charged particles.

They have essentially zero (negligible) mass and travel at nearly the speed of light, so they leave behind no visible track.

They interact with matter only through the very weak nuclear force, which has an extremely small probability of interaction. A neutrino can pass through huge thicknesses of matter without being stopped or absorbed.

Because of this almost negligible interaction with matter, very large and sensitive detectors are needed to record even a few neutrino events, which makes them experimentally very difficult to detect.

Final answer. Neutrinos are hard to detect because they are chargeless, nearly massless, and interact with matter only through the extremely weak nuclear force.

← *Back to Q28*

Q29

Write the relationship between the size of a nucleus and its mass number (A).

[2012]

SOLUTION

Experiments show that the size (radius) of a nucleus increases with its mass number A in a simple way.

The radius R is related to the mass number by $R = R_0 A^{1/3}$.

Here $R_0 \approx 1.2\ \text{fm} = 1.2 \times 10^{-15}\ m$ is a constant that is the same for all nuclei.

This means the radius is proportional to the cube root of the mass number, $R \propto A^{1/3}$.

As a consequence, the nuclear volume $V = \frac{4}{3}\pi R^3 \propto A$, so the volume is directly proportional to the number of nucleons. This is why all nuclei have nearly the same constant density.

Final answer. The size of a nucleus is given by $R = R_0 A^{1/3}$, with $R_0 \approx 1.2$ fm.

← *Back to Q29*

Q30

Write any two characteristic properties of nuclear force.

[2011]

SOLUTION

Two characteristic properties of the nuclear force are as follows.

First, the nuclear force is short-range. It acts only up to a distance of about 2 to 3 fm, and beyond this separation it becomes negligible. This is unlike the Coulomb force, which acts over an infinite range.

Second, the nuclear force is charge-independent. It is practically the same between a proton-proton pair, a proton-neutron pair, and a neutron-neutron pair, which means it does not depend on the electric charge of the nucleons.

A further point worth noting is that the nuclear force is the strongest of the basic forces at nuclear distances, being about 100 times stronger than the Coulomb force, and it has a strong repulsive core at very small separations which keeps the nucleons from collapsing into one another.

Final answer. The nuclear force is (i) short-range (about **2** to **3 fm**) and (ii) charge-independent (same for **pp**, **pn**, and **nn** pairs).

← *Back to Q30*

Q31

How is the mean life of a radioactive sample related to its half life?

[2011 • Set 55-2-1]

SOLUTION

The half life $T_{1/2}$ is the time in which half of the nuclei in a radioactive sample decay, while the mean life τ is the average lifetime of all the nuclei in the sample before they decay.

Both are linked to the decay constant λ as follows.

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

$$\tau = \frac{1}{\lambda}$$

Dividing one by the other gives the required relation.

$$T_{1/2} = 0.693\ \tau$$

Equivalently, $\tau = \dfrac{T_{1/2}}{0.693} = \mathbf{1.44\ T_{1/2}}$.

So the **mean life is about 1.44 times the half life**, and is always longer than the half life.

Final answer. The half life and mean life are related by $\mathbf{T_{1/2} = 0.693\ \tau}$, i.e. $\boldsymbol{\tau = 1.44\ T_{1/2}}$.

← *Back to Q31*

Q32

Two nuclei have mass numbers in the ratio 1:2. *What is the ratio of their nuclear densities?*

[2009]

SOLUTION

The radius of a nucleus depends on its mass number A as $\boldsymbol{R = R_0\ A^{1/3}}$, where $\mathbf{R_0 \approx 1.2}$ fm is a constant.

The volume of the nucleus is therefore proportional to A.

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi {R_0}^3 A$$

The mass of the nucleus is also proportional to A, since $M \approx A\,u$ (where u is one atomic mass unit).

Hence the nuclear density is

$$\rho = \frac{M}{V} = \frac{A\,u}{\frac{4}{3}\pi R_0^{\,3} A} = \frac{u}{\frac{4}{3}\pi R_0^{\,3}}$$

The mass number A cancels out, so nuclear density is **independent of A** and is the same for every nucleus.

Therefore, even though the mass numbers are in the ratio 1:2, the ratio of their nuclear densities is

$$\frac{\rho_1}{\rho_2} = 1{:}1$$

CONCEPT

Nuclear density comes out to about 2.3×10^{17} kg/m^3 for all nuclei, which is why this is a favourite one mark trap.

Final answer. The ratio of their nuclear densities is **1:1**, because nuclear density is independent of mass number.

← *Back to Q32*

Q33

State the reason, why heavy water is generally used as a moderator in a nuclear reactor.

[2008]

SOLUTION

The job of a moderator is to slow down the fast neutrons released in fission (born with energy about 2 MeV) down to thermal energies (about 0.025 eV), so that they can efficiently cause further fission of ^{235}U.

Heavy water (D_2O) is preferred for two reasons.

First, the deuterium nucleus has a mass very close to that of a neutron. In an elastic collision between two nearly equal masses, the neutron loses a large fraction of its kinetic energy in each hit, so the neutrons are slowed down quickly and effectively.

Second, heavy water absorbs very few neutrons. Its neutron absorption is much lower than that of ordinary water, so most neutrons survive the slowing-down collisions instead of being captured.

Because of this combination of good slowing and very low neutron loss, **heavy water lets a reactor run even on natural (unenriched) uranium**, as in India's PHWR reactors.

Final answer. Heavy water slows neutrons efficiently (D has mass close to a neutron) while absorbing very few neutrons, so it is an excellent moderator.

← *Back to Q33*

Q34

What is the nuclear radius of ^{125}Fe*, if that of* ^{27}Al *is* 3.6 *fermi?*

[2008]

SOLUTION

The nuclear radius is related to the mass number by $R = R_0 A^{1/3}$, which means $R \propto A^{1/3}$.

Taking the ratio for the two nuclei removes the constant R_0.

$$\frac{R_{Fe}}{R_{Al}} = \left(\frac{A_{Fe}}{A_{Al}}\right)^{1/3}$$

Substituting the mass numbers $A_{Fe} = 125$ and $A_{Al} = 27$,

$$\frac{R_{Fe}}{R_{Al}} = \left(\frac{125}{27}\right)^{1/3} = \frac{5}{3}$$

Therefore

$$R_{Fe} = \frac{5}{3} \times R_{Al} = \frac{5}{3} \times 3.6 \text{ fm}$$

$$R_{Fe} = 6.0 \text{ fm}$$

So the required nuclear radius is **6.0 fermi**.

Final answer. The nuclear radius of the $A = 125$ nucleus is **6.0** fermi.

← *Back to Q34*

Q35

The radioactive isotope D decays according to the sequence D α−particle D_1 β−particle D_2. *If the mass number and atomic number of* D_2 *are 176 and 71 respectively, what is (i) the mass number (ii) atomic number of D?*

[2007]

SOLUTION

We work backwards from D_2 through the decay chain, using the conservation rules for mass number A and atomic number Z.

In β decay, the mass number stays the same and the atomic number increases by 1. So going backwards from D_2 to D_1, the mass number is unchanged and the atomic number decreases by 1.

$$D_1: A = 176,\ Z = 71 - 1 = 70$$

In α decay, the mass number decreases by 4 and the atomic number decreases by 2. So going backwards from D_1 to D, we add 4 to the mass number and 2 to the atomic number.

$$D: A = 176 + 4 = 180,\ Z = 70 + 2 = 72$$

Hence for the parent nucleus D:

(i) Mass number = **180**

(ii) Atomic number = **72**

Final answer. The parent nucleus ***D*** has mass number **180** and atomic number **72**.

← Back to Q35

Q36

Define the term 'activity' of a radionuclide. Write its SI unit.

[2007]

SOLUTION

The **activity** of a radionuclide is the rate of disintegration of its nuclei, that is, the number of nuclei that decay per unit time.

By the law of radioactive decay, it is given by

$$A = \left|\frac{dN}{dt}\right| = \lambda N$$

where λ is the decay constant and N is the number of undecayed nuclei present at that instant.

The **SI unit of activity is the becquerel (Bq)**, where one becquerel equals one disintegration per second.

$$1\ \text{Bq} = 1\ \text{disintegration per second}$$

An older, larger unit is the curie, with $\mathbf{1\ Ci = 3.7 \times 10^{10}\ Bq}$.

Final answer. Activity is the rate of disintegration, $A = \lambda N$; its SI unit is the becquerel (Bq), equal to one disintegration per second.

← Back to Q36

Q37

Write the nuclear decay process for β^--decay of ${}_{15}{}^{32}\text{P}$.

[2004]

SOLUTION

In β^- decay, a neutron inside the nucleus converts into a proton, releasing an electron and an antineutrino. As a result, the mass number A stays the same while the atomic number Z increases by 1.

For $^{32}_{15}P$, we have $\boldsymbol{A = 32}$ and $\boldsymbol{Z = 15}$. After β^- decay, the daughter has $\boldsymbol{A = 32}$ and $\boldsymbol{Z = 16}$, which is sulphur.

The complete decay process is

$$^{32}_{15}P \longrightarrow {}^{32}_{16}S + e^- + \bar{\nu}_e$$

Here e^- is the emitted **electron** and $\bar{\nu}_e$ is the **antineutrino**.

COMMON MISTAKE

Always include the antineutrino $\bar{\nu}_e$ in β^- decay; leaving it out is a common one mark loss.

Final answer. The decay is $\boldsymbol{^{32}_{15}P \rightarrow {}^{32}_{16}S + e^- + \bar{\nu}_e}$.

← *Back to Q37*

Q38

'Heavy water is often used as a moderator in thermal nuclear reactors.' Give reason.

[2004]

SOLUTION

A moderator is used to slow down the fast neutrons produced in fission (born with energy about 2 MeV) to thermal energies (about 0.025 eV), where they can efficiently cause further fission of ^{235}U.

Heavy water (D_2O) is an excellent moderator for two reasons.

First, a deuterium nucleus has a mass close to that of a neutron. In an elastic collision between nearly equal masses, the neutron transfers a large fraction of its kinetic energy in each collision, so it slows down quickly and effectively.

Second, heavy water has a very small neutron absorption cross section. So neutrons can undergo many slowing-down collisions in D_2O without being

absorbed and lost.

This combination of **efficient slowing with very low neutron loss** makes heavy water a good moderator, and it even allows the reactor to run on natural, unenriched uranium.

Final answer. Heavy water slows neutrons efficiently because deuterium's mass is close to a neutron's, while absorbing very few neutrons, making it an ideal moderator.

← *Back to Q38*

Q39

Draw the plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from this plot.

[2026 • Set 55-1-1]

SOLUTION

The graph of potential energy $U(r)$ of a pair of nucleons against their separation r has the following shape. Plot U on the vertical axis and r on the horizontal axis. Starting from very small r, the curve lies high above the axis in the positive region, showing a steep repulsive core for r below about 0.8 fm. As r increases the curve falls and crosses the axis, reaching a minimum (most negative value of U, about −100 MeV) at $\boldsymbol{r = r_0 \approx 0.8}$ fm. Beyond this minimum the curve rises again, approaching zero from below, and becomes effectively zero for r greater than about 3 fm. Mark the equilibrium separation $\mathbf{r_0 \approx 0.8}$ fm at the bottom of the well and label the axes $U(r)$ in MeV and r in fm.

Two important conclusions from this plot are as follows.

(1) The nuclear force is **attractive** for separations greater than $\mathbf{r_0 \approx 0.8}$ fm (here U rises with r, so the force pulls the nucleons together), and **repulsive** for separations less than r_0 (the steep positive core), which keeps the nucleons from collapsing into one another.

(2) The nuclear force is **short range**: the potential energy becomes practically zero beyond about 3 fm, so a nucleon interacts only with its nearest

neighbours.

Final answer. The U vs r curve shows a repulsive core below **0.8** fm and an attractive well that vanishes beyond about **3** fm, proving the nuclear force has a repulsive core, is attractive at larger r, and is short range.

← *Back to Q39*

Q40

Draw a graph showing variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei **(A > 170)** *decreases with increase in mass number. Explain its significance.*

[2026 • Set 55-1-2]

SOLUTION

The graph of binding energy per nucleon $\frac{B}{A}$ against mass number A is drawn with $\frac{B}{A}$ in MeV on the vertical axis and A on the horizontal axis. The curve rises steeply for light nuclei (with sharp peaks at ^{4}He, ^{12}C, ^{16}O), reaches a broad maximum of about 8.8 MeV near $A = 56$ (iron), and then falls slowly to about 7.6 MeV for the heaviest nuclei. Label the peak region around $A = 56$ as the most stable, and show the gentle decline for large A.

For heavy nuclei with $A > 170$, the binding energy per nucleon decreases because the long range Coulomb repulsion between the many protons (which grows roughly as Z^2) increases faster than the short range nuclear attraction (which grows roughly as A). This extra repulsion reduces the net binding per nucleon.

The significance of this decrease is as follows.

Heavy nuclei are **less tightly bound and less stable** than medium mass nuclei. Many of them are radioactive and decay by α or β emission to become more stable.

Most importantly, it means that **nuclear fission releases energy**. When a heavy nucleus splits into two medium mass fragments, the fragments have a higher

binding energy per nucleon, so the system moves to a more tightly bound state and the surplus energy (about 200 MeV per fission) is released.

Final answer. B/A peaks near **8.8** MeV at A**=56** and falls for heavy nuclei because Coulomb repulsion grows faster than nuclear binding, which makes heavy nuclei unstable and allows fission to release energy.

← *Back to Q40*

Q41

Suppose a nucleus with mass number A **= 240** *and B.E. per nucleon* **= 7.6** *MeV, breaks into two nuclei, each of mass number* A **= 120** *with B.E. per nucleon* **= 8.5** *MeV. Calculate the energy released in the process.*

[2026 • Set 55-1-3]

SOLUTION

The energy released in fission equals the increase in total binding energy, because the fragments are more tightly bound than the parent.

The total binding energy of the parent nucleus is (*B.E.* per nucleon) × A.

$$B.E._{\text{before}} = 7.6 \times 240 = 1824 \text{ MeV}$$

After fission we have two identical nuclei, each of mass number 120 with B.E. per nucleon 8.5 MeV.

$$B.E._{\text{after}} = 2 \times (8.5 \times 120) = 2 \times 1020 = 2040 \text{ MeV}$$

The energy released is the gain in binding energy.

$$Q = B.E._{\text{after}} - B.E._{\text{before}}$$

$$Q = 2040 - 1824$$

$$Q = 216 \text{ MeV}$$

Since the daughter nuclei have a higher binding energy per nucleon, they are more stable, and this extra binding energy of **216 MeV** appears as the kinetic energy of the fragments and the emitted neutrons.

Final answer. The energy released in the fission process is 216 MeV.

← Back to Q41

Q42

Explain the terms mass defect and binding energy. How are they related?

[2026 • Set 55-3-1]

SOLUTION

Mass defect (Δm) is the difference between the sum of the masses of the free nucleons (protons and neutrons) that make up a nucleus and the actual measured mass of that nucleus.

For a nucleus ${}^{A}_{Z}\mathrm{X}$ containing Z protons and $(A-Z)$ neutrons,

$$\Delta m = [\, Z\, m_p + (A-Z)\, m_n \,] - M_{\text{nucleus}}$$

The measured nuclear mass is always smaller than the sum of the constituent masses, so Δm is positive.

Binding energy (*B.E.*) is the energy required to break a nucleus completely into its individual free protons and neutrons. Equivalently, it is the energy released when the free nucleons combine to form the nucleus.

The two quantities are connected through Einstein's mass-energy equivalence relation, $\boldsymbol{E = mc^2}$. The mass that disappears (the mass defect) is converted into the binding energy.

$$B.E. = \Delta m \cdot c^2$$

In practical units, if Δm is expressed in atomic mass units (u), then$\boldsymbol{B.E.}$**(in MeV)** $= \boldsymbol{\Delta m}$**(in** $\boldsymbol{u}$**)** $\times$ **931.5**

A larger binding energy per nucleon means a more tightly bound and more stable nucleus.

Final answer. Mass defect and binding energy are related by B.E. = $\boldsymbol{\Delta m \cdot c^2}$.

← Back to Q42

Q43

Prove that the density of nuclear matter is same for all nuclei.

[2026 • Set 55-4-2]

SOLUTION

Density of nuclear matter is defined as the nuclear mass divided by the nuclear volume.

Experiments show that the radius of a nucleus depends only on its mass number A through the relation

$$R = R_0 A^{1/3}$$

where $\mathbf{R_0 \approx 1.2 \times 10^{-15}\ m}$ is a constant.

Treating the nucleus as a sphere, its volume is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi (R_0 A^{1/3})^3 = \frac{4}{3}\pi R_0{}^3 A$$

The mass of a nucleus of mass number A is approximately$\boldsymbol{M = A\,m}$, where m is the average mass of one nucleon, about 1.66×10^{-27} kg.

Therefore the density is

$$\rho = \frac{M}{V} = \frac{A\,m}{\frac{4}{3}\pi R_0{}^3 A}$$

$$\rho = \frac{m}{\frac{4}{3}\pi R_0{}^3} = \frac{3\,m}{4\pi R_0{}^3}$$

The mass number A cancels out, so the density does not depend on A. Hence the **density of nuclear matter is the same for all nuclei**.

Substituting the values gives a constant value of about $\boldsymbol{\rho \approx 2.3 \times 10^{17}\ kg/m^3}$, which is the same for every nucleus.

CONCEPT

Nucleons are packed equally closely in every nucleus, so adding more of them grows the volume in exact proportion to the mass. That is the physical reason density stays constant.

Final answer. Since A cancels in $\rho = \dfrac{3m}{4\pi R_0^{\,3}} \approx 2.3 \times 10^{17}$ kg/m^3, nuclear density is the same for all nuclei.

← *Back to Q43*

Q44

Calculate the mass of an α-particle in atomic mass unit (u). Given,
Mass of a normal helium atom = **4.002603 *u***
Mass of carbon atom = **1.9926×10^{-26} kg**

[2025 • Set 55-4-1]

SOLUTION

First we find the value of one atomic mass unit in kilograms. By definition, 1 u is one-twelfth of the mass of a ^{12}C atom.

$$1\ u = \frac{1}{12} \times m(^{12}\mathrm{C})$$

$$1\ u = \frac{1.9926 \times 10^{-26}}{12}$$

$$1\ u = 1.6605 \times 10^{-27}\ \mathrm{kg}$$

An α-particle is a helium nucleus. So its mass is found by removing the mass of the two orbital electrons from the mass of a neutral helium atom.

$$m_\alpha = m(\text{He atom}) - 2\, m_e$$

Taking the electron mass as $\mathbf{m_e = 0.000549\ u}$,

$$m_\alpha = 4.002603 - 2(0.000549)$$

$$m_\alpha = 4.002603 - 0.001098$$

$$m_\alpha = 4.001505\ u$$

Thus the mass of an α-particle is **4.001505 *u***, which in SI units is about **4.001505 × 1.6605 × 10^{-27} ≈ 6.644 × 10^{-27} kg**.

Final answer. The mass of an **α**-particle is about 4.0015 u (**≈ 6.64 × 10^{-27}** kg).

← *Back to Q44*

Q45

When a neutron collides with $^{235}_{92}U$*, the nucleus gives* $^{140}_{54}Xe$ *and* $^{94}_{38}Sr$ *as fission products and two neutrons are ejected. Calculate the mass defect and the energy released (in MeV) in the process. Given:*
m*($^{235}_{92}U$) = 235.04393 *u*, *m*($^{140}_{54}Xe$) = 139.92164 *u
m*($^{94}_{38}Sr$) = 93.91536 *u*, m_n = 1.00866 *u
1 *u* = 931 MeV/c^2

[2025 • Set 55-5-1]

SOLUTION

The fission reaction is

$$^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sr + 2\ {}^{1}_{0}n$$

The mass defect is the difference between the total mass before the reaction and the total mass after it.

Total mass of the reactants (before):

$$M_i = m(^{235}_{92}U) + m_n = 235.04393 + 1.00866 = 236.05259\ u$$

Total mass of the products (after), noting two neutrons are released:

$$M_f = m(^{140}_{54}Xe) + m(^{94}_{38}Sr) + 2\ m_n$$

$$M_f = 139.92164 + 93.91536 + 2(1.00866)$$

$$M_f = 139.92164 + 93.91536 + 2.01732 = 234.85432\ u$$

The mass defect is

$$\Delta m = M_i - M_f = 236.05259 - 234.85432 = 0.19827\ u$$

The energy released is obtained using $\mathbf{1\ u = 931\ MeV/c^2}$.

$$E = \Delta m \times 931$$

$$E = 0.19827 \times 931 = 184.6\ \text{MeV}$$

So the **mass defect is 0.19827 *u* and the energy released is about 184.6 MeV**.

Final answer. Mass defect = **0.19827** u and energy released ≈ **184.6** MeV.

← *Back to Q45*

Q46

(a) *Why is the mass of a nucleus always less than the sum of the masses of its constituents, i.e. free neutrons and free protons?*
(b) *How is Coulomb repulsion between protons in a nucleus overcome? Explain.*

[2025 • Set 55-5-2]

SOLUTION

(a) When free protons and neutrons come together to form a bound nucleus, they release energy known as the binding energy. This release happens because a bound system has lower energy than its separated parts.

By Einstein's mass-energy equivalence, $\boldsymbol{E = mc^2}$, this released energy corresponds to a loss of mass of the system.

$$M_{\text{nucleus}}\, c^2 = [\, Z\, m_p + (A-Z)\, m_n\,]c^2 - B.E.$$

Therefore

$$M_{\text{nucleus}} = [\, Z\, m_p + (A-Z)\, m_n\,] - \Delta m,\ \Delta m = \frac{B.E.}{c^2}$$

Since the mass equivalent of the binding energy (Δm) is always subtracted, the mass of the nucleus is always less than the total mass of its free constituent nucleons.

(b) Inside the nucleus the protons repel one another through the electrostatic (Coulomb) force, which varies as $F_C \propto \frac{1}{r^2}$. Despite this repulsion the nucleus stays together because of the **strong nuclear force**, which has these properties.

It is strongly attractive at nuclear distances and is about 100 times stronger than the Coulomb force over a separation of about 1 fm.

It is very short ranged, acting only up to about 2 to 3 fm, and is charge independent, acting equally between proton-proton, proton-neutron and neutron-neutron pairs.

Because the strong nuclear force is much larger than the Coulomb force at these tiny distances, it easily overcomes the proton-proton repulsion. The neutrons add extra attractive nuclear force without adding any Coulomb repulsion, which gives heavier nuclei the additional binding they need.

Final answer. The nuclear mass is reduced by $\Delta m = B.E./c^2$, and the short-range, much stronger strong nuclear force overcomes the Coulomb repulsion.

← *Back to Q46*

Q47

Calculate the binding energy per nucleon (in MeV) of a helium nucleus ($_2^4$He).
Given: $m(_2^4\text{He}) = 4.002603\ u$, $m_n = 1.008665\ u$, $m_p = 1.007825\ u$, $1\ u = 931.5\ \text{MeV}/c^2$.

[2025 • Set 55-7-2]

SOLUTION

The binding energy per nucleon is found by first calculating the mass defect, then converting it to energy, and finally dividing by the number of nucleons.

For $^4_2$He there are $Z = 2$ protons and $(A-Z) = 2$ neutrons. The mass defect is

$$\Delta m = [\ 2\, m_p + 2\, m_n\ ] - m(^4_2\text{He})$$

$$\Delta m = [\ 2(1.007825) + 2(1.008665)\ ] - 4.002603$$

$$\Delta m = (2.015650 + 2.017330) - 4.002603$$

$$\Delta m = 4.032980 - 4.002603 = 0.030377\ u$$

The total binding energy is obtained using $\mathbf{1\ u = 931.5\ MeV/c^2}$.

$$B.E. = \Delta m \times 931.5$$

$$B.E. = 0.030377 \times 931.5 = 28.30\ \text{MeV}$$

The binding energy per nucleon is the total binding energy divided by the mass number $A = 4$.

$$\frac{B.E.}{A} = \frac{28.30}{4}$$

$$\frac{B.E.}{A} = 7.07\ \text{MeV per nucleon}$$

Hence the binding energy per nucleon of the helium nucleus is about **7.07 MeV/nucleon**, which is why ^{4}He is unusually stable compared with its neighbours.

Final answer. The binding energy per nucleon of $^4_2$**He** is about 7.07 MeV/nucleon.

← *Back to Q47*

Q48

The carbon isotope $_6{}^{12}$C *has a nuclear mass of 12.000000 u. Calculate the binding energy of its nucleus.*
Given $\mathbf{m_p = 1.007825\ u;\ m_n = 1.008665\ u}$.

[2024 • Set 55-2-1]

SOLUTION

The binding energy is found from the mass defect using $\mathbf{B.E. = \Delta m \cdot c^2}$, where the mass defect is

$$\Delta m = [\, Z\, m_p + (A - Z)\, m_n \,] - M_{\text{nucleus}}$$

For $^{12}_{6}C$ we have $Z = 6$ protons, $(A-Z) = 6$ neutrons, and a nuclear mass of 12.000000 u.

$$\Delta m = [\ 6\ m_p + 6\ m_n\ ] - 12.000000$$

$$\Delta m = [\ 6(1.007825) + 6(1.008665)\ ] - 12.000000$$

$$\Delta m = (6.046950 + 6.051990) - 12.000000$$

$$\Delta m = 12.098940 - 12.000000 = 0.098940\ u$$

Converting this mass defect into energy with $\mathbf{1\ u = 931\ MeV/c^2}$,

$$B.E. = \Delta m \times 931$$

$$B.E. = 0.098940 \times 931 = 92.1\ \text{MeV}$$

Hence the binding energy of the carbon-12 nucleus is about **92.1 MeV**, corresponding to a binding energy per nucleon of about **92.1/12 ≈ 7.7 MeV**.

Final answer. The binding energy of the $^{12}_{6}C$ nucleus is about 92.1 MeV.

← *Back to Q48*

Q49

Calculate the energy released/absorbed in the following nuclear reaction:
$_{6}^{12}C + _{6}^{12}C \longrightarrow _{10}^{20}Ne + _{2}^{4}He$
Given: $\boldsymbol{m(_{6}^{12}C) = 12.000000\ u}$
$\boldsymbol{m(_{10}^{20}Ne) = 19.992439\ u}$
$\boldsymbol{m(_{2}^{4}He) = 4.002603\ u}$
$\boldsymbol{1\ u = 931\ MeV/c^2}$

[2024 • Set 55-2-2]

SOLUTION

The energy released or absorbed in a nuclear reaction is the Q-value, given by $Q = \Delta m \cdot c^2$, where $\Delta m = M_{\text{reactants}} - M_{\text{products}}$.

The reaction is

$$^{12}_{6}C + {}^{12}_{6}C \rightarrow {}^{20}_{10}Ne + {}^{4}_{2}He$$

Total mass of the reactants:

$$M_{reactants} = 2 \times m(^{12}_{6}C) = 2 \times 12.000000 = 24.000000\ u$$

Total mass of the products:

$$M_{products} = m(^{20}_{10}Ne) + m(^{4}_{2}He) = 19.992439 + 4.002603 = 23.995042\ u$$

The mass change is

$$\Delta m = M_{reactants} - M_{products}$$

$$\Delta m = 24.000000 - 23.995042 = 0.004958\ u$$

Since **1 u = 931 MeV/c^2**, the energy is

$$Q = \Delta m \times 931$$

$$Q = 0.004958 \times 931 = 4.62\ \text{MeV}$$

As **$\Delta m > 0$**, mass is lost and energy is given out. The reaction is **exothermic, releasing about 4.62 MeV**.

TIP

A positive Δm (reactant mass greater than product mass) means energy is released. A negative Δm would mean energy must be supplied for the reaction to occur.

Final answer. The reaction releases about 4.62 MeV (exothermic).

← *Back to Q49*

Q50

Deuterium undergoes the following fusion reaction:
$_{1}^{2}H + {}_{1}^{2}H \longrightarrow {}_{2}^{3}He + {}_{0}^{1}n + 3.27\ \text{MeV}$

How long an electric bulb of 200 W will glow by using the energy released in 2 g of deuterium?

[2024 • Set 55-5-3]

SOLUTION

First find the number of deuterium atoms in 2 g. The molar mass of deuterium ($^2_1$H) is 2 g/mol, so 2 g is exactly one mole.

$$N = \frac{2}{2} \times 6.022 \times 10^{23} = 6.022 \times 10^{23} \text{ atoms}$$

Each fusion reaction consumes two deuterons and releases 3.27 MeV. So the number of reactions is half the number of deuterons.

$$\text{Number of reactions} = \frac{N}{2} = \frac{6.022 \times 10^{23}}{2} = 3.011 \times 10^{23}$$

The total energy released is

$$E = 3.011 \times 10^{23} \times 3.27 = 9.85 \times 10^{23} \text{ MeV}$$

Converting to joules using $\mathbf{1\ MeV = 1.6 \times 10^{-13}\ J}$,

$$E = 9.85 \times 10^{23} \times 1.6 \times 10^{-13} = 1.575 \times 10^{11}\ J$$

The bulb consumes power $\boldsymbol{P = 200\ W}$, so the time for which it can glow is

$$t = \frac{E}{P} = \frac{1.575 \times 10^{11}}{200}$$

$$t = 7.88 \times 10^{8}\ s$$

Expressing this in years using $\mathbf{1\ year \approx 3.16 \times 10^{7}\ s}$,

$$t = \frac{7.88 \times 10^{8}}{3.16 \times 10^{7}} \approx 24.9 \text{ years}$$

Hence the 200 W bulb would glow for about **7.88×10^{8} s, that is roughly 25 years**.

Final answer. The bulb would glow for about **7.88×10^{8}** s, that is nearly 25 years.

← Back to Q50

Q51

Define the term, mass defect. How is it related to stability of the nucleus?

[2023 • Set 55-2-1]

SOLUTION

The mass defect of a nucleus is the difference between the total mass of its free constituent nucleons (the Z protons and the $(A-Z)$ neutrons) and the actual measured mass of the nucleus.

If $M_{nucleus}$ is the rest mass of a nucleus of atomic number Z and mass number A, then

$$\Delta m = [\, Z\, m_p + (A-Z)\, m_n \,] - M_{nucleus}$$

The measured nuclear mass is always less than the sum of the masses of the separate nucleons, so Δm is positive. When nucleons bind together to form a nucleus, this missing mass is converted into the **binding energy** of the nucleus, released according to Einstein's mass-energy relation

$$E_b = \Delta m\, c^2$$

This same energy must be supplied from outside to break the nucleus back into free nucleons.

A larger mass defect means a larger binding energy, and dividing by A gives the **binding energy per nucleon** ($\frac{E_b}{A}$), which is the true measure of stability. A greater $\frac{E_b}{A}$ means the nucleons are held more tightly, so the nucleus is more stable. Nuclei near $A \approx 56$ (iron region) have the highest binding energy per nucleon and are therefore the most stable.

CONCEPT

Stability is judged by binding energy per nucleon (E_b/A), not by the total binding energy, since total E_b keeps rising with A even for unstable heavy nuclei.

Final answer. Mass defect $\Delta m = [Z\,m_p + (A-Z)m_n] - M_{nucleus}$; a larger Δm gives a larger binding energy per nucleon and hence a more stable nucleus.

← *Back to Q51*

Q52

*(a) Draw a graph showing the variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei (**A > 170**) decreases with the increase in mass number. Explain.*

OR

(b) Using Bohr's postulates, obtain the expression for radius of nth *stable orbit in a hydrogen atom.*

[2023 • Set 55-3-1]

SOLUTION

(a) The graph plots binding energy per nucleon $\frac{E_b}{A}$ (in MeV, vertical axis) against mass number A (horizontal axis).

Description of the curve: starting from $A=2$ it rises steeply, showing sharp peaks at ^{4}He, ^{12}C and ^{16}O. It reaches a flat maximum of about 8.8 MeV near $A \approx 56$ (iron region), then falls slowly and reaches roughly 7.6 MeV near $A = 240$. The curve is broad and almost flat in the middle.

Reason for the decrease for $A > 170$: In a heavy nucleus there are a large number of protons. The Coulomb (electrostatic) repulsion acts between every pair of protons and is long-range, so it grows roughly as Z^2. The attractive nuclear force, however, is short-range and a given nucleon only attracts its immediate neighbours, so the total nuclear attraction grows only as A.

As A increases beyond about 170, the rapidly increasing Coulomb repulsion offsets more and more of the nuclear attraction. This reduces the net binding, so the **binding energy per nucleon decreases** with increasing mass number. This is also why very heavy nuclei tend to undergo α-decay or fission.

OR

(b) Consider an electron of mass m and charge $-e$ revolving in the nth circular orbit of radius r_n with speed v_n around a nucleus of charge $+e$.

By Bohr's postulate, the electrostatic force of attraction provides the necessary centripetal force:

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n^2} = \frac{m\,v_n^2}{r_n}$$

$$\Rightarrow m\,v_n^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n} \quad ...(1)$$

By Bohr's quantisation of angular momentum:

$$m\,v_n\,r_n = \frac{nh}{2\pi}$$

$$\Rightarrow v_n = \frac{nh}{2\pi\,m\,r_n} \quad ...(2)$$

Substituting (2) in (1):

$$m\left(\frac{nh}{2\pi\,m\,r_n}\right)^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n}$$

$$\frac{n^2h^2}{4\pi^2\,m\,r_n^2} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_n}$$

Solving for r_n:

$$r_n = \frac{n^2h^2\varepsilon_0}{\pi\,m\,e^2}$$

Thus the radius of the nth orbit is proportional to n^2, giving $\mathbf{r_n = n^2a_0}$ with $\mathbf{a_0 = 0.53\ \text{Å}}$ for the smallest (ground state) orbit.

Final answer. (a) Coulomb repulsion ($\propto \mathbf{Z^2}$) grows faster than the short-range nuclear attraction ($\propto A$), so $\mathbf{E_b/A}$ falls for $\boldsymbol{A>170}$; OR (b) $\mathbf{r_n = \frac{n^2h^2\varepsilon_0}{\pi\,m\,e^2} = n^2a_0}$.

← *Back to Q52*

Q53

Draw the graph showing the variation of binding energy per nucleon with mass number A of nuclei ($2 \le A \le 170$). *Use this graph to explain the release of energy in nuclear fission.*

[2023 • Set 55-4-1]

SOLUTION

The graph plots binding energy per nucleon $\frac{E_b}{A}$ (in MeV, vertical axis) against mass number A (horizontal axis) for $2 \le A \le 170$.

Description of the curve: it starts low near $A=2$, then rises steeply, showing sharp peaks at ^{4}He (about 7.07 MeV), ^{12}C and ^{16}O. The curve flattens and reaches its maximum of about 8.8 MeV near $A \approx 56$, and then decreases slowly up to $A = 170$.

Use of the graph to explain fission: A very heavy nucleus (large A) lies on the falling part of the curve, so it has a relatively **low** binding energy per nucleon. When it splits into two medium-mass fragments, these fragments lie nearer the peak of the curve and so have a **higher** binding energy per nucleon.

Higher binding energy per nucleon means the nucleons in the fragments are bound more tightly, so the products are in a more stable, lower-energy state than the parent. The total binding energy after fission is greater than before fission.

Numerically, if a parent with $A \approx 240$ at $\frac{E_b}{A} \approx 7.6$ **MeV** splits into fragments at about 8.5 MeV per nucleon:

$$\text{Energy released} = (240)(8.5) - (240)(7.6) = (240)(0.9) \approx 216 \text{ MeV}$$

This difference in binding energy appears as the kinetic energy of the fragments, neutrons and gamma radiation. Hence energy is released in fission because the system moves from a lower E_b/A to a **higher E_b/A (more stable) configuration**.

Final answer. Heavy nuclei have lower E_b/A than the mid-mass fragments they split into; the gain in binding energy (around **200** MeV) is released as the energy

of fission.

← Back to Q53

Q54

Briefly describe the multi-step process involved in the generation of energy in the Sun.

[2023 • Set 55-4-3]

SOLUTION

The Sun's energy is produced by the **nuclear fusion** of hydrogen into helium in its hot, dense core, mainly through the proton-proton (p-p) chain. It is a multi-step process because four protons cannot collide and fuse simultaneously.

The steps are:

$$^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + \mathrm{e}^{+} + \nu_e$$

Two protons fuse to form a deuteron, releasing a positron and a neutrino. This is the slowest, rate-controlling step as it depends on the weak interaction.

$$^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma$$

A deuteron fuses with another proton to give a helium-3 nucleus and a gamma-ray photon.

$$^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H}$$

Two helium-3 nuclei fuse to form a stable helium-4 nucleus and release two protons.

The net effect of the whole chain is:

$$4\ {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\mathrm{e}^{+} + 2\nu_e + 26.7\ \mathrm{MeV}$$

So four protons combine to give one helium-4 nucleus, two positrons, two neutrinos and an energy of about 26.7 MeV per cycle. The positrons annihilate with electrons, adding more energy. This released energy is the source of the Sun's heat and light.

Final answer. Through the proton-proton chain four hydrogen nuclei fuse step by step into one ^{4}**He** nucleus, releasing about **26.7 MeV** per cycle as the Sun's energy.

← *Back to Q54*

Q55

(i) Distinguish between isotopes and isobars.
(ii) Two nuclei have different mass numbers A_1 *and* A_2*. Are these nuclei necessarily the isotopes of the same element? Explain.*

[2022 • Set 55-3-1]

SOLUTION

(i) Distinction between isotopes and isobars:

Isotopes are nuclei of the **same element** having the **same atomic number *Z*** but **different mass numbers *A*** (they have the same number of protons but different numbers of neutrons). Since Z decides chemical behaviour, isotopes have identical chemical properties but different masses. Example: $^{12}_{6}C$ and $^{14}_{6}C$.

Isobars are nuclei having the **same mass number *A*** but **different atomic numbers *Z***. They belong to different elements and so have different chemical properties but nearly the same mass. Example: $^{40}_{18}Ar$ and $^{40}_{20}Ca$.

(ii) Two nuclei have different mass numbers A_1 and A_2. Are they necessarily isotopes of the same element?

No. Different mass numbers alone do not make two nuclei isotopes. Nuclei are isotopes of the same element only if they have the **same atomic number *Z*** (same number of protons). If the two nuclei with different A_1 and A_2 also have different Z, they are nuclei of different elements, not isotopes.

For example, $^{12}_{6}C$ (**Z=6, A=12**) and $^{14}_{7}N$ (**Z=7, A=14**) have different mass numbers but different atomic numbers, so they are different elements and not isotopes.

Final answer. Isotopes have same Z but different A; isobars have same A but different Z. Two nuclei with different A are isotopes only if their Z is the same,

so it is not necessary.

← Back to Q55

Q56

Define the term 'Half-life' of a radioactive substance. Two different radioactive substances have half-lives T_1 *and* T_2 *and number of undecayed atoms at an instant,* N_1 *and* N_2*, respectively. Find the ratio of their activities at that instant.*

[2020 • Set 55-1-1]

SOLUTION

Half-life: The half-life of a radioactive substance is the time in which half the radioactive nuclei present in a sample disintegrate (decay). In other words, after one half-life the number of undecayed nuclei falls from N_0 to $N_0/2$.

It is related to the decay (disintegration) constant λ by

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Ratio of activities: The activity of a sample is $A = \lambda N$, where $\lambda = \frac{\ln 2}{T}$.

For the first substance:

$$A_1 = \lambda_1 N_1 = \frac{\ln 2}{T_1} N_1$$

For the second substance:

$$A_2 = \lambda_2 N_2 = \frac{\ln 2}{T_2} N_2$$

Taking the ratio, the factor ln 2 cancels:

$$\frac{A_1}{A_2} = \frac{(\ln 2/T_1)\, N_1}{(\ln 2/T_2)\, N_2} = \frac{N_1 T_2}{N_2 T_1}$$

Hence the required ratio of activities is $\mathbf{A_1 : A_2 = N_1 T_2 : N_2 T_1}$.

Final answer. Half-life is the time for half the nuclei to decay; the activity ratio is $\frac{A_1}{A_2} = \frac{N_1 T_2}{N_2 T_1}$.

← Back to Q56

Q57

Define activity of a sample of a radioactive substance. The value of the disintegration constant of a radioactive substance is $0{\cdot}0693\ h^{-1}$. *Find the time after which the activity of a sample of this substance reduces to one-half that of its present value.*

[2020 • Set 55-1-2]

SOLUTION

Activity: The activity of a radioactive sample is the rate at which its nuclei disintegrate, that is, the number of disintegrations per unit time. It is given by

$$A = \lambda N$$

where λ is the decay constant and N is the number of undecayed nuclei. Its SI unit is the becquerel (**1 Bq = 1** disintegration per second).

The activity is directly proportional to the number of undecayed nuclei, so the activity falls to half its value in exactly one half-life. Hence the required time is the half-life $T_{1/2}$.

Using

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Substituting $\lambda = 0.0693\ h^{-1}$:

$$T_{1/2} = \frac{0.693}{0.0693\ h^{-1}}$$

$$T_{1/2} = 10\ h$$

Therefore the activity of the sample reduces to one-half of its present value after **10 hours**.

Final answer. The activity halves in one half-life, $\mathbf{T_{1/2} = \frac{0.693}{0.0693} = 10\ hours.}$

← *Back to Q57*

Q58

(a) For a radioactive substance, show the variation of the total mass disintegrated as a function of time t graphically.
(b) Initial mass of a radioactive substance is 3.2 *mg. It has a half-life of* 4 *h. Find the mass of the substance left undecayed after* 8 *h.*

[2020 • Set 55-2-2]

SOLUTION

(a) Mass disintegrated versus time:

If m_0 is the initial mass and m is the mass remaining at time t, then by the law of radioactive decay

$$m = m_0\, e^{-\lambda t}$$

So the mass that has disintegrated up to time t is

$$m_{dis} = m_0 - m = m_0(1 - e^{-\lambda t})$$

Graph description: plot m_{dis} on the vertical axis against time t on the horizontal axis. The curve starts at the origin ($\mathbf{m_{dis} = 0}$ at $\boldsymbol{t = 0}$), rises steeply at first and then bends and flattens, approaching the horizontal line $\mathbf{m_{dis} = m_0}$ as $\boldsymbol{t \to \infty}$ (a rising saturation curve that asymptotically reaches m_0).

(b) Mass left undecayed after 8 h:

Given initial mass $\mathbf{m_0 = 3.2\ mg}$ and half-life $\mathbf{T_{1/2} = 4\ \boldsymbol{h}}$.

Number of half-lives in 8 h:

$$n = \frac{t}{T_{1/2}} = \frac{8}{4} = 2$$

The mass remaining after n half-lives is

$$m = m_0(\frac{1}{2})^n = 3.2 \times (\frac{1}{2})^2$$

$$m = 3.2 \times \frac{1}{4}$$

$$m = 0.8 \text{ mg}$$

Therefore **0.8 mg** of the substance remains undecayed after 8 hours.

Final answer. After **8** h (**2** half-lives) the mass left is $\mathbf{3.2 \times (1/2)^2 = 0.8}$ **mg**.

← *Back to Q58*

Q59

(a) Write the SI unit of activity of a radioactive substance.
(b) Find the half-life of a radioactive substance if the activity of the substance drops to $\frac{1}{16}$ *of its initial value in* 15 *years.*

[2020 • Set 55-2-3]

SOLUTION

(a) SI unit of activity:

The SI unit of activity of a radioactive substance is the **becquerel (Bq)**, defined as one disintegration (decay) per second.

$$1 \text{ Bq} = 1 \text{ disintegration per second}$$

(A larger, older unit is the curie: $\mathbf{1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}}$.)

(b) Half-life from the drop in activity:

Since activity is proportional to the number of undecayed nuclei, after n half-lives the activity becomes $(\frac{1}{2})^n$ of its initial value.

The activity drops to $\frac{1}{16}$ of its initial value, and

$$\frac{1}{16} = (\frac{1}{2})^4$$

So $n = 4$ half-lives have elapsed. The total time is 15 years, hence

$$n\,T_{1/2} = 15 \text{ years}$$

$$4\,T_{1/2} = 15 \text{ years}$$

$$T_{1/2} = \frac{15}{4} \text{ years}$$

$$T_{1/2} = 3.75 \text{ years}$$

Therefore the half-life of the substance is **3.75 years**.

Final answer. (a) The SI unit is the becquerel (**1 Bq = 1** decay/s); (b) $\frac{1}{16} = (1/2)^4$ means **4** half-lives in **15** years, so $\mathbf{T_{1/2} = 3.75}$ years.

← *Back to Q59*

Q60

A heavy nucleus P of mass number 240 *and binding energy* 7.6 *MeV per nucleon splits in to two nuclei Q and R of mass numbers* 110, 130 *and binding energy per nucleon* 8.5 *MeV and* 8.4 *MeV, respectively. Calculate the energy released in the fission.*

[2020 • Set 55-5-1]

SOLUTION

The energy released in fission equals the increase in total binding energy, that is, the total binding energy of the product nuclei minus the total binding energy of the parent nucleus.

Given data: parent P has $A = 240$ and binding energy per nucleon 7.6 MeV; daughter Q has $A = 110$ and 8.5 MeV per nucleon; daughter R has $A = 130$ and 8.4 MeV per nucleon.

Total binding energy of the parent:

$$BE_P = 240 \times 7.6 = 1824 \text{ MeV}$$

Total binding energy of the products:

$$\mathrm{BE}_Q = 110 \times 8.5 = 935\ \mathrm{MeV}$$

$$\mathrm{BE}_R = 130 \times 8.4 = 1092\ \mathrm{MeV}$$

$$\mathrm{BE}_Q + \mathrm{BE}_R = 935 + 1092 = 2027\ \mathrm{MeV}$$

Energy released in fission:

$$Q = (\mathrm{BE}_Q + \mathrm{BE}_R) - \mathrm{BE}_P$$

$$Q = 2027 - 1824$$

$$Q = 203\ \mathrm{MeV}$$

Therefore the energy released in the fission is **203 MeV**. The products are more tightly bound than the parent, so this binding-energy difference is liberated as kinetic energy of the fragments and radiation.

TIP

Energy released = (total BE of products) - (BE of parent); compute each total BE as (mass number) x (binding energy per nucleon).

Final answer. Energy released $Q = (935 + 1092) - 1824 = 203$ **MeV**.

← *Back to Q60*

Q61

(a) Define one Becquerel.
(b) A radioactive substance disintegrates into two types of daughter nuclei, one type with disintegration constant λ_1 and the other type with disintegration constant λ_2. Determine the half-life of the radioactive substance.

[2020 • Set 55-5-2]

SOLUTION

(a) The activity of a radioactive sample is the number of nuclei that disintegrate per unit time. Its SI unit is the becquerel.

One becquerel is the activity of a radioactive source in which one nucleus disintegrates per second.

$$1\ \text{Bq} = 1\ \text{disintegration per second}$$

(The older unit, the curie, is related by $\mathbf{1\ Ci = 3.7 \times 10^{10}\ Bq}$.)

(b) Here a single radioactive species can decay by two independent routes, with decay constants λ_1 and λ_2. Each route removes nuclei from the same sample, so the rates add.

The rate of disintegration by the first mode is $\frac{dN_1}{dt} = -\lambda_1 N$ and by the second mode is $\frac{dN_2}{dt} = -\lambda_2 N$.

The total rate of decrease of nuclei is the sum of both.

$$\frac{dN}{dt} = -\lambda_1 N - \lambda_2 N = -(\lambda_1 + \lambda_2)N$$

Comparing with the standard law $\frac{dN}{dt} = -\lambda N$, the effective decay constant of the substance is the sum of the two.

$$\lambda = \lambda_1 + \lambda_2$$

The half-life is related to the decay constant by $T_{1/2} = \frac{\ln 2}{\lambda}$. Substituting the effective decay constant gives the required result.

$$T_{1/2} = \frac{\ln 2}{\lambda_1 + \lambda_2} = \frac{0.693}{\lambda_1 + \lambda_2}$$

So the **effective half-life** of the substance is $\frac{0.693}{\lambda_1 + \lambda_2}$, which is shorter than the half-life for either decay mode taken alone.

TIP

When a nucleus has more than one decay channel, the decay constants add (not the half-lives). The half-lives combine like resistors in parallel.

Final answer. 1 Bq is one disintegration per second, and the half-life is $\mathbf{T_{1/2} = \frac{0.693}{\lambda_1 + \lambda_2}}$.

← Back to Q61

Q62

Calculate for how many years will the fusion of 2.0 *kg deuterium keep* 800 *W electric lamp glowing. Take the fusion reaction as*

$^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \longrightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 3.27\ \mathrm{MeV}$

[2020 • Set 55-5-3]

SOLUTION

The lamp draws energy from the fusion of deuterium nuclei. First find how many deuterium nuclei are present in 2.0 kg, then how much fusion energy they release, then divide by the lamp power to get the time.

The molar mass of deuterium $^{2}_{1}\mathrm{H}$ is 2 g/mol, so the number of deuterium nuclei in 2000 g is found using Avogadro's number $\mathbf{N_A = 6.022 \times 10^{23}\ mol^{-1}}$.

$$\mathrm{N}_D = \frac{2000}{2} \times 6.022 \times 10^{23} = 1000 \times 6.022 \times 10^{23}$$

$$\mathrm{N}_D = 6.022 \times 10^{26}\ \text{nuclei}$$

Each fusion reaction consumes 2 deuterium nuclei and releases 3.27 MeV. So the number of reactions is half the number of nuclei.

$$\text{Number of reactions} = \frac{\mathrm{N}_D}{2} = \frac{6.022 \times 10^{26}}{2} = 3.011 \times 10^{26}$$

The total energy released is the number of reactions times the energy per reaction.

$$E = 3.011 \times 10^{26} \times 3.27\ \mathrm{MeV} = 9.846 \times 10^{26}\ \mathrm{MeV}$$

Converting to joules using $\mathbf{1\ MeV = 1.6 \times 10^{-13}\ J}$.

$$E = 9.846 \times 10^{26} \times 1.6 \times 10^{-13} = 1.575 \times 10^{14}\ J$$

The lamp consumes power $\boldsymbol{P = 800\ W = 800\ J/s}$. The time for which this energy keeps it glowing is $t = \dfrac{E}{P}$.

$$t = \frac{1.575 \times 10^{14}}{800} = 1.969 \times 10^{11}\ s$$

Converting to years using **1 year = 3.16×10^7 s**.

$$t = \frac{1.969 \times 10^{11}}{3.16 \times 10^{7}} \approx 6.23 \times 10^{3}\ \text{years}$$

The fusion energy of 2.0 kg of deuterium would keep the 800 *W* lamp glowing for about **6230 years**.

COMMON MISTAKE

Remember each reaction uses TWO deuterium nuclei, so divide the number of nuclei by 2 before multiplying by the energy per fusion.

Final answer. The lamp would glow for about **6.23×10^3** years (nearly 6230 years).

← *Back to Q62*

Q63

Plot a graph showing the variation of undecayed nuclei N versus time t. From the graph, find out how one can determine the half-life and average life of the radioactive nuclei.

[2019 • Set 55-3-1]

SOLUTION

According to the law of radioactive decay, the rate of disintegration at any instant is proportional to the number *N* of undecayed nuclei present at that instant.

$$\frac{dN}{dt} = -\lambda N$$

Here λ is the decay constant, the probability per unit time of decay of a nucleus. Integrating this from $t = 0$, when $N = N_0$, to time *t*, when the number is *N*, gives the exponential decay law.

$$N = N_0\, e^{-\lambda t}$$

The graph of N versus t is a smooth curve that starts at N_0 on the y-axis and falls exponentially, getting closer and closer to the time axis but never touching it (asymptotic to the t-axis).

Description of the graph: the vertical axis is labelled N (number of undecayed nuclei) with N_0 marked at the top; the horizontal axis is labelled t (time). The curve is steepest at the start and flattens as t increases. Dashed horizontal lines are drawn at $N = \frac{N_0}{2}$ and at $N = \frac{N_0}{e} \approx 0.37\, N_0$, each dropping vertically to the time axis to mark the corresponding times.

Finding the half-life $T_{1/2}$ from the graph: locate the point $N = \frac{N_0}{2}$ on the y-axis, draw a horizontal line to meet the curve, then drop a vertical line to the t-axis. The time read off there is the **half-life $T_{1/2}$** (it satisfies $T_{1/2} = \frac{0.693}{\lambda}$).

Finding the average (mean) life τ from the graph: locate the point $N = \frac{N_0}{e} = 0.37\, N_0$ on the y-axis, draw a horizontal line to the curve and drop to the t-axis. The time read off there is the **mean life τ** (it satisfies $\tau = \frac{1}{\lambda}$).

The two are related by $\tau = 1.44\, T_{1/2}$.

Final answer. $N = N_0\, e^{-\lambda t}$ is a decaying exponential; $T_{1/2}$ is read at $N = N_0/2$ and τ at $N = N_0/e$.

← *Back to Q63*

Q64

(a) Write two distinguishing features of nuclear forces.
(b) Complete the following nuclear reactions for α and β decay:
(i) $^{238}_{92}U \longrightarrow ? + {}^{4}_{2}He + Q$
(ii) $^{22}_{11}Na \longrightarrow {}^{22}_{10}Ne + ? + \nu$

[2019 • Set 55-3-1]

SOLUTION

(a) Two distinguishing features of nuclear forces:

First, nuclear forces are short-range forces. They act only over a very small distance of about 2 to 3 fm and become negligible beyond this. (In contrast, the Coulomb force has infinite range.)

Second, nuclear forces are charge-independent. The force between two protons, between a proton and a neutron, and between two neutrons is the same: $\mathbf{F_{pp} = F_{pn} = F_{nn}}$. (The Coulomb force, on the other hand, acts only between charged particles.)

(b) The reactions are completed by balancing the mass number A (top) and the atomic number Z (bottom) on both sides.

(i) For the alpha decay $^{238}_{92}\mathrm{U} \rightarrow \mathbf{?} + {}^{4}_{2}\mathrm{He} + Q$, let the unknown be $^{A}_{Z}\mathrm{X}$.

Balancing mass number: $A = 238 - 4 = 234$.

Balancing atomic number: $Z = 92 - 2 = 90$.

The element with $Z = 90$ is thorium, so $X = {}^{234}_{90}\mathrm{Th}$.

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He} + Q$$

(ii) For the reaction $^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{10}\mathrm{Ne} + \mathbf{?} + \nu$, the emission of a neutrino ν (not an antineutrino) signals β^+ decay.

Balancing mass number: $22 = 22 + A \Rightarrow A = 0$.

Balancing atomic number: $11 = 10 + Z \Rightarrow Z = +1$.

A particle of zero mass number and charge +1 is a positron $^{0}_{+1}\mathrm{e}$ (i.e. e^+).

$$^{22}_{11}\mathrm{Na} \rightarrow {}^{22}_{10}\mathrm{Ne} + \mathrm{e}^+ + \nu$$

So the missing particles are $\mathbf{^{234}_{90}Th}$ in (i) and a **positron $\mathbf{e^+}$** in (ii).

Final answer. (i) the missing nucleus is $\mathbf{^{234}_{90}Th}$; (ii) the missing particle is a positron $\mathbf{e^+}$.

← Back to Q64

Q65

Distinguish between nuclear fusion and nuclear fission. Give one example for each.

[2019 • Set 55-3-2]

SOLUTION

Nuclear fission and nuclear fusion are two different ways of releasing nuclear energy, and they differ in several ways.

In nuclear fission a heavy nucleus splits into two lighter nuclei of comparable mass, usually after being struck by a slow neutron, along with the release of a few neutrons and a large amount of energy.

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} + 200\ \mathrm{MeV}$$

In nuclear fusion two light nuclei combine (fuse) together to form a single heavier nucleus, with the release of energy.

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

The key points of difference are:

1. Process: fission is the splitting of a heavy nucleus into lighter ones, while fusion is the joining of light nuclei into a heavier one.

2. Conditions: fission can be triggered at ordinary temperature by a bombarding neutron, while fusion requires extremely high temperature ($\sim 10^7$ *K*) and pressure to overcome the Coulomb repulsion between the nuclei.

3. Energy per nucleon: the energy released per nucleon in fusion (about 3.5 MeV) is much greater than that in fission (about 0.9 MeV), so fusion is more efficient per unit mass.

Thus **fission splits a heavy nucleus** (example: ^{235}U) while **fusion combines light nuclei** (example: ^{2}H and ^{3}H).

Final answer. Fission splits a heavy nucleus (e.g. ^{235}U fission); fusion joins light nuclei (e.g. $^{2}\mathrm{H} + {}^{3}\mathrm{H} \rightarrow {}^{4}\mathrm{He}$).

← *Back to Q65*

Q66

(a) Which property of nuclear force explains the constancy of binding energy per nucleon $\left(\frac{B}{A}\right)$ *for nuclei in the range* $\mathbf{20 < A < 170}$*?*

(b) Complete the following nuclear reactions:

(i) $^{32}_{15}\mathrm{P} \longrightarrow {}^{A}_{Z}\mathrm{X} + \mathrm{e}^{-} + \bar{\nu}$

(ii) $^{12}_{6}\mathrm{C} + {}^{12}_{6}\mathrm{C} \longrightarrow {}^{A}_{Z}\mathrm{Y} + {}^{4}_{2}\mathrm{He}$

[2019 • Set 55-3-3]

SOLUTION

(a) The binding energy per nucleon stays almost constant (about 8.5 MeV) for nuclei in the mass-number range $\mathbf{20 < A < 170}$ because of the saturation property of nuclear forces.

Since nuclear forces are short-ranged, a given nucleon interacts (attracts) only with its few immediate neighbours, not with every other nucleon in the nucleus. The force is therefore said to be saturated.

Because each nucleon binds only to a fixed number of neighbours, the total binding energy is proportional to the number of nucleons, $\boldsymbol{B.E. \propto A}$.

Hence the binding energy per nucleon, $\frac{B}{A}$, remains nearly constant over this range. So the **saturation of nuclear forces** is the property responsible.

(b) The reactions are completed by conserving mass number A and atomic number Z.

(i) For $^{32}_{15}\mathbf{P} \rightarrow {}^{A}_{Z}\mathbf{X} + \mathbf{e}^{-} + \bar{\nu}$, the emission of an electron e^{-} and an antineutrino $\bar{\nu}$ shows this is β^{-} decay.

Balancing mass number: $\boldsymbol{A = 32}$ (unchanged).

Balancing atomic number: $\mathbf{15 = Z + (-1) \Rightarrow Z = 16}$.

The element with $\boldsymbol{Z = 16}$ is sulphur, so $\boldsymbol{X = {}^{32}_{16}S}$.

$$^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + \mathrm{e}^{-} + \bar{\nu}$$

(ii) For $^{12}_{6}\mathbf{C} + {}^{12}_{6}\mathbf{C} \rightarrow {}^{A}_{Z}\mathbf{Y} + {}^{4}_{2}\mathbf{He}$:

Balancing mass number: $\mathbf{12 + 12 = A + 4 \Rightarrow A = 20}$.

Balancing atomic number: $6 + 6 = Z + 2 \Rightarrow Z = 10$.

The element with $Z = 10$ is neon, so $Y = {}^{20}_{10}\text{Ne}$.

$$^{12}_{6}\text{C} + {}^{12}_{6}\text{C} \rightarrow {}^{20}_{10}\text{Ne} + {}^{4}_{2}\text{He}$$

So the missing nuclei are $\mathbf{^{32}_{16}S}$ in (i) and $\mathbf{^{20}_{10}Ne}$ in (ii).

Final answer. Saturation of nuclear force explains constant B/A; missing nuclei are $^{32}_{16}\text{S}$ and $^{20}_{10}\text{Ne}$.

← *Back to Q66*

Q67

Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions which you can draw regarding the nature of nuclear forces.

OR

Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei, $2 \le A \le 240$. *How do you explain the constancy of binding energy per nucleon in the range* $\mathbf{30 < A < 170}$ *using the property that nuclear force is short-ranged?*

[2012]

SOLUTION

Potential energy versus separation of a pair of nucleons:

Description of the graph: the vertical axis is the potential energy $U(r)$ (in MeV) and the horizontal axis is the separation r (in fm). For very small separation, below about $\mathbf{r_0 \approx 0.8\ fm}$, the curve lies high above the axis in the positive (repulsive) region and rises steeply. As r increases past r_0, the curve dips down and crosses the axis into the negative region, reaching a minimum of about −100 MeV at $\boldsymbol{r \approx 1\ fm}$ (the position of strongest attraction). Beyond this minimum the curve rises again toward zero, becoming negligibly small for r greater than about 2 to 3 fm. The minimum point and the distances r_0 and the attractive well should be labelled.

Two important conclusions about nuclear forces from this plot:

First, nuclear forces are strongly attractive but short-ranged. The deep negative well shows a strong attraction around $r \approx 1$ fm, but the force becomes negligible beyond about 2 to 3 fm.

Second, the force has a repulsive core. For separations smaller than about 0.8 fm the potential energy is positive and rising, meaning the nucleons repel each other strongly at very small distances. This prevents the nucleus from collapsing and keeps nuclear density nearly constant.

OR (alternative part):

Binding energy per nucleon versus mass number A:

Description of the graph: the vertical axis is the binding energy per nucleon $\frac{B}{A}$ (in MeV) and the horizontal axis is the mass number A (from 2 to 240). The curve rises steeply for light nuclei, reaches a maximum of about 8.75 MeV near $A = 56$ (iron), and then falls slowly for heavy nuclei, dropping to about 7.6 MeV near $A = 240$. The flat plateau region $30 < A < 170$ should be marked, with the peak at $A = 56$ labelled.

Explanation of the constancy of $\frac{B}{A}$ in the range $30 < A < 170$: because the nuclear force is short-ranged and saturates, each nucleon interacts only with its nearest neighbours rather than with all the other nucleons. So the total binding energy is proportional to the number of nucleons, $B.E. \propto A$, and therefore the binding energy per nucleon $\frac{B}{A}$ stays nearly constant in this mid-mass plateau.

Final answer. The U-r curve shows strong short-range attraction with a repulsive core; the B/A curve is flat in **30 due to force saturation.**

← *Back to Q67*

Q68

In a given sample, two radioisotopes, A and B, are initially present in the ratio of 1:4. *The half lives of A and B are respectively* 100 *years and* 50 *years. Find the time after which the amounts of A and B become equal.*

[2012]

SOLUTION

Let the initial amounts be in the ratio 1:4, so take $\mathbf{N_A = N_0}$ and $\mathbf{N_B = 4N_0}$ at $\boldsymbol{t = 0}$. We must find the time t at which the remaining amounts are equal.

Using the half-life form of the decay law, the amount left after time t is $\boldsymbol{N = N_0(\frac{1}{2})^{t/T_{1/2}}}$.

For isotope A, with $\mathbf{T_{1/2} = 100}$ **years**:

$$N_A(t) = N_0(\frac{1}{2})^{t/100}$$

For isotope B, with $\mathbf{T_{1/2} = 50}$ **years**:

$$N_B(t) = 4N_0(\frac{1}{2})^{t/50}$$

Setting the two amounts equal:

$$N_0(\frac{1}{2})^{t/100} = 4N_0(\frac{1}{2})^{t/50}$$

Cancelling N_0 and dividing both sides:

$$\frac{(1/2)^{t/100}}{(1/2)^{t/50}} = 4$$

$$(\frac{1}{2})^{\frac{t}{100} - \frac{t}{50}} = 4$$

Simplifying the exponent, $\boldsymbol{\frac{t}{100} - \frac{t}{50} = \frac{t}{100} - \frac{2t}{100} = -\frac{t}{100}}$.

$$(\frac{1}{2})^{-t/100} = 4$$

$$2^{t/100} = 2^2$$

Comparing exponents:

$$\frac{t}{100} = 2 \Rightarrow t = 200 \text{ years}$$

The amounts of the two isotopes become equal after **200 years**.

Final answer. The two isotopes have equal amounts after $\boldsymbol{t = 200}$ years.

← Back to Q68

Q69

Using the curve for the binding energy per nucleon as a function of mass number A, state clearly how the release in energy in the processes of nuclear fission and nuclear fusion can be explained.

[2011]

SOLUTION

The binding energy per nucleon $\frac{B}{A}$ measures how tightly nucleons are bound: a larger $\frac{B}{A}$ means a more stable, lower-energy nucleus.

Description of the graph: the vertical axis is the binding energy per nucleon $\frac{B}{A}$ (in MeV) and the horizontal axis is the mass number A. The curve rises steeply for light nuclei from near zero at $\boldsymbol{A = 1}$ to a maximum of about 8.75 MeV near $\boldsymbol{A = 56}$ (iron), then decreases slowly to about 7.6 MeV for heavy nuclei near $\boldsymbol{A = 240}$. The peak near $\boldsymbol{A = 56}$ should be labelled as the most stable region.

Energy release in nuclear fission: a heavy nucleus lies on the right of the peak where $\frac{B}{A}$ is lower (about 7.6 MeV). When it splits into two medium-mass fragments, those fragments lie nearer the peak where $\frac{B}{A}$ is higher (about 8.5 MeV). Since the products are more tightly bound, their total binding energy is greater than that of the parent. This increase in binding energy (about 200 MeV per fission) is released as energy.

Energy release in nuclear fusion: very light nuclei lie on the steeply rising left part of the curve where $\frac{B}{A}$ is small. When two such light nuclei combine to form a heavier nucleus, the product lies higher up the curve with a larger $\frac{B}{A}$. Again the products are more tightly bound than the reactants, so the increase in total binding energy is released as energy (a few MeV up to about 25 MeV per fusion).

In both processes the system moves toward the **peak of higher $\frac{B}{A}$ (greater stability)**, and the difference in binding energy is released as energy.

Final answer. Both fission and fusion move nuclei toward higher B/A (greater stability), and the gain in binding energy is released as energy.

← *Back to Q69*

Q70

When four hydrogen nuclei combine to form a helium nucleus, estimate the amount of energy in MeV released in this process of fusion. (Neglect the masses of electrons and neutrinos)
Given:
(i) mass of $^1_1\text{H} = 1.007825\ u$
(ii) mass of helium nucleus = 4.002603 u, 1 u = 931 MeV/c^2

[2011 • Set 55-2-1]

SOLUTION

When four hydrogen nuclei (protons) fuse to form one helium nucleus, the energy released equals the mass defect converted to energy using $E = \Delta m\, c^2$, with the conversion $1\ u = 931\ \text{MeV/c}^2$.

The net fusion reaction (neglecting the masses of electrons and neutrinos) is $4\ ^1_1\text{H} \rightarrow\ ^4_2\text{He}$.

The mass defect is the difference between the total mass of the four hydrogen nuclei and the mass of the helium nucleus.

$$\Delta m = 4\, m(^1_1\text{H}) - m(^4_2\text{He})$$

Substituting the given masses, $m(^1_1\text{H}) = 1.007825\ u$ and $m(^4_2\text{He}) = 4.002603\ u$:

$$\Delta m = 4 \times 1.007825 - 4.002603$$

$$\Delta m = 4.031300 - 4.002603$$

$$\Delta m = 0.028697\ u$$

The energy released is this mass defect multiplied by 931 MeV/u.

$$Q = \Delta\, m \times 931 \text{ MeV}/u$$

$$Q = 0.028697 \times 931$$

$$Q \approx 26.72 \text{ MeV}$$

So the energy released in forming one helium nucleus from four hydrogen nuclei is about **26.72 MeV**.

Final answer. The energy released is about **26.72** MeV per helium nucleus formed.

← *Back to Q70*

Q71

A heavy nucleus X of mass number 240 and binding energy per nucleon 7.6 MeV is split into two fragments Y and Z of mass numbers 110 and 130. The binding energy of nucleons in Y and Z is 8.5 MeV per nucleon. Calculate the energy Q released per fission in MeV.

[2010]

SOLUTION

In nuclear fission, the energy released, called the Q-value, equals the total binding energy of the product fragments minus the binding energy of the parent nucleus, because the fragments are more tightly bound than the parent.

For the parent nucleus X, mass number $A = 240$ and binding energy per nucleon = **7.6 MeV**, so its total binding energy is

$$\text{BE}_X = 240 \times 7.6 = 1824 \text{ MeV}$$

For the fragments Y and Z, the mass numbers are 110 and 130 and each has a binding energy per nucleon of 8.5 MeV, so

$$\text{BE}_Y + \text{BE}_Z = (110 + 130) \times 8.5 = 240 \times 8.5 = 2040 \text{ MeV}$$

The energy released per fission is the gain in binding energy:

$$Q = (BE_Y + BE_Z) - BE_X$$

$$Q = 2040 - 1824$$

$$Q = 216 \text{ MeV}$$

Hence the energy released per fission is **216 MeV**, which is typical of the energy liberated in heavy-nucleus fission.

Final answer. The energy released per fission is Q = 216 MeV.

← *Back to Q71*

Q72

A radioactive nucleus A undergoes a series of decays according to the following scheme:

$A \ \alpha \ A_1 \ \beta \ A_2 \ \gamma \ A_3 \ \gamma \ A_4$

The mass number and atomic number of A are 180 *and* 72 *respectively. What are these numbers for* A_4*?*

[2009]

SOLUTION

Each type of decay changes the mass number A and atomic number Z in a fixed way. In α-decay the nucleus emits a ${}^4_2\text{He}$ nucleus, so A decreases by 4 and Z decreases by 2. In β^--decay a neutron converts to a proton, so A stays the same and Z increases by 1. In γ-decay only a photon is emitted, so both A and Z remain unchanged.

The parent nucleus A has $\mathbf{A = 180}$ and $\mathbf{Z = 72}$.

After α-decay, A_1 has $\mathbf{A = 180 - 4 = 176}$ and $\mathbf{Z = 72 - 2 = 70}$.

After β^--decay, A_2 has $\mathbf{A = 176}$ and $\mathbf{Z = 70 + 1 = 71}$.

After the first γ-decay, A_3 has $\mathbf{A = 176}$ and $\mathbf{Z = 71}$ (unchanged).

After the second γ-decay, A_4 has $\mathbf{A = 176}$ and $\mathbf{Z = 71}$ (unchanged).

Hence for A_4, the **mass number is 176** and the **atomic number is 71**.

Final answer. For A4, mass number = 176 and atomic number = 71.

← Back to Q72

Q73

(a) The mass of a nucleus in its ground state is always less than the total mass of its constituents - neutrons and protons. Explain.
(b) Plot a graph showing the variation of potential energy of a pair of nucleons as a function of their separation.

[2009]

SOLUTION

(a) When free protons and neutrons combine to form a nucleus, energy is released as binding energy. By Einstein's mass-energy relation $\boldsymbol{E = mc^2}$, this released energy comes from a loss of mass. So the mass of the bound nucleus is less than the sum of the masses of its free constituent nucleons.

This missing mass is called the mass defect Δm, and the binding energy of the nucleus is

$$\text{BE} = \Delta m \, c^2$$

To hold the nucleons together, this much energy was given up by the system, which appears as the reduced mass of the nucleus. For example, a deuteron has mass 2.014102 u, while a free proton and neutron together have mass $\mathbf{1.007825 + 1.008665 = 2.016490\ u}$. The defect $\boldsymbol{\Delta m = 0.002388\ u}$ corresponds to a binding energy of about 2.22 MeV.

(b) Potential energy versus separation r for a pair of nucleons: Plot r (in femtometres) on the x-axis and potential energy U (in MeV) on the y-axis. Starting from very small r, the curve lies high in the positive (repulsive) region. As r increases past about $\mathbf{r_0 \approx 0.8\ fm}$, the curve crosses the x-axis and dips into the negative region, reaching a minimum of about −100 MeV near $\boldsymbol{r \approx 1\ fm}$. Beyond this minimum the curve rises back towards zero and flattens to $\boldsymbol{U = 0}$ for $\boldsymbol{r > 3\ fm}$.

The graph shows that the **nuclear force is strongly repulsive at very short range** ($r < 0.8$ fm), **strongly attractive at moderate range**, and **short-ranged**, vanishing beyond about 3 fm.

Final answer. Mass defect converts to binding energy via E = mc^2, so the nucleus is lighter; the PE curve is repulsive below 0.8 fm, attractive (minimum about −100 MeV near 1 fm) and zero beyond 3 fm.

← *Back to Q73*

Q74

A nucleus $^{23}_{10}\text{Ne}$ *undergoes* β*-decay and becomes* $^{23}_{11}\text{Na}$. *Calculate the maximum kinetic energy of electrons emitted assuming that the daughter nucleus and anti-neutrino carry negligible kinetic energy.*
Mass of $^{23}_{10}\text{Ne}$ **= 22.994455** ***u***
Mass of $^{23}_{11}\text{Na}$ **= 22.989770** ***u***
1 ***u*** **= 931.5 MeV/c^2**

[2008]

SOLUTION

In β^--decay, the maximum kinetic energy of the emitted electron equals the Q-value of the reaction, which happens when the daughter nucleus and the antineutrino carry negligible kinetic energy. The decay is

$$^{23}_{10}\text{Ne} \rightarrow {}^{23}_{11}\text{Na} + e^- + \bar{\nu}$$

The Q-value is found from the mass difference. Using atomic masses, the mass of the extra electron created is automatically balanced because the sodium atom has one more orbital electron than neon, so the difference of atomic masses already gives the correct Q:

$$\Delta m = m(^{23}_{10}\text{Ne}) - m(^{23}_{11}\text{Na})$$

$$\Delta m = 22.994455 - 22.989770$$

$$\Delta m = 0.004685\ u$$

Converting this mass defect to energy using $\mathbf{1\ u = 931.5\ MeV/c^2}$:

$$KE_{max} = Q = \Delta m \times 931.5\ \text{MeV}$$

$$KE_{max} = 0.004685 \times 931.5$$

$$KE_{max} = 4.364\ \text{MeV}$$

Hence the maximum kinetic energy of the emitted electrons is about **4.36 MeV**.

Final answer. The maximum kinetic energy of the emitted electrons is about 4.36 MeV.

← Back to Q74

Q75

State the law of radioactive decay. If N_0 is the number of radioactive nuclei in the sample at some initial time t_0, find out the relation to determine the number N present at a subsequent time. Draw a plot of N as a function of time.

[2008]

SOLUTION

Law of radioactive decay: The rate of disintegration of a radioactive sample at any instant is directly proportional to the number N of undecayed nuclei present at that instant. In symbols,

$$\frac{dN}{dt} = -\lambda N$$

where λ is the decay constant, the probability per unit time for any one nucleus to decay, and the negative sign shows that N decreases with time.

To find N at a later time, rearrange and integrate:

$$\frac{dN}{N} = -\lambda\, dt$$

$$\int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_{0}^{t} dt$$

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t$$

$$N = N_0 e^{-\lambda t}$$

This is the relation giving the number of undecayed nuclei present at time t.

Plot of N versus t: Take time t on the x-axis and number of nuclei N on the y-axis. The curve starts at $\boldsymbol{N = N_0}$ on the y-axis and falls as a smooth exponential decay, dropping to $N_0/2$ at $\boldsymbol{t = T_{1/2}}$, to $N_0/4$ at $\boldsymbol{t = 2T_{1/2}}$, and so on. It approaches the time axis asymptotically but never quite reaches zero.

Hence the number present at time t is $\boldsymbol{N = N_0 e^{-\lambda t}}$.

Final answer. N = N0 e^(-lambda t), an exponentially decaying curve starting from N0.

← *Back to Q75*

Q76

Draw a plot of the binding energy per nucleon as a function of mass number for a large number of nuclei. Explain the energy release in the process of nuclear fission from the above plot. Write a typical nuclear reaction in which a large amount of energy is released in the process of nuclear fission.

[2008]

SOLUTION

Binding energy per nucleon versus mass number graph: Take mass number A on the x-axis and binding energy per nucleon B/A (in MeV) on the y-axis. The curve rises steeply for light nuclei (with sharp peaks at ^{4}He, ^{12}C and ^{16}O), reaches a broad maximum of about 8.75 MeV near $\boldsymbol{A = 56}$ (the iron region), and then falls slowly for heavier nuclei, reaching about 7.6 MeV at $\boldsymbol{A \approx 240}$.

Energy release in fission from the graph: A heavy nucleus with $\boldsymbol{A \approx 235}$ has $\boldsymbol{B/A \approx 7.6}$ **MeV**, whereas the mid-mass fragments with $\boldsymbol{A \approx 120}$ lie higher on the curve at $\boldsymbol{B/A \approx 8.5}$ **MeV**. When the heavy nucleus splits, each nucleon becomes more tightly bound by about

$$\Delta(B/A) \approx 8.5 - 7.6 = 0.9 \text{ MeV}$$

The total energy released is roughly

$$Q \approx 0.9 \times 235 \approx 200 \text{ MeV}$$

which appears mainly as the kinetic energy of the fragments and the emitted neutrons and γ-rays.

A typical fission reaction in which a large amount of energy is released is

$$^{1}_{0}\text{n} + ^{235}_{92}\text{U} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3\,^{1}_{0}\text{n} + Q$$

with $\boldsymbol{Q \approx 200}$ **MeV** per fission.

Final answer. Fragments are more tightly bound (8.5 vs 7.6 MeV/nucleon), so fission of A about 235 releases roughly 200 MeV, e.g. n + U-235 to Ba-141 + Kr-92 + 3n.

← *Back to Q76*

Q77

Draw a plot of potential energy of a pair of nucleons as a function of their separation. What is the significance of negative potential energy in the graph drawn?

[2007]

SOLUTION

Potential energy versus separation graph for a pair of nucleons: Plot the separation r (in femtometres) on the x-axis and the potential energy U (in MeV) on the y-axis. For very small r (below about 0.8 fm) the curve lies high in the positive region, showing a steep repulsive core. As r increases past $\mathbf{r_0 \approx 0.8\ fm}$, the curve crosses the x-axis and falls into the negative region, reaching a minimum of about −100 MeV near $\boldsymbol{r \approx 1}$ **fm**. Beyond the minimum the curve rises back towards zero and becomes essentially $\boldsymbol{U = 0}$ for $\boldsymbol{r > 3}$ **fm**, showing the short range of the force.

Significance of negative potential energy: In the region where U is negative (for $\boldsymbol{r > r_0 \approx 0.8}$ **fm**), the nuclear force is attractive and the two nucleons form a

bound system. A negative potential energy means that external work must be done against the attractive nuclear force to pull the nucleons apart to infinity, where $U = 0$. In other words, the nucleons cannot separate unless energy at least equal to $|U|$ is supplied. This negative potential energy is therefore what provides the **binding energy** that holds the nucleons together and makes the nucleus a stable bound system.

Final answer. Negative PE means the nucleons are bound; energy equal to |U| must be supplied to separate them, and this provides the binding energy of the nucleus.

← *Back to Q77*

Q78

Draw a graph showing the variation of potential energy between a pair of nucleons as a function of their separation. Indicate the regions in which the nuclear force is (i) attractive, (ii) repulsive.

[2007]

SOLUTION

Potential energy versus separation graph for a pair of nucleons: Plot the separation r (in femtometres) on the x-axis and the potential energy U (in MeV) on the y-axis. For very small r (below about 0.8 fm) the curve lies high in the positive region, forming a steep repulsive core. As r increases beyond $\mathbf{r_0 \approx 0.8}$ **fm**, the curve crosses the x-axis and dips into the negative region, reaching a minimum of about −100 MeV near $\boldsymbol{r \approx 1}$ **fm**. Beyond this minimum the curve rises back towards zero and flattens to $\boldsymbol{U = 0}$ for $\boldsymbol{r > 3}$ **fm**.

Indicating the regions of the nuclear force:

(i) Attractive region: for $\boldsymbol{r > r_0 \approx 0.8}$ **fm** (where the potential energy is negative and the force pulls the nucleons together). The force is strongest near $\boldsymbol{r \approx 1}$ **fm**, the position of the minimum.

(ii) Repulsive region: for $\boldsymbol{r < r_0 \approx 0.8}$ **fm** (the steep positive core), where the nucleons strongly repel one another and resist being pushed closer.

Thus the nuclear force is **repulsive below about 0.8 fm and attractive above it**, vanishing beyond about 3 fm.

Final answer. The force is repulsive for r below about 0.8 fm and attractive for r above 0.8 fm, becoming negligible beyond 3 fm.

← *Back to Q78*

Q79

Draw the graph showing the variation of binding energy per nucleon with mass number. Give the reason for the decrease of binding energy per nucleon for nuclei with high mass numbers.

[2004]

SOLUTION

Binding energy per nucleon versus mass number graph: Take mass number A on the x-axis and binding energy per nucleon B/A (in MeV) on the y-axis. The curve starts near zero at $\boldsymbol{A = 1}$ (a single nucleon has no binding), rises steeply for light nuclei with prominent peaks at the alpha-particle nuclei ^{4}He, ^{12}C and ^{16}O, reaches a broad maximum of about 8.8 MeV near $\boldsymbol{A = 56}$ (the iron region), and then decreases gradually for heavier nuclei, falling to about 7.6 MeV at $\boldsymbol{A \approx 240}$.

Reason for the decrease of B/A for heavy nuclei: The attractive nuclear force is short-ranged, so each nucleon binds only with its nearest neighbours, and the total nuclear attraction grows roughly in proportion to A. The Coulomb repulsion between protons, however, is long-ranged, so every proton repels every other proton, and this disruptive Coulomb energy grows roughly as Z^2, much faster than the attractive binding. In heavy nuclei the large number of protons makes this Coulomb repulsion increasingly important, so it reduces the net binding and $\boldsymbol{B/A}$ **decreases for nuclei with high mass numbers**. This is also why heavy nuclei tend to undergo α-decay and fission.

Final answer. B/A decreases for heavy nuclei because the long-range Coulomb repulsion (growing as Z^2) increasingly offsets the short-range nuclear

attraction.

← Back to Q79

Q80

Name the reaction which takes place when a slow neutron beam strikes ${}_{92}{}^{235}\mathrm{U}$ nuclei. Write the nuclear reaction involved.

[2003]

SOLUTION

When a slow (thermal) neutron beam strikes ${}^{235}{}_{92}\mathrm{U}$ nuclei, the reaction that takes place is **nuclear fission**. The uranium nucleus captures the slow neutron and forms the highly unstable compound nucleus ${}^{236}{}_{92}\mathrm{U}$, which then splits into two medium-mass fragments, releasing two or three fast neutrons along with a large amount of energy as γ-rays and kinetic energy of the products.

A typical nuclear reaction involved is

$$ {}^{1}{}_{0}\mathrm{n} + {}^{235}{}_{92}\mathrm{U} \rightarrow {}^{236}{}_{92}\mathrm{U} \rightarrow {}^{141}{}_{56}\mathrm{Ba} + {}^{92}{}_{36}\mathrm{Kr} + 3\,{}^{1}{}_{0}\mathrm{n} + Q $$

with energy release $Q \approx 200$ **MeV** per fission. The secondary neutrons produced can cause further fissions in nearby ${}^{235}\mathrm{U}$ nuclei, leading to a self-sustaining **chain reaction**, which is the basis of nuclear reactors.

Final answer. It is nuclear fission: n + U-235 to Ba-141 + Kr-92 + 3n + about 200 MeV.

← Back to Q80

Q81

In the series of radioactive disintegration of ${}_{92}{}^{238}\mathrm{U}$, first an alpha-particle and then a beta-particle is emitted. What is the atomic number and mass number of the new nucleus formed by these successive disintegrations?

[2003]

SOLUTION

When a nucleus emits an alpha particle ($_2^4\text{He}$), its mass number A falls by 4 and its atomic number Z falls by 2. When it emits a beta particle ($_{-1}^0\text{e}$), the mass number A stays the same while the atomic number Z rises by 1.

We start from $_{92}^{238}\text{U}$ and first emit an α particle.

$$_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_{2}^{4}\text{He}$$

Here $A = 238 - 4 = 234$ and $Z = 92 - 2 = 90$, giving the daughter $_{90}^{234}\text{Th}$ (thorium).

Next a β^- particle is emitted from this thorium nucleus.

$$_{90}^{234}\text{Th} \rightarrow {}_{91}^{234}\text{Pa} + {}_{-1}^{0}\text{e}$$

Now $A = 234$ (unchanged) and $Z = 90 + 1 = 91$, giving the final nucleus $_{91}^{234}\text{Pa}$ (protactinium).

So after the two successive decays the new nucleus has **mass number $A = 234$ and atomic number $Z = 91$**.

COMMON MISTAKE

Track A and Z together at every step. An easy slip is changing Z but forgetting that beta decay leaves A unchanged.

Final answer. The new nucleus has mass number 234 and atomic number 91 (it is $_{91}^{234}\textbf{Pa}$).

← *Back to Q81*

Q82

(a) Consider the following nuclides: $^{12}_{6}\text{C}$, $^{198}_{80}\text{Hg}$, $^{14}_{6}\text{C}$, $^{197}_{79}\text{Au}$. *Group them into isotopes and isotones.*
(b) How does the size of a nucleus depend on its mass number A? Hence prove that the density of nucleus is a constant, independent of A, for all nuclei.

[2026 • Set 55-2-1]

SOLUTION

(a) Isotopes have the same atomic number Z but different mass number A. Isotones have the same number of neutrons $N = A - Z$ but different Z.

For the given nuclides: ${}_{6}^{12}\mathrm{C}$ has $N = 12 - 6 = 6$; ${}_{80}^{198}\mathrm{Hg}$ has $N = 198 - 80 = 118$; ${}_{6}^{14}\mathrm{C}$ has $N = 14 - 6 = 8$; ${}_{79}^{197}\mathrm{Au}$ has $N = 197 - 79 = 118$.

Isotopes (same $Z = 6$): ${}_{6}^{12}\mathrm{C}$ and ${}_{6}^{14}\mathrm{C}$.

Isotones (same $N = 118$): ${}_{80}^{198}\mathrm{Hg}$ and ${}_{79}^{197}\mathrm{Au}$.

(b) Experiments on nuclear scattering show that the radius of a nucleus depends on its mass number through the relation

$$R = \mathrm{R_0}\, \mathrm{A}^{1/3}$$

where $\mathrm{R_0} \approx 1.2\ \mathrm{fm} = 1.2 \times 10^{-15}\ m$ is a constant. So the nuclear radius grows as the cube root of A.

Treating the nucleus as a sphere, its volume is

$$V = \frac{4}{3}\pi\, \mathrm{R}^3 = \frac{4}{3}\pi\, (\mathrm{R_0}\, \mathrm{A}^{1/3})^3 = \frac{4}{3}\pi\, \mathrm{R_0}^3\, A$$

The mass of the nucleus is approximately $M = A\, m$, where m is the average mass of one nucleon.

The nuclear density is therefore

$$\rho = \frac{M}{V} = \frac{A\, m}{\frac{4}{3}\pi\, \mathrm{R_0}^3\, A}$$

$$\rho = \frac{3\, m}{4\pi\, \mathrm{R_0}^3}$$

The mass number A has cancelled completely. Since m and $\mathrm{R_0}$ are constants, ρ comes out the same for every nucleus (about $2.3 \times 10^{17}\ \mathrm{kg\ m^{-3}}$). Hence the **density of a nucleus is constant and independent of its mass number A.**

Final answer. Isotopes: ${}_{6}^{12}\mathrm{C}$, ${}_{6}^{14}\mathrm{C}$; isotones: ${}_{80}^{198}\mathrm{Hg}$, ${}_{79}^{197}\mathrm{Au}$. Since $R = \mathrm{R_0}\, \mathrm{A}^{1/3}$, the A cancels in $\rho = M/V$, so nuclear density is independent of A.

← *Back to Q82*

Q83

Differentiate between nuclear fission and nuclear fusion. Give one example for each with nuclear reaction.

[2026 • Set 55-3-1]

SOLUTION

Nuclear fission and nuclear fusion are the two ways a nucleus can release large amounts of energy, but they are opposite processes.

In nuclear fission a heavy nucleus splits into two lighter nuclei of comparable mass, usually after capturing a neutron, along with a few free neutrons. In nuclear fusion two very light nuclei combine to form a single heavier nucleus.

The energy source is different in spirit. Fission is triggered easily by a slow neutron and can occur at ordinary temperatures, whereas fusion needs extremely high temperature (of the order of $10^7\ K$) and pressure so that the light nuclei can overcome their mutual Coulomb repulsion and get close enough for the nuclear force to act.

The energy released per nucleon is much larger in fusion (about 3.5 MeV per nucleon) than in fission (about 0.9 MeV per nucleon), so fusion is the more efficient energy source. Both happen because the products have a higher binding energy per nucleon than the reactants.

Example of fission:

$$_{92}{}^{235}\mathrm{U} + {}_{0}{}^{1}\mathrm{n} \rightarrow {}_{56}{}^{141}\mathrm{Ba} + {}_{36}{}^{92}\mathrm{Kr} + 3\,{}_{0}{}^{1}\mathrm{n} + Q\ (\approx 200\ \mathrm{MeV})$$

Example of fusion:

$$_{1}{}^{2}\mathrm{H} + {}_{1}{}^{3}\mathrm{H} \rightarrow {}_{2}{}^{4}\mathrm{He} + {}_{0}{}^{1}\mathrm{n} + Q\ (\approx 17.6\ \mathrm{MeV})$$

Thus **fission splits a heavy nucleus while fusion joins light nuclei**, and both release energy because the products are more tightly bound.

Final answer. Fission: a heavy nucleus splits into lighter ones (e.g. $\mathbf{_{92}{}^{235}U + \mathit{n} \rightarrow {}_{56}{}^{141}Ba + {}_{36}{}^{92}Kr + 3n + 200\ MeV}$); fusion: light nuclei combine (e.g. $\mathbf{_{1}{}^{2}H + {}_{1}{}^{3}H \rightarrow {}_{2}{}^{4}He + \mathit{n} + 17.6\ MeV}$).

← Back to Q83

Q84

Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation (r). Using this plot, show that the nuclear force is (i) attractive for $r > r_0$, and (ii) repulsive for $r < r_0$.

[2026 • Set 55-3-2]

SOLUTION

The plot is potential energy U of a pair of nucleons on the vertical axis against their separation r on the horizontal axis.

Description of the curve: for very small r the curve lies high up in the positive U region and falls steeply as r increases. It crosses the zero line, reaches a minimum (most negative value, about −100 MeV) at a separation $r = r_0 \approx 0.8$ fm, and then rises back towards zero as r increases further, becoming essentially zero beyond a few femtometres. So the curve has a sharp potential well with its lowest point at $r = r_0$.

The force between the nucleons is related to the potential energy by

$$F = -\frac{dU}{dr}$$

(i) For $r > r_0$: to the right of the minimum the curve climbs upward, so the slope is positive, $\frac{dU}{dr} > 0$. Then

$$F = -\frac{dU}{dr} < 0$$

A negative force points towards smaller r, i.e. it pulls the two nucleons together. Hence the nuclear force is **attractive for $r > r_0$**.

(ii) For $r < r_0$: to the left of the minimum the curve drops steeply, so the slope is negative, $\frac{dU}{dr} < 0$. Then

$$F = -\frac{dU}{dr} > 0$$

A positive force points towards larger r, i.e. it pushes the nucleons apart. Hence the nuclear force is **repulsive for $r < r_0$**.

At $r = r_0$ the slope is zero, so the force is zero; this is the equilibrium separation where the potential energy is minimum and the nucleon pair is most stable.

Final answer. From $F = -dU/dr$: for $r > r_0$ the slope $dU/dr > 0$ so $F < 0$ (attractive), and for $r < r_0$ the slope $dU/dr < 0$ so $F > 0$ (repulsive).

← *Back to Q84*

Q85

In the nuclear reaction:
$^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{A}_{2}\mathrm{X} + {}^{1}_{0}\mathrm{n}$
(a) Find the value of A.
(b) Calculate the amount of energy released in the reaction.
Given: $m(^{2}_{1}\mathrm{H}) = 2{\cdot}014102\ u$, $m(^{A}_{2}\mathrm{X}) = 3{\cdot}016049\ u$, $m(^{1}_{0}\mathrm{n}) = 1{\cdot}008665\ u$, $1\ u = 931{\cdot}5\ \mathrm{MeV}/c^2$

[2026 • Set 55-4-1]

SOLUTION

The reaction is $^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{A}_{2}\mathrm{X} + {}^{1}_{0}\mathrm{n}$.

(a) Both mass number and charge number are conserved in a nuclear reaction.

Conserving mass number: $2 + 2 = A + 1$

$$A = 4 - 1 = 3$$

(Charge check: $1 + 1 = 2 + 0$, which is satisfied.) So X is the helium isotope $^{3}_{2}\mathrm{He}$, and $A = 3$.

(b) The energy released equals the mass defect times c^2. First find the total mass before and after the reaction.

Total initial mass:

$$M_i = 2 \times m(^{2}_{1}\mathrm{H}) = 2 \times 2.014102 = 4.028204\ u$$

Total final mass:

$$M_f = m(_2^3\text{He}) + m(_0^1\text{n}) = 3.016049 + 1.008665 = 4.024714\ u$$

Mass defect:

$$\Delta m = M_i - M_f = 4.028204 - 4.024714 = 0.003490\ u$$

Energy released, using $\mathbf{1\ u = 931.5\ MeV/c^2}$:

$$Q = \Delta m \times 931.5\ \text{MeV}$$

$$Q = 0.003490 \times 931.5$$

$$Q \approx 3.25\ \text{MeV}$$

Since $\Delta m > 0$, energy is released. The reaction gives out about **3.25 MeV** of energy.

Final answer. $A = 3$ (so $X = {}_2^3\text{He}$) and the energy released is $Q \approx 3.25$ **MeV**.

← *Back to Q85*

Q86

(a) Distinguish between isotopes and isobars, giving one example for each.
(b) Derive the relation between atomic mass unit (u) and electron volt (eV).

[2026 • Set 55-4-3]

SOLUTION

(a) Isotopes are nuclei of the same element that have the same atomic number Z but different mass number A. They have the same chemical properties but different masses. Example: ${}_1^1\text{H}$, ${}_1^2\text{H}$ and ${}_1^3\text{H}$ (all have $Z = 1$).

Isobars are nuclei that have the same mass number A but different atomic number Z. They belong to different elements, so they have different chemical properties but nearly the same mass. Example: ${}_{18}^{40}\text{Ar}$ and ${}_{20}^{40}\text{Ca}$ (both have $A = 40$).

(b) One atomic mass unit is defined as one-twelfth of the mass of a ${}_6^{12}\text{C}$ atom.

$$1\ u = 1.6605 \times 10^{-27}\ \text{kg}$$

Its energy equivalent follows from Einstein's mass-energy relation $E = mc^2$, with $c = 3 \times 10^8\ m\ s^{-1}$:

$$E = (1.6605 \times 10^{-27}) \times (3 \times 10^8)^2$$

$$E = 1.6605 \times 10^{-27} \times 9 \times 10^{16}$$

$$E = 1.4924 \times 10^{-10}\ J$$

To express this in electron volt, divide by the charge $1.6 \times 10^{-19}\ C$ (since $1\ eV = 1.6 \times 10^{-19}\ J$):

$$E = \frac{1.4924 \times 10^{-10}}{1.6 \times 10^{-19}}\ \text{eV}$$

$$E = 9.327 \times 10^8\ \text{eV}$$

$$E \approx 931.5 \times 10^6\ \text{eV} = 931.5\ \text{MeV}$$

Hence the required relation is $1\ u = 931.5\ MeV/c^2$, i.e. a mass of one atomic mass unit is equivalent to about 931.5 MeV of energy.

Final answer. Isotopes have same Z (e.g. ${}_1{}^1H$, ${}_1{}^2H$, ${}_1{}^3H$), isobars have same A (e.g. ${}_{18}{}^{40}Ar$, ${}_{20}{}^{40}Ca$); and using $E = mc^2$, $1\ u = 931.5\ MeV/c^2$.

← *Back to Q86*

Q87

(a) (i) Write any two features of nuclear forces.
(ii) If both the number of protons and the neutrons are conserved in each nuclear reaction, in what way is mass converted into energy (or vice versa) in a nuclear reaction? Explain.

[2026 • Set 55-5-1]

SOLUTION

(a)(i) Two features of nuclear forces:

First, nuclear forces are very short ranged. They act strongly only when the separation between two nucleons is of the order of a few femtometres (about r 3

fm) and become negligible beyond that. They are also much stronger than the electrostatic and gravitational forces within this range.

Second, nuclear forces are charge independent and show saturation. The force between two nucleons does not depend on their charge, so it is the same for a proton-proton, neutron-neutron or proton-neutron pair. Because of saturation, a given nucleon interacts only with its nearest neighbours, which keeps the binding energy per nucleon nearly constant for medium and heavy nuclei.

(a)(ii) Even though the total number of protons and the total number of neutrons are each conserved in a nuclear reaction, the total rest mass of the products is generally not equal to the total rest mass of the reactants. The reason is that the nucleons are bound together with different binding energies before and after the reaction.

A nucleon bound inside a nucleus has a slightly smaller mass than a free nucleon, the deficit being the mass defect Δm linked to the binding energy by $\mathbf{BE = \Delta m\, c^2}$. When the binding rearranges during the reaction, this small difference in total mass appears as energy according to Einstein's relation

$$E = \Delta m\, c^2$$

If the products are more tightly bound (their total mass is less), the missing mass is released as energy (kinetic energy of products and gamma rays). If the products are less tightly bound, energy must be supplied and is converted into mass. So **mass is converted into energy (or vice versa) through the change in binding energy, governed by $E = \Delta m\, c^2$**, not by any change in the number of protons or neutrons.

Final answer. Nuclear forces are short ranged and charge independent (with saturation); and even with proton and neutron numbers conserved, the change in binding energy makes the total mass change, so mass converts to energy via $E = \Delta m\, c^2$.

← *Back to Q87*

Q88

(a) *Define 'Mass defect' and 'Binding energy' of a nucleus. Describe 'Fission process' on the basis of binding energy per nucleon.*
(b) *A deuteron contains a proton and a neutron and has a mass of* 2.013553 *u. Calculate the mass defect for it in u and its energy equivalence in* MeV. *(***m_p = 1.007277 *u*, m_n = 1.008665 *u*, 1 *u* = 931.5 MeV/c^2***)***

[2025 • Set 55-1-1]

SOLUTION

(a) Mass defect is the difference between the sum of the masses of the free constituent nucleons (protons and neutrons) and the actual mass of the nucleus formed from them. For a nucleus with Z protons and $(A - Z)$ neutrons,

$$\Delta m = [Z\, m_p + (A - Z)\, m_n] - m_{\text{nucleus}}$$

Binding energy is the energy equivalent of this mass defect. It is the energy needed to split the nucleus completely into its free nucleons (or, equally, the energy released when the free nucleons combine to form the nucleus):

$$BE = \Delta m\, c^2$$

Fission on the basis of binding energy per nucleon: the binding energy per nucleon (BE/A) is a measure of the stability of a nucleus. Very heavy nuclei such as ${}_{92}{}^{235}U$ have a relatively low BE/A (about 7.6 MeV), while medium-mass nuclei have a higher BE/A (about 8.5 MeV). When a heavy nucleus splits into two medium-mass fragments, the fragments are more tightly bound, so BE/A increases. The gain in binding energy, multiplied by the number of nucleons, is released as energy (about 200 MeV per fission of ${}_{92}{}^{235}U$). This is why the fission of a heavy nucleus liberates energy.

(b) For the deuteron, m_p = 1.007277 *u*, m_n = 1.008665 *u*, and the nuclear mass is 2.013553 *u*.

$$\Delta m = (m_p + m_n) - m_{\text{deuteron}}$$

$$\Delta m = (1.007277 + 1.008665) - 2.013553$$

$$\Delta m = 2.015942 - 2.013553$$

$$\Delta m = 0.002389\ u$$

Energy equivalence, using $\mathbf{1\ u = 931.5\ MeV/c^2}$:

$$\text{BE} = \Delta m \times 931.5\ \text{MeV}$$

$$\text{BE} = 0.002389 \times 931.5$$

$$\text{BE} \approx 2.225\ \text{MeV}$$

So the mass defect of the deuteron is $\mathbf{\Delta m = 0.002389\ u}$ and its binding energy is about **2.23 MeV**.

Final answer. $\mathbf{\Delta m = 0.002389\ u}$ and the binding energy is about **2.23 MeV**.

← *Back to Q88*

Q89

(a) Show the variation of binding energy per nucleon with mass number. Write the significance of the binding energy curve.
(b) Two nuclei with lower binding energy per nucleon form a nuclei with more binding energy per nucleon.
(i) What type of nuclear reaction is it?
(ii) Whether the total mass of nuclei increases, decreases or remains unchanged?
(iii) Does the process require energy or produce energy?

[2025 • Set 55-2-1]

SOLUTION

(a) The graph has binding energy per nucleon BE/A (in MeV) on the vertical axis and mass number A on the horizontal axis.

Description of the curve: starting from light nuclei, the curve rises steeply from $\boldsymbol{A = 2}$ (deuterium, $\mathbf{BE/A \approx 1.1\ MeV}$) up to a broad flat maximum of about 8.75 MeV near $\boldsymbol{A = 56}$ (the iron region). Beyond this peak it falls slowly, reaching about 7.6 MeV for very heavy nuclei near $\boldsymbol{A = 238}$. The curve is almost flat (around 8.5 MeV) over the middle range of mass numbers.

Significance of the curve:

The maximum near $A = 56$ shows that iron-region nuclei are the most stable, because they have the highest binding energy per nucleon.

Heavy nuclei (right of the peak) can release energy by fission, splitting into medium-mass nuclei of higher BE/A, while light nuclei (left of the peak) can release energy by fusion, joining to form heavier nuclei of higher BE/A.

The near-constant value of BE/A over the middle range indicates the saturation (short-range) property of nuclear forces, since each nucleon binds only to its nearest neighbours.

(b) Two nuclei with lower binding energy per nucleon combine to form a single nucleus with higher binding energy per nucleon.

(i) Since two lighter nuclei join into a heavier one, this is a nuclear fusion reaction.

(ii) The product is more tightly bound, so the total mass of the product is less than the total mass of the two reactant nuclei; the total mass decreases (a positive mass defect appears).

(iii) The lost mass is converted into energy by $E = \Delta m\, c^2$, so the process produces (releases) energy.

Final answer. The curve peaks at **BE/$A \approx 8.75$ MeV** near $A = 56$ (most stable); the described process is nuclear fusion, the total mass decreases, and the process produces (releases) energy.

← *Back to Q89*

Q90

(a) Consider the so-called 'D-T reaction' (Deuterium-Tritium reaction). In a thermonuclear fusion reactor, the following nuclear reaction occurs:

$${}_1^2H + {}_1^3H \longrightarrow {}_2^4\mathrm{He} + {}_0^1n + Q$$

Find the amount of energy released in the reaction.

Given:

$m({}_1^2H) = 2{\cdot}014102\ u$

$m({}_1^3H) = 3{\cdot}016049\ u$

$m({}_2^4\mathrm{He}) = 4{\cdot}002603\ u$

$m({}_0^1 n) = 1{\cdot}008665\ u$

$1\ u = 931\ \text{MeV/c}^2$

OR

(b) Show that the nuclear density is independent of mass number.

[2025 • Set 55-6-1]

SOLUTION

(a) The energy released Q equals the mass defect of the reaction multiplied by c^2. First find the total mass on each side of

$$_1^2\text{H} + {}_1^3\text{H} \rightarrow {}_2^4\text{He} + {}_0^1\text{n} + Q$$

Total mass of reactants:

$$M_i = m({}_1^2\text{H}) + m({}_1^3\text{H}) = 2.014102 + 3.016049 = 5.030151\ u$$

Total mass of products:

$$M_f = m({}_2^4\text{He}) + m({}_0^1\text{n}) = 4.002603 + 1.008665 = 5.011268\ u$$

Mass defect:

$$\Delta m = M_i - M_f = 5.030151 - 5.011268 = 0.018883\ u$$

Energy released, using **$1\ u = 931\ \text{MeV/c}^2$**:

$$Q = \Delta m \times 931\ \text{MeV}$$

$$Q = 0.018883 \times 931$$

$$Q \approx 17.58\ \text{MeV}$$

So the D-T fusion reaction releases about **17.58 MeV** of energy.

(b) OR: To show that nuclear density is independent of mass number, use the experimental radius relation

$$R = R_0\, A^{1/3},\ R_0 \approx 1.2\ \text{fm}$$

Volume of the (spherical) nucleus:

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0{}^3 A$$

Mass of the nucleus, with m the average nucleon mass:

$$M = A\,m$$

Density:

$$\rho = \frac{M}{V} = \frac{A\,m}{\frac{4}{3}\pi\,R_0{}^3\,A} = \frac{3\,m}{4\pi\,R_0{}^3}$$

The mass number A cancels, so ρ depends only on the constants m and R_0. Hence the nuclear density is the same for all nuclei, i.e. **independent of mass number A**.

Final answer. The D-T reaction releases $Q \approx 17.58$ **MeV**; alternatively, since $R = R_0\,A^{1/3}$ the A cancels in $\rho = M/V$, so nuclear density is independent of A.

← *Back to Q90*

Q91

Find the Q value of the following nuclear reaction:

$_6{}^{12}C + {}_6{}^{12}C \longrightarrow {}_{10}{}^{20}Ne + {}_2{}^4He$

Is this reaction exothermic or endothermic?

Given:

$m(_6{}^{12}C) = 12{\cdot}000000\ u$

$m(_{10}{}^{20}Ne) = 19{\cdot}992439\ u$

$m(_2{}^4He) = 4{\cdot}002603\ u$

$1\ u = 931\ MeV/c^2$

[2025 • Set 55-6-2]

SOLUTION

The Q value of a nuclear reaction equals the total rest-mass of the reactants minus the total rest-mass of the products, converted to energy. A positive Q means energy is released (exothermic), while a negative Q means energy must be supplied (endothermic).

The given reaction is $_6{}^{12}C + {}_6{}^{12}C \longrightarrow {}_{10}{}^{20}Ne + {}_2{}^4He$.

First find the mass defect, the difference between the initial and final masses.

$$\Delta m = [\, 2\, m(_{6}^{12}\mathrm{C}) \,] - [\, m(_{10}^{20}\mathrm{Ne}) + m(_{2}^{4}\mathrm{He}) \,]$$

$$\Delta m = [\, 2 \times 12.000000 \,] - [\, 19.992439 + 4.002603 \,]\, u$$

$$\Delta m = 24.000000 - 23.995042\, u$$

$$\Delta m = 0.004958\, u$$

Now convert this mass defect into energy using $\mathbf{1\ u = 931\ MeV/c^2}$.

$$Q = \Delta m \times 931\ \mathrm{MeV}$$

$$Q = 0.004958 \times 931\ \mathrm{MeV}$$

$$Q = 4.62\ \mathrm{MeV}$$

Since Q is positive, energy is released, so the reaction is **exothermic** (exoergic). The total mass of the products is less than that of the reactants, and this missing mass appears as the released energy. This carbon-burning reaction occurs in the cores of heavy stars during the late stages of stellar evolution.

COMMON MISTAKE

Always keep the masses to all the given decimal places until the final subtraction. Rounding early can wipe out the tiny mass defect and give a wrong Q.

Final answer. $Q = +4.62$ **MeV**; since $Q>0$ the reaction is exothermic.

← *Back to Q91*

Q92

(a) Differentiate between 'nuclear fission' and 'nuclear fusion'. Briefly discuss one example of each.
(b) Draw a graph of potential energy between a pair of nucleons as a function of their separation.

[2025 • Set 55-6-3]

SOLUTION

(a) Nuclear fission and nuclear fusion both release energy but are opposite processes on the binding-energy-per-nucleon curve.

In **nuclear fission**, a heavy nucleus splits into two lighter nuclei of comparable mass, along with a few neutrons and a large release of energy. It is usually triggered when a heavy nucleus captures a slow (thermal) neutron, and it can sustain a chain reaction.

Example of fission: ${}_{92}^{235}\mathrm{U} + {}_{0}^{1}n \longrightarrow {}_{56}^{141}\mathrm{Ba} + {}_{36}^{92}\mathrm{Kr} + 3\,{}_{0}^{1}n + Q$, releasing about 200 MeV of energy per fission.

In **nuclear fusion**, two light nuclei combine to form a single heavier nucleus, releasing energy. It requires extremely high temperatures, of the order of $10^7\ K$, so that the nuclei have enough kinetic energy to overcome their mutual Coulomb repulsion and come within the range of the attractive nuclear force.

Example of fusion: ${}_{1}^{2}\mathrm{H} + {}_{1}^{3}\mathrm{H} \longrightarrow {}_{2}^{4}\mathrm{He} + {}_{0}^{1}n + 17.6\ \mathrm{MeV}$.

In both processes the products have a higher binding energy per nucleon than the reactants, so energy is liberated.

(b) Description of the potential energy versus separation graph for a pair of nucleons.

Draw the horizontal axis as the separation r between the two nucleons (in femtometres) and the vertical axis as the potential energy $U(r)$ (in MeV), with the value $\boldsymbol{U = 0}$ marked on the vertical axis. The curve behaves as follows. Starting from large r on the right, U is nearly zero. As r decreases, the curve dips below the zero line into a negative minimum at a separation $\mathbf{r_0 \approx 0.8\ fm}$, where $\boldsymbol{U \approx -100\ \mathrm{MeV}}$ (this minimum is the most stable, attractive region). For separations smaller than r_0, the curve turns sharply upward and rises steeply to large positive values, crossing the zero line and shooting up. This steep rising part for $\boldsymbol{r < r_0}$ represents a strong repulsive core. So the graph is a deep negative well at about 0.8 fm, with a steep repulsive wall on its left and a tail rising to zero on its right.

Final answer. Fission splits a heavy nucleus while fusion joins light nuclei; the PE curve is a deep well at $\mathbf{r_0 \approx 0.8\ fm}$ with a repulsive core at smaller $\boldsymbol{r}$.

← Back to Q92

Q93

Briefly explain how energy is produced in stars, giving two examples of the nuclear reactions involved.

[2025 • Set 55-7-1]

SOLUTION

Stars produce energy by **nuclear fusion** in their extremely hot and dense cores. Under the enormous temperature (of the order of $10^7\ K$) and pressure, light nuclei such as hydrogen acquire enough kinetic energy to overcome their Coulomb repulsion and fuse into heavier nuclei. Because the heavier product nucleus has a larger binding energy per nucleon than the light reactant nuclei, a part of the mass is converted into energy, which is radiated away as the star's light and heat. The net effect is the conversion of four protons into one helium nucleus with the release of energy.

Example 1: the proton-proton (pp) chain, which powers stars like the Sun (core temperature about $1.5 \times 10^7\ K$). It proceeds in steps:

$$_1{}^1\mathrm{H} + {}_1{}^1\mathrm{H} \longrightarrow {}_1{}^2\mathrm{H} + \mathrm{e}^+ + \nu_e$$

$$_1{}^2\mathrm{H} + {}_1{}^1\mathrm{H} \longrightarrow {}_2{}^3\mathrm{He} + \gamma$$

$$_2{}^3\mathrm{He} + {}_2{}^3\mathrm{He} \longrightarrow {}_2{}^4\mathrm{He} + 2\,{}_1{}^1\mathrm{H}$$

The overall result is $4\,{}_1{}^1\mathrm{H} \longrightarrow {}_2{}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e + 26.7\ \mathrm{MeV}$.

Example 2: the carbon-nitrogen-oxygen (CNO) cycle, which dominates in stars more massive than the Sun. Here $^{12}\mathrm{C}$ acts as a catalyst, capturing protons in successive steps and finally being released, while four protons are again converted into one helium nucleus:

$$4\,{}_1{}^1\mathrm{H} \longrightarrow {}_2{}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e + 26.7\ \mathrm{MeV}$$

Both chains give the same net reaction and release the same energy per helium nucleus formed. Once the core hydrogen is used up, heavier elements up to iron are built in later fusion stages of massive stars.

Final answer. Stars shine by fusion; two examples are the proton-proton chain and the CNO cycle, each turning **4 $_1{}^1$H** into **$_2{}^4$He** with **26.7 MeV** released.

← *Back to Q93*

Q94

(a) State any two properties of a nucleus.
(b) Why is the density of a nucleus much more than that of an atom?
(c) Show that the density of the nuclear matter is the same for all nuclei.

[2024 • Set 55-1-1]

SOLUTION

(a) Two properties of a nucleus.

First, the nucleus is extremely small: its radius is of the order of 10^{-15} m (a few femtometres), compared with the atomic radius of about 10^{-10} m. Yet it carries almost the entire mass of the atom, since the electrons are nearly massless by comparison.

Second, the nucleus is made of protons and neutrons, together called nucleons. The number of protons is the atomic number Z, and the number of neutrons is $N = A - Z$, where A is the mass number. The nucleus is positively charged, with charge +Ze.

(b) Why the density of a nucleus is much more than that of an atom.

Almost all the mass of an atom is packed into the nucleus, but the nucleus occupies only a very tiny fraction of the atomic volume. Since volume goes as the cube of the radius, the ratio of nuclear to atomic volume is about

$$\frac{V_{\text{nucleus}}}{V_{\text{atom}}} \sim \left(\frac{10^{-15}}{10^{-10}}\right)^3 = 10^{-15}$$

So nearly all the mass sits in 10^{-15} of the atom's volume. Concentrating the same mass into so small a volume makes the nuclear density roughly 10^{14} times the average atomic density. Hence nuclear density is enormously larger than atomic density.

(c) The density of nuclear matter is the same for all nuclei.

The radius of a nucleus of mass number A is given by

$$R = R_0 A^{1/3},\ R_0 \approx 1.2 \times 10^{-15}\ m$$

Its volume is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi {R_0}^3 A$$

Each nucleon has mass nearly equal to m_n, so the nuclear mass is $M \approx A\, m_n$. The density is therefore

$$\rho = \frac{M}{V} = \frac{A\, m_n}{\frac{4}{3}\pi {R_0}^3 A}$$

$$\rho = \frac{3\, m_n}{4\pi {R_0}^3}$$

The mass number A cancels completely, so ρ does not depend on A. Putting in numbers gives $\rho \approx 2.3 \times 10^{17}$ kg/m^3, **the same for every nucleus**.

Final answer. Since $R = R_0 A^{1/3}$, the factor A cancels and $\rho = \dfrac{3\, m_n}{4\pi {R_0}^3} \approx 2.3\times 10^{17}$ kg/m^3 for all nuclei.

← Back to Q94

Q95

(a) Write two characteristic properties of nuclear force.
(b) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from the plot.

[2024 • Set 55-1-2]

SOLUTION

(a) Two characteristic properties of nuclear force.

First, the nuclear force is **short-ranged**. It is appreciable only when nucleons are very close, within about 3 fm, and it falls rapidly to zero beyond a few

femtometres. This is why a nucleon interacts only with its nearest neighbours (the saturation property).

Second, the nuclear force is **strongly attractive** at separations of about 1 to 2 fm, but it becomes **repulsive** at very small separations (below about 0.8 fm). This repulsive core prevents the nucleus from collapsing and keeps nuclear density constant. The force is also charge-independent: the force between two protons, two neutrons, or a proton and a neutron is essentially the same (after removing the Coulomb effect).

(b) Description of the potential energy versus separation plot and conclusions.

Take the horizontal axis as the separation r between the two nucleons (in femtometres) and the vertical axis as the potential energy $U(r)$ (in MeV), with $\boldsymbol{U = 0}$ marked. Starting at large r, the curve sits at nearly zero. As r decreases the curve dips into a negative minimum at $\mathbf{r_0 \approx 0.8\ fm}$, where U reaches its lowest value, about −100 MeV. For $\boldsymbol{r < r_0}$ the curve turns sharply upward and rises steeply to large positive values. So the plot is a deep potential well with its minimum at about 0.8 fm, a steep repulsive wall on the left, and a tail approaching zero on the right.

Two important conclusions from the plot:

First, the nuclear force is **attractive only over a narrow range** of about 1 to 2 fm and decays to zero beyond a few femtometres, confirming that it is a short-range force.

Second, for separations less than $\mathbf{r_0 \approx 0.8\ fm}$ the potential energy rises steeply, meaning the force becomes strongly **repulsive at very small distances**. This repulsive core stops nucleons from overlapping and explains why nuclear matter has a nearly constant density.

Final answer. Nuclear force is short-ranged and charge-independent; the PE well (minimum at $\mathbf{r_0 \approx 0.8\ fm}$) shows attraction over **1** to **2 fm** and repulsion for **r0**.

← *Back to Q95*

Q96

(a) Plot a graph depicting potential energy of a pair of nucleons in a nucleus as a function of their separation.
(b) Identify the regions where the nuclear force is (i) attractive, and (ii) repulsive. Justify your answer.

[2024 • Set 55-2-2]

SOLUTION

(a) Description of the potential energy versus separation plot for a pair of nucleons.

Take the horizontal axis as the separation r between the two nucleons (in femtometres) and the vertical axis as the potential energy $U(r)$ (in MeV), with the line $\boldsymbol{U = 0}$ shown. From large r on the right, the curve is almost zero. As r decreases, it dips below zero into a negative minimum at a separation $\mathbf{r_0 \approx 0.8\ fm}$, where U reaches its lowest value (about −100 MeV). For separations smaller than r_0, the curve turns and rises very steeply to large positive values. Thus the graph is a deep well with its minimum near 0.8 fm, a steep wall to its left, and a tail rising back to zero on its right.

(b) Identifying the attractive and repulsive regions.

The force between the nucleons is related to the potential energy by

$$F = -\frac{dU}{dr}$$

(i) For $\boldsymbol{r > r_0 \approx 0.8\ fm}$, the potential energy increases as r increases, so $\boldsymbol{\frac{dU}{dr} > 0}$ and $\boldsymbol{F = -\frac{dU}{dr} < 0}$. A negative force points toward decreasing r, that is, the nucleons are pulled together. Hence in this region the nuclear force is **attractive**. This is the region where the strong nuclear interaction binds the nucleons into the nucleus.

(ii) For $\boldsymbol{r < r_0 \approx 0.8\ fm}$, the potential energy rises steeply as r decreases, so $\boldsymbol{\frac{dU}{dr} < 0}$ and $\boldsymbol{F = -\frac{dU}{dr} > 0}$. A positive force points toward increasing r, that is, the nucleons push apart. Hence in this region the nuclear force is **repulsive**. This

repulsive core prevents the nucleons from coming too close and collapsing, keeping the nuclear density constant.

> **TIP**
>
> The sign of the force comes straight from the slope of the $U(r)$ curve. Where U rises with r the force is attractive; where U rises as r falls, the force is repulsive.

Final answer. Using $F=-dU/dr$: the force is attractive for $r>r_0\approx 0.8$ fm and repulsive for $r<r_0$.

← *Back to Q96*

Q97

(a) Briefly discuss three characteristics of the forces between nucleons.
(b) Which out of $_8{}^{16}\mathrm{O}$ *and* $_7{}^{14}\mathrm{N}$ *nuclei is more stable and why?*

[2024 • Set 55-2-3]

SOLUTION

(a) Three characteristics of the forces between nucleons.

First, the nuclear force is **short-ranged**. It acts strongly only within about 3 fm and becomes negligible beyond a few femtometres. This leads to saturation, so a nucleon interacts only with its nearest neighbours.

Second, it is the **strongest of the fundamental forces** at nuclear distances and is attractive at separations of about 1 to 2 fm, but it has a repulsive core at very small separations (below about 0.8 fm), which prevents the nucleus from collapsing.

Third, the nuclear force is **charge-independent**: the force between a pair of protons, a pair of neutrons, or a proton and a neutron is essentially the same once the Coulomb effect between protons is removed.

(b) Comparing the stability of $_8{}^{16}\mathrm{O}$ and $_7{}^{14}\mathrm{N}$.

The stability of a nucleus is decided by its binding energy per nucleon (BE/A): the larger this value, the more tightly the nucleons are held and the more stable the nucleus.

For ${}_{8}^{16}\mathrm{O}$: binding energy ≈ **127.6 MeV**, so

$$\frac{\mathrm{BE}}{A} \approx \frac{127.6}{16} \approx 7.98 \text{ MeV/nucleon}$$

For ${}_{7}^{14}\mathrm{N}$: binding energy ≈ **104.7 MeV**, so

$$\frac{\mathrm{BE}}{A} \approx \frac{104.7}{14} \approx 7.48 \text{ MeV/nucleon}$$

Since ${}_{8}^{16}\mathrm{O}$ has the larger binding energy per nucleon, it is the **more stable** nucleus. It is also a doubly-magic nucleus, with $Z = N = 8$, which adds to its stability.

Final answer. **${}_{8}^{16}\mathrm{O}$** is more stable because its BE/A (**≈ 7.98** MeV) exceeds that of **${}_{7}^{14}\mathrm{N}$** (**≈ 7.48** MeV).

← *Back to Q97*

Q98

(a) Define atomic mass unit (u).
(b) Calculate the energy required to separate a deuteron into its constituent parts (a proton and a neutron). Given: $\boldsymbol{m(D) = 2.014102}$ *u;* $\mathbf{m_p = 1.007825}$ *u;* $\mathbf{m_n = 1.008665}$ *u.*

[2024 • Set 55-3-1]

SOLUTION

(a) Atomic mass unit (u).

One atomic mass unit is defined as exactly **one-twelfth of the mass of a neutral carbon-12 (${}_{6}^{12}\mathrm{C}$) atom in its ground state**.

$$1\ u = 1.6605 \times 10^{-27}\ \mathrm{kg} \equiv 931.5\ \mathrm{MeV/c^2}$$

(b) Energy required to separate a deuteron into a proton and a neutron.

The energy needed to break a deuteron into its free constituents equals its binding energy, which comes from the mass defect Δm (the difference between the total mass of the separate nucleons and the mass of the bound deuteron).

$$\Delta m = (m_p + m_n) - m(D)$$

$$\Delta m = (1.007825 + 1.008665) - 2.014102\ u$$

$$\Delta m = 2.016490 - 2.014102\ u$$

$$\Delta m = 0.002388\ u$$

Convert this mass defect into energy using $\mathbf{1\ u = 931.5\ MeV/c^2}$.

$$\text{BE} = \Delta m \times 931.5\ \text{MeV}$$

$$\text{BE} = 0.002388 \times 931.5\ \text{MeV}$$

$$\text{BE} = 2.224\ \text{MeV}$$

So about 2.22 MeV of energy must be supplied to separate the deuteron into a free proton and a free neutron.

Final answer. Energy required = **BE of deuteron ≈ 2.22 MeV**.

← *Back to Q98*

Q99

(a) Differentiate between nuclear fission and fusion.
(b) The fission properties of $_{94}^{239}\text{Pu}$ *are very similar to those of* $_{92}^{235}\text{U}$. *How much energy (in MeV), is released if all the atoms in* 1 *g of pure* $_{94}^{239}\text{Pu}$ *undergo fission? The average energy released per fission is* 180 MeV.

[2024 • Set 55-4-1]

SOLUTION

(a) Difference between nuclear fission and nuclear fusion.

In **nuclear fission**, a heavy nucleus splits into two lighter nuclei of nearly equal mass, releasing a few neutrons and a large amount of energy (about 200 MeV per fission). It is usually started by the capture of a slow neutron and can sustain a chain reaction, as in a nuclear reactor or atomic bomb.

In **nuclear fusion**, two light nuclei combine to form a single heavier nucleus, releasing energy (about 17 MeV per deuterium-tritium event). It requires extremely high temperatures, of the order of $10^7\ K$, to overcome the Coulomb repulsion between the nuclei, as in stars and the hydrogen bomb.

(b) Energy released when all atoms in 1 g of ${}_{94}^{239}\text{Pu}$ undergo fission.

First find the number of atoms in 1 g. The molar (atomic) mass of ${}_{94}^{239}\text{Pu}$ is 239 g, which contains Avogadro's number of atoms.

$$N = \frac{\text{mass}}{\text{atomic mass}} \times \text{N}_A$$

$$N = \frac{1}{239} \times 6.022 \times 10^{23}$$

$$N = 2.52 \times 10^{21} \text{ atoms}$$

Each fission releases 180 MeV, so the total energy is

$$E = N \times 180 \text{ MeV}$$

$$E = 2.52 \times 10^{21} \times 180 \text{ MeV}$$

$$E = 4.54 \times 10^{23} \text{ MeV}$$

Converting to joules using $\mathbf{1\ MeV = 1.6 \times 10^{-13}\ J}$,

$\boldsymbol{E = 4.54 \times 10^{23} \times 1.6 \times 10^{-13}\ J \approx 7.3 \times 10^{10}\ J}$.

So the energy released is about 4.54×10^{23} MeV.

Final answer. $\boldsymbol{E = 2.52\times 10^{21} \times 180 \approx 4.54 \times 10^{23}}$ **MeV** $\boldsymbol{(\approx 7.3\times 10^{10}\ J)}$.

← *Back to Q99*

Q100

Draw a diagram to show the variation of binding energy per nucleon with mass number for different nuclei and mention its two features. Why do lighter nuclei usually undergo nuclear fusion?

[2023 • Set 55-1-1]

SOLUTION

Description of the binding-energy-per-nucleon graph.

Plot the mass number A on the horizontal axis and the binding energy per nucleon (BE/A) in MeV on the vertical axis. The curve rises steeply for light nuclei (with sharp peaks at ^{4}He, ^{12}C and ^{16}O), reaches a broad maximum of about 8.8 MeV near $\boldsymbol{A = 56}$ (iron), and then slowly decreases for heavier nuclei, falling to about 7.5 MeV near $\boldsymbol{A = 240}$. The curve is fairly flat (around 8 MeV) over the middle range of mass numbers.

Two features of the curve.

First, the binding energy per nucleon is maximum (about 8.8 MeV) near $\boldsymbol{A = 56}$, so iron-group nuclei are the most tightly bound and the most stable. This is the endpoint of energy-releasing fusion in stars.

Second, BE/A is nearly constant (about 8 MeV) over the wide range $\mathbf{30 < A < 170}$. This near-constancy is a consequence of the saturation property of the nuclear force, since each nucleon interacts only with its immediate neighbours and not with all the other nucleons.

Why lighter nuclei usually undergo nuclear fusion.

Light nuclei have low mass numbers and lie on the steeply rising left part of the curve, where BE/A is small. When two light nuclei fuse, they form a heavier nucleus that lies higher on the curve, with a larger binding energy per nucleon. The increase in total binding energy means the product is more tightly bound than the reactants, and this extra binding energy is released as kinetic energy of the products and as gamma photons. Hence fusion of light nuclei is energetically favourable and releases energy.

Final answer. BE/A peaks (≈ **8.8** MeV) near $\boldsymbol{A}$**=56** and is roughly constant for **30; light nuclei fuse because the heavier product has higher BE/A, releasing energy.**

← Back to Q100

Q101

(i) Distinguish between nuclear fission and fusion giving an example of each.
(ii) Explain the release of energy in nuclear fission and fusion on the basis of binding energy per nucleon curve.

[2023 • Set 55-2-1]

SOLUTION

Nuclear fission and nuclear fusion are two distinct nuclear processes, and they differ as follows.

In fission, a heavy nucleus splits into two lighter nuclei of comparable mass, usually after absorbing a slow neutron, along with the release of a few neutrons and energy. A typical example is $^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} + \textbf{energy}$ (about 200 MeV).

In fusion, two very light nuclei combine to form a heavier nucleus, with the release of energy. This requires extremely high temperature so that the nuclei have enough kinetic energy to overcome their mutual Coulomb repulsion. A typical example is $^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$.

(ii) The release of energy is explained using the binding energy per nucleon (BE/A) curve, which rises to a maximum of about 8.8 MeV near $A \approx 56$ (iron) and then falls gradually for heavier nuclei.

In fission, a heavy nucleus lying to the right of the peak (low BE/A, around 7.6 MeV) breaks into medium mass fragments which lie closer to the peak and so have a higher BE/A (around 8.4 MeV). The products are more tightly bound, so the gain in total binding energy is released as energy.

In fusion, light nuclei lying to the left of the peak have low BE/A. When they combine, the heavier product nucleus formed has a higher BE/A, so again the products are more tightly bound and the extra binding energy is released.

In **both processes the system moves towards the iron peak** of greater stability, and the increase in binding energy appears as the energy released.

Final answer. Fission splits a heavy nucleus and fusion joins light nuclei; in both, the products have higher binding energy per nucleon, and this gain is released as energy.

← Back to Q101

Q102

(i) How is the size of a nucleus found experimentally? Write the relation between the radius and mass number of a nucleus.
(ii) Prove that the density of a nucleus is independent of its mass number.

[2023 • Set 55-2-1]

SOLUTION

(i) The size of a nucleus is found experimentally by **scattering high energy particles off the nucleus**, most directly by fast electron scattering (and also by alpha particle scattering, as in the Rutherford and Geiger-Marsden experiments). Since electrons are not affected by the strong nuclear force, the distance of closest approach and the diffraction pattern of the scattered electrons reveal the charge distribution and hence the radius of the nucleus.

These experiments give the relation between the nuclear radius R and the mass number A as

$$R = R_0 A^{1/3}$$

where $\mathbf{R_0 \approx 1.2 \times 10^{-15}\ m = 1.2\ fm}$ is a constant.

(ii) To prove that the nuclear density is independent of mass number, consider a nucleus of mass number A, treated as a sphere of radius R.

Its volume is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi (R_0 A^{1/3})^3 = \frac{4}{3}\pi R_0{}^3 A$$

Its mass is approximately

$$M = A\, m_n$$

where m_n is the average mass of a nucleon. Therefore the density is

$$\rho = \frac{M}{V} = \frac{A\, m_n}{\frac{4}{3}\pi R_0{}^3 A}$$

$$\rho = \frac{3\,m_n}{4\pi\,R_0^{\,3}}$$

The mass number A cancels completely. Since m_n and R_0 are constants, ρ has the same value (about 2.3×10^{17} kg m^{-3}) for every nucleus. **Hence the nuclear density is independent of the mass number A.**

Final answer. Experimentally $R = R_0\,A^{1/3}$ with $R_0 \approx 1.2$ fm, and since $\rho = \dfrac{3\,m_n}{4\pi\,R_0^{\,3}}$ is free of A, the nuclear density is the same for all nuclei.

← *Back to Q102*

Q103

(a) (i) Prove that the nuclear density is same for all nuclei.
(ii) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Draw two inferences from this plot.
OR
(b) (i) Draw a graph to show the variation of the number of scattered particles detected (N) in Geiger-Marsden experiment as a function of scattering angle (θ).
(ii) Discuss briefly two conclusions that can be drawn from this graph and how they lead to the discovery of nucleus in an atom.

[2023 • Set 55-3-1]

SOLUTION

(a)(i) Consider a nucleus of mass number A as a sphere of radius R, where experimentally $R = R_0\,A^{1/3}$ with $R_0 \approx 1.2$ fm.

Its volume is

$$V = \frac{4}{3}\pi\,R^3 = \frac{4}{3}\pi\,R_0^{\,3}\,A$$

Its mass is

$$M = A\,m_n$$

where m_n is the mass of one nucleon. Hence the density is

$$\rho = \frac{M}{V} = \frac{A\, m_n}{\frac{4}{3}\pi R_0^3 A} = \frac{3\, m_n}{4\pi R_0^3}$$

As A cancels, ρ is a constant (about 2.3×10^{17} kg m^{-3}), the **same for all nuclei**.

(a)(ii) Potential energy versus separation plot for a pair of nucleons: Take the separation r along the horizontal axis and the potential energy $U(r)$ along the vertical axis. For very small r (below about 0.8 fm) the curve lies high above the axis and rises steeply, showing a strong positive (repulsive) potential, the hard core. As r increases, the curve dips down through zero and reaches a minimum negative value of about −100 MeV at $\boldsymbol{r = r_0 \approx 0.8}$ **fm**. Beyond r_0 the curve rises again towards zero and becomes negligible for r greater than about 2 to 3 fm. Mark the equilibrium separation $\mathbf{r_0 \approx 0.8}$ **fm** at the minimum.

Two inferences from this plot:

1. The nuclear force is **short range**; it is appreciable only up to a few femtometres and is negligible beyond about 2 to 3 fm.

2. The force is **strongly attractive** near r_0 (the minimum), but becomes **repulsive** for separations smaller than r_0 (the steeply rising core), which prevents the nucleus from collapsing.

Final answer. Since $\boldsymbol{\rho = \frac{3\, m_n}{4\pi R_0^3}}$ is independent of $\boldsymbol{A}$, nuclear density is the same for all nuclei; the $\boldsymbol{U(r)}$ plot shows the nuclear force is short range with a repulsive core below $\mathbf{r_0 \approx 0.8}$ fm and attraction beyond it.

← *Back to Q103*

Q104

Draw a plot showing the variation of potential energy of two nucleons as a function of distance between them. Identify the regions in which the force between the nucleons is (i) attractive, and (ii) repulsive. Justify your answers.

[2023 • Set 55-4-3]

SOLUTION

Potential energy versus distance plot for two nucleons: Take the separation r between the two nucleons along the horizontal axis and the potential energy $U(r)$ along the vertical axis. For very small separation (below about 0.8 fm) the curve is high above the axis and rises very steeply, showing a strong positive (repulsive) potential, the hard core. As r increases the curve falls steeply, crosses zero, and reaches a minimum negative value of about −100 MeV at the equilibrium separation $\mathbf{r_0 \approx 0.8\ fm}$. Beyond r_0 the curve rises back towards zero and the potential energy is practically zero for r greater than about 2 to 3 fm. Label the minimum point $\mathbf{r_0 \approx 0.8\ fm}$.

The force between the nucleons is found from the slope of this curve, since

$$F = -\frac{dU}{dr}$$

(i) Region where the force is **attractive**: for $r > r_0$ (that is, $r > 0.8$ fm). Here the potential energy increases as r increases, so $\frac{dU}{dr} > 0$, which gives $F = -\frac{dU}{dr} < 0$. A negative force points towards smaller separation, that is, it pulls the nucleons together, so the force is attractive. This is the region in which the strong nuclear force binds the nucleons.

(ii) Region where the force is **repulsive**: for $r < r_0$ (that is, $r < 0.8$ fm). Here the potential energy decreases sharply as r increases, so $\frac{dU}{dr} < 0$, which gives $F = -\frac{dU}{dr} > 0$. A positive force pushes the nucleons apart, so the force is repulsive. This hard repulsive core prevents the nucleons from being squeezed into one another and keeps the nucleus from collapsing.

TIP

When the question links a force to a PE graph, always quote $F = -\frac{dU}{dr}$ and read the sign of the slope; that single relation earns the justification marks.

Final answer. From $F = -\frac{dU}{dr}$, the nuclear force is attractive for $r > r_0 \approx 0.8$ fm and strongly repulsive for $r < r_0$.

← *Back to Q104*

Q105

(a) Calculate the binding energy of an alpha particle in MeV. Given
mass of a proton = **1.007825 *u***
mass of a neutron = **1.008665 *u***
mass of He nucleus = **4.002800 *u***
1u = 931 MeV/c^2
OR
(b) A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits into two nuclei Q and R of mass number 110 and 130 and binding energy per nucleon 8.5 MeV and 8.4 MeV respectively. Calculate the energy released in the fission.

[2023 • Set 55-5-3]

SOLUTION

(a) An alpha particle is a helium nucleus $^4_2$He, made of 2 protons and 2 neutrons. The binding energy equals the mass defect (in energy units) when the nucleus is built from its separate nucleons.

First find the total mass of the free constituents:

$$2m_p + 2m_n = 2(1.007825) + 2(1.008665)$$

$$2m_p + 2m_n = 2.015650 + 2.017330 = 4.032980\ u$$

The mass defect is the difference between this and the actual nuclear mass:

$$\Delta m = 4.032980 - 4.002800 = 0.030180\ u$$

Converting to energy using **1 *u* = 931 MeV/c^2**:

$$\text{BE} = \Delta m \times 931 = 0.030180 \times 931$$

$$\text{BE} \approx 28.1\ \text{MeV}$$

So the alpha particle is very tightly bound, with **binding energy about 28.1 MeV**.

(b) The heavy nucleus P (***A* = 240**) splits into Q (***A* = 110**) and R (***A* = 130**). The energy released equals the increase in total binding energy.

Binding energy of P:

$$\mathrm{BE}_P = 240 \times 7.6 = 1824 \text{ MeV}$$

Binding energy of Q:

$$\mathrm{BE}_Q = 110 \times 8.5 = 935 \text{ MeV}$$

Binding energy of R:

$$\mathrm{BE}_R = 130 \times 8.4 = 1092 \text{ MeV}$$

Total binding energy of the products:

$$\mathrm{BE}_Q + \mathrm{BE}_R = 935 + 1092 = 2027 \text{ MeV}$$

Energy released in fission:

$$Q = (\mathrm{BE}_Q + \mathrm{BE}_R) - \mathrm{BE}_P = 2027 - 1824$$

$$Q = 203 \text{ MeV}$$

So the **energy released in the fission is 203 MeV**.

Final answer. (a) Binding energy of the alpha particle is about **28.1** MeV; (b) the fission releases **203** MeV.

← *Back to Q105*

Q106

Calculate the energy released in MeV in the following reaction:

$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + n$

Given: **$m(^{2}_{1}\mathrm{H}) = 2.014102\ u$**

$m(^{3}_{1}\mathrm{H}) = 3.016049\ u$

$m(^{4}_{2}\mathrm{He}) = 4.002603\ u$

$\mathrm{m}_n = 1.008665\ u$

[2022 • Set 55-1-1]

SOLUTION

This is the deuterium-tritium fusion reaction $^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$. The energy released ($Q$ value) equals the mass defect of the reaction converted into

energy.

Total mass of the reactants (left hand side):

$$M_i = m(^2_1\text{H}) + m(^3_1\text{H}) = 2.014102 + 3.016049$$

$$M_i = 5.030151\ u$$

Total mass of the products (right hand side):

$$M_f = m(^4_2\text{He}) + m_n = 4.002603 + 1.008665$$

$$M_f = 5.011268\ u$$

Mass defect:

$$\Delta m = M_i - M_f = 5.030151 - 5.011268$$

$$\Delta m = 0.018883\ u$$

Energy released, using $\mathbf{1\ u = 931.5\ MeV/c^2}$:

$$Q = \Delta m \times 931.5 = 0.018883 \times 931.5$$

$$Q \approx 17.6\ \text{MeV}$$

Since $\Delta m > 0$, the reaction is exothermic. The **energy released is about 17.6 MeV**.

Final answer. The energy released in the reaction is about **17.6** MeV.

← *Back to Q106*

Q107

(i) Depict the variation of the potential energy of a pair of nucleons with the separation between them.
(ii) Imagine the fission of a $^{56}_{26}\text{Fe}$ *into two equal fragments of* $^{28}_{13}\text{Al}$ *nucleus. Is the fission energetically possible? Justify your answer by working out Q value of the*

process.

Given: $m(^{56}_{26}\text{Fe}) = 55.93494\ u$, $m(^{28}_{13}\text{Al}) = 27.98191\ u$.

[2022 • Set 55-2-1]

SOLUTION

(i) Potential energy versus separation plot for a pair of nucleons: Take the separation r along the horizontal axis and the potential energy $U(r)$ along the vertical axis. For separation below about 0.8 fm the curve is high above the axis and rises steeply, a strong positive (repulsive) potential, the hard core. As r increases the curve dips down, crosses zero, and reaches a minimum of about −100 MeV at the equilibrium separation $\mathbf{r_0 \approx 0.8\ fm}$. Beyond r_0 the curve rises back towards zero and becomes negligible for r greater than about 2 to 3 fm. This shows the nuclear force is short range, attractive near r_0 and repulsive for $\boldsymbol{r < r_0}$.

(ii) Imagine $^{56}_{26}\text{Fe}$ splitting into two equal $^{28}_{13}\text{Al}$ fragments. The reaction is feasible only if its Q value is positive.

Mass before (initial):

$$M_i = m(^{56}_{26}\text{Fe}) = 55.93494\ u$$

Mass after (final):

$$M_f = 2\ m(^{28}_{13}\text{Al}) = 2 \times 27.98191 = 55.96382\ u$$

Mass defect:

$$\Delta m = M_i - M_f = 55.93494 - 55.96382$$

$$\Delta m = -0.02888\ u$$

Q value, using $\mathbf{1\ u = 931\ MeV/c^2}$:

$$Q = \Delta m \times 931 = -0.02888 \times 931$$

$$Q \approx -26.9\ \text{MeV}$$

Since Q is **negative, the fission is not energetically possible**; energy would have to be supplied rather than released. This is expected because ^{56}Fe lies at

the peak of the binding energy per nucleon curve and is one of the most stable nuclei, so it cannot spontaneously break into lighter fragments.

CONCEPT

A negative Q value means the reaction absorbs energy, so it cannot happen spontaneously; always state this conclusion explicitly for the feasibility mark.

Final answer. $Q \approx -26.9$ MeV; being negative, the fission of ^{56}Fe into two ^{28}Al nuclei is not energetically possible.

← Back to Q107

Q108

(a) James Chadwick, in 1932, studied the emission of neutral radiations when Beryllium nuclei were bombarded with alpha particles. He concluded that emitted radiations were neutrons and not photons. Explain.
(b) Two nuclei may have the same radius, even though they contain different number of protons and neutrons. Explain.

[2022 • Set 55-3-3]

SOLUTION

(a) When alpha particles strike beryllium, a highly penetrating, electrically neutral radiation is emitted in the reaction

$$^{9}_{4}\mathrm{Be} + {}^{4}_{2}\mathrm{He} \rightarrow {}^{12}_{6}\mathrm{C} + {}^{1}_{0}\mathrm{n}$$

Chadwick let this radiation strike hydrogen (paraffin) and nitrogen targets and measured the recoil energies of the knocked-out protons and nitrogen nuclei.

If the radiation were **gamma photons**, then applying conservation of energy and momentum to the elastic recoil of protons and of the much heavier nitrogen nuclei would require photon energies of the order of 50 MeV. No alpha source available at that time could produce photons of such enormous energy, so the gamma photon explanation was physically unreasonable.

If instead the radiation consisted of **neutral particles of mass nearly equal to that of a proton**, then the same momentum and energy conservation gave

recoil energies that agreed consistently for both the hydrogen and the nitrogen targets. So Chadwick concluded that the radiation was a new neutral particle of mass about equal to the proton mass, which he named the **neutron**.

(b) The radius of a nucleus depends only on its mass number A through

$$R = R_0 A^{1/3}$$

It does not depend separately on the number of protons Z or the number of neutrons N. Therefore two nuclei that have the **same mass number A but different Z and N** (called isobars) will have the same radius, even though their proton and neutron contents differ.

For example, ${}^{40}_{18}\text{Ar}$ and ${}^{40}_{20}\text{Ca}$ both have $A = 40$, so both have the same nuclear radius although their proton and neutron numbers are different.

Final answer. Chadwick showed the radiation was neutrons because a photon model demanded an impossible ~ **50** MeV while a neutral proton-mass particle fit the recoil data; and two nuclei share a radius when they have the same A (since $R = R_0 A^{1/3}$) despite different Z and N.

← *Back to Q108*

Q109

(a) Differentiate between nuclear fission and nuclear fusion.
(b) Deuterium undergoes fusion as per the reaction: ${}_1^2\text{H} + {}_1^2\text{H} \rightarrow {}_2^3\text{He} + {}_0^1\text{n} + 3.27$ **MeV**. *Find the duration for which an electric bulb of 500 W can be kept glowing by the fusion of 100 g of deuterium.*

[2022 • Set 55-4-1]

SOLUTION

(a) Nuclear fission and nuclear fusion differ as follows. In fission a heavy nucleus splits into two lighter nuclei of comparable mass (with a few neutrons released), for example ${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\,{}^1_0\text{n}$. In fusion two light nuclei combine to form a heavier nucleus, for example ${}^2_1\text{H} + {}^3_1\text{H} \rightarrow {}^4_2\text{He} + {}^1_0\text{n}$. Fission is triggered by a slow neutron and occurs at ordinary temperature, whereas fusion needs very high temperature to overcome the Coulomb

repulsion of the nuclei. Per nucleon, fusion of light nuclei releases more energy than fission.

(b) The fusion reaction is $^{2}_{1}\mathbf{H} + {}^{2}_{1}\mathbf{H} \rightarrow {}^{3}_{2}\mathbf{He} + {}^{1}_{0}\mathbf{n} + \mathbf{3.27\ MeV}$, in which two deuterium nuclei are consumed per reaction.

Number of deuterium atoms in 100 g (atomic mass ≈ **2** g/mol):

$$N = \frac{100}{2} \times 6.0 \times 10^{23} = 50 \times 6.0 \times 10^{23}$$

$$N = 3.0 \times 10^{25} \text{ atoms}$$

Each reaction uses 2 deuterium nuclei, so the number of reactions is

$$n = \frac{N}{2} = \frac{3.0 \times 10^{25}}{2} = 1.5 \times 10^{25}$$

Energy released per reaction in joules:

$$3.27 \text{ MeV} = 3.27 \times 1.6 \times 10^{-13}\, J = 5.232 \times 10^{-13}\, J$$

Total energy released:

$$E = n \times 5.232 \times 10^{-13} = 1.5 \times 10^{25} \times 5.232 \times 10^{-13}$$

$$E \approx 7.85 \times 10^{12}\, J$$

The bulb consumes power $\boldsymbol{P = 500\ W}$, so the time for which it can glow is

$$t = \frac{E}{P} = \frac{7.85 \times 10^{12}}{500}$$

$$t \approx 1.57 \times 10^{10}\, s$$

Converting to years ($\mathbf{1\ year = 3.15 \times 10^{7}\ s}$):

$$t = \frac{1.57 \times 10^{10}}{3.15 \times 10^{7}} \approx 498 \text{ years}$$

So the bulb can be kept glowing for about $1.57 \times 10^{10}\, s$, that is, **nearly 498 years**.

Final answer. The **500** W bulb can glow for about $\mathbf{1.57 \times 10^{10}}$ s, which is roughly **498** years.

← Back to Q109

Q110

In a fission event of $^{238}_{92}\text{U}$ *by fast moving neutrons, no neutrons are emitted and final products, after the beta decay of the primary fragments, are* $^{140}_{58}\text{Ce}$ *and* $^{99}_{44}\text{Ru}$. *Calculate Q for this process. Neglect the masses of electrons/positrons emitted during the intermediate steps.*

Given: $\boldsymbol{m(^{238}_{92}\text{U}) = 238.05079\ u;\ m(^{140}_{58}\text{Ce}) = 139.90543\ u;\ m(^{99}_{44}\text{Ru}) = 98.90594\ u;\ m(^{1}_{0}\text{n}) = 1.008665\ u.}$

[2022 • Set 55-5-1]

SOLUTION

In this fission of $^{238}_{92}\text{U}$ by a fast neutron, no neutrons are emitted, and after the beta decays of the primary fragments the final products are $^{140}_{58}\text{Ce}$ and $^{99}_{44}\text{Ru}$ together with electrons:

$$^{238}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{140}_{58}\text{Ce} + {}^{99}_{44}\text{Ru} + x\,\text{e}^{-}$$

Mass number balance: **238 + 1 = 140 + 99 = 239**, which checks. Charge balance: the protons go from 92 to **58 + 44 = 102**, so **102 − 92 = 10** beta particles (e^{-}) are emitted, that is $\boldsymbol{x = 10}$. As stated, the electron masses are neglected.

Mass before the reaction (initial):

$$\text{M}_i = m(^{238}_{92}\text{U}) + \text{m}_n = 238.05079 + 1.008665$$

$$\text{M}_i = 239.059455\ u$$

Mass after the reaction (final):

$$\text{M}_f = m(^{140}_{58}\text{Ce}) + m(^{99}_{44}\text{Ru}) = 139.90543 + 98.90594$$

$$\text{M}_f = 238.811370\ u$$

Mass defect:

$$\Delta m = \text{M}_i - \text{M}_f = 239.059455 - 238.811370$$

$$\Delta m = 0.248085\ u$$

Energy released (Q value), using $\mathbf{1\ u = 931.5\ MeV/c^2}$:

$$Q = \Delta m \times 931.5 = 0.248085 \times 931.5$$

$$Q \approx 231.1\ \text{MeV}$$

So the **Q value of this fission process is about 231.1 MeV**, and being positive it confirms the process is energetically favourable.

Final answer. The Q value of the fission process is about **231.1** MeV.

← Back to Q110

Q111

The nucleus $^{235}_{92}\text{U}$ *initially at rest, decays into* $^{231}_{90}\text{X}$ *by emitting an* α*-particle*
$^{235}_{92}\text{U} \longrightarrow {}^{231}_{90}\text{X} + {}^{4}_{2}\text{He} + \text{energy}$.
The binding energies per nucleon of the parent nucleus, the daughter nucleus and α*-particle are* 7·8 MeV, 7·835 MeV *and* 7·07 MeV, *respectively. Assuming the daughter nucleus to be formed in the unexcited state and neglecting its share in the energy of the reaction, find the speed of the emitted* α*-particle. (Mass of* α*-particle* **= 6·68 × 10^{-27} kg**)

[2020 • Set 55-1-1]

SOLUTION

In an α-decay the energy released (the Q-value) equals the gain in total binding energy of the products over the parent nucleus, because a more tightly bound system has lower mass.

$$Q = (\text{BE}_X + \text{BE}_\alpha) - \text{BE}_U$$

The binding energy of each nucleus is (binding energy per nucleon) × (number of nucleons).

$$\text{BE}_X = 231 \times 7.835 = 1809.885\ \text{MeV}$$

$$\text{BE}_\alpha = 4 \times 7.07 = 28.28\ \text{MeV}$$

$$\mathrm{BE}_U = 235 \times 7.8 = 1833 \text{ MeV}$$

$$Q = (1809.885 + 28.28) - 1833 = 5.165 \approx 5.17 \text{ MeV}$$

We are told to neglect the recoil energy of the heavy daughter nucleus X. Hence the entire released energy appears as the kinetic energy of the emitted α-particle.

$$\mathrm{E}_\alpha = 5.17 \text{ MeV} = 5.17 \times 1.6 \times 10^{-13}\, J = 8.27 \times 10^{-13}\, J$$

The kinetic energy gives the speed through $\mathbf{E_\alpha = \frac{1}{2} m v^2}$.

$$v = \sqrt{\frac{2\mathrm{E}_\alpha}{m}}$$

$$v = \sqrt{\frac{2 \times 8.27 \times 10^{-13}}{6.68 \times 10^{-27}}}$$

$$v = \sqrt{2.476 \times 10^{14}} = 1.57 \times 10^{7}\ m/s$$

Hence the emitted α-particle moves with a speed of about $\mathbf{1.57 \times 10^7\ m/s}$.

TIP

The Q-value of a decay can be found either from binding energies (products minus parent) or from masses (parent minus products). Both routes give the same energy; pick whichever the data fits.

Final answer. The speed of the emitted α-particle is about $\mathbf{1.57 \times 10^7\ m/s}$.

← *Back to Q111*

Q112

(a) Differentiate between half-life and average life of a radioactive substance.
(b) A radioactive substance decays for an interval of time equal to its mean life. Find the fraction of the amount of the substance which is left undecayed after this time interval.

[2020 • Set 55-2-1]

SOLUTION

(a) Both quantities measure how fast a radioactive sample decays, but they are defined differently.

The half-life $T_{1/2}$ is the time in which half of the radioactive nuclei present in a sample disintegrate. It is related to the decay constant by $\mathbf{T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}}$.

The average (mean) life τ is the average of the lifetimes of all the nuclei of the sample before they decay. It is given by $\boldsymbol{\tau = \frac{1}{\lambda}}$.

The two are connected by $\mathbf{T_{1/2} = 0.693\ \tau}$, so the mean life is always longer than the half-life.

(b) The number of undecayed nuclei after time t follows the radioactive decay law.

$$N = N_0\, e^{-\lambda t}$$

Here the time interval equals the mean life, so $\boldsymbol{t = \tau = \frac{1}{\lambda}}$, which gives $\boldsymbol{\lambda t = 1}$.

$$\frac{N}{N_0} = e^{-\lambda \tau} = e^{-1}$$

$$\frac{N}{N_0} = \frac{1}{e} = 0.368$$

Hence the fraction left undecayed after one mean life is $\boldsymbol{\frac{1}{e} \approx 0.368}$, i.e. about 36.8% of the original amount remains.

Final answer. The fraction left undecayed after one mean life is $\boldsymbol{\frac{1}{e} \approx 0.368}$ (about **36.8%**).

← *Back to Q112*

Q113

(a) The density of the nuclear matter is tremendously larger than the physical density of the material. Explain.

(b) The nuclear forces are not coulomb forces between nucleons. Explain.
(c) Draw a plot of the potential energy between a pair of nucleons as a function of distance between them inside a nucleus.

[2020 • Set 55-3-1]

SOLUTION

(a) The mass of an atom is almost entirely concentrated in its tiny nucleus, while the atom as a whole is mostly empty space. The nuclear radius is about 10^{-15} *m* while the atomic radius is about 10^{-10} *m*, so the ratio of radii is roughly 10^{-5} and the ratio of volumes is about 10^{-15}.

Since nearly all the mass sits in a volume that is around 10^{15} times smaller than the atomic volume, the nuclear density comes out to about 2.3×10^{17} kg/m^3. In bulk material this same mass is spread over the much larger atomic volume, so the physical (bulk) density is only about 10^3 kg/m^3. This is why nuclear density is roughly 10^{14} times the ordinary physical density.

(b) Nuclear forces differ from Coulomb forces in several ways. Coulomb forces have an infinite range and vary as $\frac{1}{r^2}$, while nuclear forces are short-ranged and act only within about 3 fm. Coulomb forces act only between charged particles, whereas nuclear forces are charge-independent, acting equally between proton-proton, neutron-neutron and proton-neutron pairs. The neutron, being neutral, experiences no Coulomb force yet is strongly bound inside the nucleus, which itself proves that the binding force is not Coulombic. Nuclear forces are also about 100 times stronger than the electrostatic force at a separation of 1 fm.

(c) The graph is a plot of potential energy *U* (vertical axis, in MeV) against the separation *r* (horizontal axis, in fm) between two nucleons. Starting from very small *r* the curve lies high above the axis at large positive *U* (a steep repulsive core for $r < r_0$, where $r_0 \approx 0.8$ fm). As *r* increases the curve falls, crosses the *r*-axis at $r = r_0$, and dips to a minimum of about −100 MeV near $r \approx 1$ fm. Beyond this minimum the curve rises back toward zero and flattens onto the *r*-axis for *r* greater than a few fm. The potential is positive (repulsive) for $r < r_0$ and negative (attractive) for $r > r_0$.

Final answer. Nuclear matter is extremely dense because all the atomic mass is packed into the tiny nucleus; nuclear forces are short-range, charge-

independent and far stronger than Coulomb forces, with a repulsive core at small r and an attractive region beyond $\mathbf{r_0}$.

← *Back to Q113*

Q114

(a) Define the term decay constant of a radioactive substance.
(b) The half life of $^{238}_{92}\mathrm{U}$ *undergoing* α *decay is* 4.5×10^9 *years. Calculate the activity of* 10 *g sample of* $^{238}_{92}\mathrm{U}$.

[2020 • Set 55-5-1]

SOLUTION

(a) The decay constant λ of a radioactive substance is the probability of decay per nucleus per unit time. Equivalently, it is the fraction of the total number of nuclei present that disintegrate per second. Its SI unit is s^{-1}.

(b) Activity is the rate of disintegration, given by $\boldsymbol{R = \lambda N}$, where N is the number of radioactive nuclei present. First find λ from the half-life.

$$\lambda = \frac{0.693}{T_{1/2}}$$

Converting the half-life into seconds using $\mathbf{1\ year = 3.16 \times 10^7\ s}$:

$$T_{1/2} = 4.5 \times 10^9 \times 3.16 \times 10^7 = 1.42 \times 10^{17}\ s$$

$$\lambda = \frac{0.693}{1.42 \times 10^{17}} = 4.88 \times 10^{-18}\ s^{-1}$$

Next find the number of nuclei in 10 *g* of $^{238}_{92}\mathrm{U}$ using Avogadro's number (238 *g* contains 6.022×10^{23} atoms).

$$N = \frac{10}{238} \times 6.022 \times 10^{23}$$

$$N = 2.53 \times 10^{22}\ \text{atoms}$$

Now the activity:

$$R = \lambda N = (4.88 \times 10^{-18}) \times (2.53 \times 10^{22})$$

$$R = 1.23 \times 10^5 \text{ disintegrations/}s$$

Hence the activity of the 10 g sample is $\boldsymbol{R \approx 1.23 \times 10^5}$ **Bq**.

Final answer. The activity of the **10 g** sample is about $\mathbf{1.23 \times 10^5}$ **Bq**.

← *Back to Q114*

Q115

Draw the curve showing the variation of binding energy per nucleon with the mass number of nuclei. Using it explain the fusion of nuclei lying on ascending part and fission of nuclei lying on descending part of this curve.

[2020 • Set 55-5-2]

SOLUTION

The required graph plots the binding energy per nucleon $\frac{\text{BE}}{A}$ (vertical axis, in MeV) against the mass number A (horizontal axis). The curve rises steeply for light nuclei, with peaks at $\boldsymbol{A = 4}$ (helium), passes through a broad maximum of about 8.8 MeV near $\boldsymbol{A = 56}$ (iron), stays almost flat over the range $\boldsymbol{A = 30}$ to $\boldsymbol{A = 120}$, and then slowly decreases for heavier nuclei, reaching about 7.6 MeV near $\boldsymbol{A = 238}$ (uranium). Higher binding energy per nucleon means a more stable, more tightly bound nucleus.

Fusion (ascending part, light nuclei): When two very light nuclei lying on the steeply rising part of the curve combine, the heavier product nucleus has a higher binding energy per nucleon than the reactants. The total binding energy therefore increases, and this gain appears as energy released. So fusion of light nuclei releases energy and moves the system toward greater stability.

Fission (descending part, heavy nuclei): When a heavy nucleus lying on the gently falling part of the curve splits into two medium-mass nuclei, the products lie nearer the peak and so have a higher binding energy per nucleon than the parent. Again the total binding energy increases, and the difference is released, about 200 MeV per fission. So fission of a heavy nucleus releases energy and yields more stable fragments.

In both processes energy is released because the products are more tightly bound (higher $\frac{\text{BE}}{A}$) than the starting nuclei.

CONCEPT

Energy release in any nuclear reaction always comes from an increase in total binding energy of the products, i.e. the products climb the binding-energy-per-nucleon curve.

Final answer. The binding energy per nucleon peaks near $A = 56$; fusion of light nuclei (ascending part) and fission of heavy nuclei (descending part) both move toward higher binding energy per nucleon and so release energy.

← *Back to Q115*

Q116

Define the term 'decay constant' of a radioactive sample. The rate of disintegration of a given radioactive nucleus is 10000 *disintegrations/s and* 5000 *disintegrations/s after* 20 *hr and* 30 *hr respectively from start. Calculate the half life and initial number of nuclei at* $t = 0$.

[2019 • Set 55-1-1]

SOLUTION

The decay constant λ of a radioactive sample is the probability of disintegration per nucleus per unit time. Equivalently, it is the fraction of the nuclei present that decay per second. Its SI unit is s^{-1}.

The activity of a sample falls exponentially with time, $R = R_0\, e^{-\lambda t}$, since activity is proportional to the number of nuclei present.

Take the ratio of the activities at the two given times ($t_1 = 20$ hr, $t_2 = 30$ hr, so $\Delta t = 10$ hr):

$$\frac{R_1}{R_2} = e^{\lambda \Delta t} = \frac{10000}{5000} = 2$$

$$\lambda \Delta t = \ln 2 = 0.693$$

$$\lambda = \frac{0.693}{10} = 0.0693\ \text{hr}^{-1}$$

The half-life follows directly:

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{0.0693} = 10\ \text{hr}$$

To find the initial number of nuclei N_0, first find the initial activity R_0 by projecting back to $t = 0$ using R_1 at $t = 20$ **hr**:

$$R_0 = R_1\, e^{\lambda t_1} = 10000 \times e^{0.0693 \times 20} = 10000 \times e^{1.386}$$

$$R_0 = 10000 \times 4 = 4 \times 10^4\ \text{disintegrations/}s$$

Since $\mathbf{R_0 = \lambda N_0}$, we need λ in s^{-1}:

$$\lambda = \frac{0.0693}{3600} = 1.925 \times 10^{-5}\ s^{-1}$$

$$N_0 = \frac{R_0}{\lambda} = \frac{4 \times 10^4}{1.925 \times 10^{-5}}$$

$$N_0 \approx 2.08 \times 10^9$$

Hence the half-life is **10 hr** and the initial number of nuclei is $\mathbf{N_0 \approx 2.08 \times 10^9}$.

Final answer. The half-life is **10 hr** and the initial number of nuclei is about $\mathbf{2.08 \times 10^9}$.

← *Back to Q116*

Q117

Write the relation between half life and average life of a radioactive nucleus.

[2019 • Set 55-1-2]

SOLUTION

Both the half-life and the average (mean) life of a radioactive nucleus are fixed by the same decay constant λ.

The half-life is the time for half the nuclei to decay:

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

The average (mean) life is the average time a nucleus survives before decaying:

$$\tau = \frac{1}{\lambda}$$

Dividing one relation by the other to eliminate λ:

$$\frac{T_{1/2}}{\tau} = \ln 2 = 0.693$$

Hence the required relation is $\mathbf{T_{1/2} = 0.693\ \tau}$, equivalently $\boldsymbol{\tau = 1.44\ T_{1/2}}$. The mean life is about 44% longer than the half-life.

Final answer. The relation is $\mathbf{T_{1/2} = 0.693\ \tau}$, equivalently $\boldsymbol{\tau = 1.44\ T_{1/2}}$.

← *Back to Q117*

Q118

In a given sample two isotopes A and B are initially present in the ratio of 1 : 2. *Their half lives are* 60 *years and* 30 *years respectively. How long will it take so that the sample has these isotopes in the ratio of* 2 : 1*?*

[2019 • Set 55-1-2]

SOLUTION

Let the initial amounts be $\mathbf{N_A = N_0}$ and $\mathbf{N_B = 2N_0}$, so initially $\mathbf{N_A : N_B = 1 : 2}$.

Each isotope decays by the half-life law $N = \mathbf{N_0}(\frac{1}{2})^{t/\mathbf{T}_{1/2}}$. With $\mathbf{T_A = 60}$ years and $\mathbf{T_B = 30}$ years:

$$N_A(t) = N_0(\frac{1}{2})^{t/60}$$

$$N_B(t) = 2N_0(\frac{1}{2})^{t/30}$$

We need the ratio to become $\mathbf{N_A : N_B = 2 : 1}$, i.e. $\mathbf{N_A = 2\ N_B}$:

$$N_0\left(\frac{1}{2}\right)^{t/60} = 2 \times 2N_0\left(\frac{1}{2}\right)^{t/30}$$

$$N_0\left(\frac{1}{2}\right)^{t/60} = 4N_0\left(\frac{1}{2}\right)^{t/30}$$

Cancel N_0 and divide both sides by $\left(\frac{1}{2}\right)^{t/30}$:

$$\left(\frac{1}{2}\right)^{\frac{t}{60} - \frac{t}{30}} = 4$$

$$\frac{t}{60} - \frac{t}{30} = \frac{t}{60} - \frac{2t}{60} = -\frac{t}{60}$$

$$\left(\frac{1}{2}\right)^{-t/60} = 4 = 2^2$$

$$2^{t/60} = 2^2 \Rightarrow \frac{t}{60} = 2$$

$$t = 120 \text{ years}$$

Hence the sample will contain the two isotopes in the ratio 2 : 1 after **120 years**.

Final answer. The isotopes reach the ratio **2 : 1** after **120** years.

← *Back to Q118*

Q119

Show that density of nucleus is independent of its mass number A.

[2019 • Set 55-1-3]

SOLUTION

The radius of a nucleus depends on its mass number through the empirical relation

$$R = R_0 A^{1/3}$$

where $\mathbf{R_0 \approx 1.2\ fm}$ is a constant and A is the mass number.

Treating the nucleus as a sphere, its volume is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0{}^3 A$$

so the volume is directly proportional to A.

The mass of the nucleus is approximately the number of nucleons times the mass of one nucleon (each nucleon has a mass of about one atomic mass unit u):

$$M = A\,u$$

so the mass is also directly proportional to A.

The nuclear density is therefore

$$\rho = \frac{M}{V} = \frac{A\,u}{\frac{4}{3}\pi R_0{}^3 A}$$

$$\rho = \frac{u}{\frac{4}{3}\pi R_0{}^3} = \frac{3u}{4\pi R_0{}^3}$$

The mass number A cancels completely, leaving an expression that contains only constants. Hence the **density of the nucleus is independent of its mass number A**, and comes out to about 2.3×10^{17} kg/m^3 for every nucleus.

CONCEPT

The cancellation of A is the whole point: density depends only on R_0 and u, both universal constants, so all nuclei share the same density.

Final answer. Since $M \propto A$ and $V \propto A$, the ratio $\rho = M/V$ is constant ($\approx 2.3 \times 10^{17}$ **kg/m^3**), independent of A.

← *Back to Q119*

Q120

Why is it difficult to detect the presence of an anti-neutrino during β-decay? Define the term decay constant of a radioactive nucleus and derive the expression for its mean life in terms of the decay constant.

OR

(a) State two distinguishing features of nuclear force.

(b) Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. Mark the regions on the graph where the force is (i) attractive, and (ii) repulsive.

[2019 • Set 55-2-1]

SOLUTION

An anti-neutrino $\bar{\nu}_e$ is hard to detect because it is electrically neutral, has negligible (almost zero) rest mass, and interacts with matter only through the extremely weak nuclear (weak) interaction. Its interaction cross-section is vanishingly small (about 10^{-44} cm^2), so it can pass through enormous thicknesses of matter, even light-years of lead, without being absorbed. This is why its presence in β-decay is so difficult to detect directly.

The decay constant λ of a radioactive nucleus is the probability of decay per nucleus per unit time. The number of undecayed nuclei follows $\boldsymbol{N = N_0\, e^{-\lambda t}}$.

To find the mean life τ, consider the number of nuclei that decay between t and $t + dt$. The decay rate is $\dfrac{dN}{dt} = -\lambda N_0\, e^{-\lambda t}$, so the number decaying in dt is $|dN| = \lambda N_0\, e^{-\lambda t}\, dt$. Each of these had a life t.

The mean life is the total of (life × number) divided by the total number:

$$\tau = \frac{1}{N_0}\int_0^\infty t\,|dN| = \frac{1}{N_0}\int_0^\infty t\,\lambda N_0\, e^{-\lambda t}\,dt$$

$$\tau = \lambda \int_0^\infty t\, e^{-\lambda t}\,dt$$

Using the standard integral $\int_0^\infty t\, e^{-\lambda t}\,dt = \dfrac{1}{\lambda^2}$:

$$\tau = \lambda \times \frac{1}{\lambda^2} = \frac{1}{\lambda}$$

Hence the **mean life** $\boldsymbol{\tau = \dfrac{1}{\lambda}}$.

OR (a) Two distinguishing features of nuclear force: (i) It is a very short-range force, effective only within about 3 fm, and is the strongest known force at that range. (ii) It is charge-independent (acts equally between proton-proton, neutron-neutron and proton-neutron pairs) and shows saturation, being

strongly attractive at normal separation but having a repulsive core at very small separation.

OR (b) The graph plots the potential energy U (vertical axis, in MeV) of a pair of nucleons against their separation r (horizontal axis, in fm). For $r < r_0$ (with $r_0 \approx 0.8$ fm) the curve is high and positive, rising steeply; here the force is repulsive. The curve crosses the r-axis at $r = r_0$ and dips to a minimum of about −100 MeV near $r \approx 1$ fm, then rises back toward zero for larger r; in the region $r > r_0$ the force is attractive. Mark (i) attractive for $r > r_0$ and (ii) repulsive for $r < r_0$.

Final answer. Anti-neutrinos are nearly massless, neutral and interact only weakly, so they are very hard to detect; the mean life is $\tau = \frac{1}{\lambda}$.

← *Back to Q120*

Q121

Draw a graph showing the variation of binding energy per nucleon with mass number of different nuclei. Write any two salient features of the curve. How does this curve explain the release of energy both in the processes of nuclear fission and fusion?

[2019 • Set 55-4-1]

SOLUTION

The graph plots binding energy per nucleon ($\frac{B}{A}$) on the y-axis against mass number A on the x-axis.

Shape of the curve: it starts at the origin for $A=1$ (since hydrogen $^{1}_{1}H$ has no binding energy), rises steeply for light nuclei, reaches a broad maximum of about 8.8 MeV near $A \approx 56$ (iron region), and then falls slowly to about 7.6 MeV for heavy nuclei up to $A \approx 240$. Mark the peak with the label ^{56}Fe and the y-axis maximum at 8.8 MeV.

Two salient features of the curve:

(1) The binding energy per nucleon is nearly **constant (about 8 MeV)** in the range $30 < A < 170$. This is because of the **saturation of nuclear forces**, where

each nucleon interacts only with its nearest neighbours, so $B \propto A$ and $\frac{B}{A}$ stays flat.

(2) The curve has a **maximum value of about 8.8 MeV at $A \approx 56$**, so iron-group nuclei are the most tightly bound and the most stable.

Explanation of energy release:

A reaction releases energy when the products have a higher $\frac{B}{A}$ than the reactants, since the extra binding energy appears as released energy.

In **nuclear fission**, a heavy nucleus (low $\frac{B}{A} \approx 7.6$ MeV) splits into two medium nuclei (higher $\frac{B}{A} \approx 8.5$ MeV). The increase in binding energy per nucleon over all the nucleons is released as energy.

In **nuclear fusion**, two very light nuclei (very low $\frac{B}{A}$) combine into a heavier nucleus with higher $\frac{B}{A}$, again releasing energy.

In both processes the system moves towards the peak of the curve (more tightly bound state), so energy is liberated.

Final answer. The B/A versus A curve peaks near $A \approx 56$; both fission and fusion form products of higher B/A, releasing the gained binding energy.

← *Back to Q121*

Q122

(a) Plot a graph showing the variation of potential energy of a pair of nucleons as a function of their separation. Also indicate the regions where the force is (i) attractive and (ii) repulsive.
(b) Write two characteristic properties of nuclear force.

[2019 • Set 55-4-2]

SOLUTION

(a) The graph plots potential energy $U(r)$ of a pair of nucleons (y-axis) against their separation r (in fm, x-axis).

Description of the curve: for very small r (less than $\mathbf{r_0 \approx 0.8}$ fm) the curve lies high in the **positive** region and rises very steeply as $\boldsymbol{r \to 0}$. As r increases past r_0, the curve dips below the axis into the **negative** region, reaching a minimum of about −100 MeV at $\boldsymbol{r = r_0 \approx 0.8}$ fm. Beyond this minimum the curve rises back towards zero and becomes negligibly small for $\boldsymbol{r > 3}$ fm. Mark the minimum point at $\mathbf{r_0 \approx 0.8}$ **fm** and the depth as about −100 MeV.

Force regions (force $\boldsymbol{F = -\frac{dU}{dr}}$):

(i) For $\boldsymbol{r > r_0}$ (i.e. $\boldsymbol{r > 0.8}$ fm), the potential energy increases with r, so the force is **attractive**.

(ii) For $\boldsymbol{r < r_0}$ (i.e. $\boldsymbol{r < 0.8}$ fm), the potential energy rises sharply as r decreases, so the force is strongly **repulsive**. This repulsive core prevents the nucleus from collapsing.

(b) Two characteristic properties of nuclear force:

(1) Nuclear force is a very **short-range force**, effective only up to about 2 to 3 fm, and it is the strongest force in nature at this range (much stronger than the Coulomb force).

(2) Nuclear force is **charge independent**, that is, the force between proton-proton, proton-neutron and neutron-neutron pairs is essentially the same; it does not depend on the charge of the nucleons.

Final answer. The $\boldsymbol{U(r)}$ curve is repulsive for $\mathbf{r < 0.8}$ fm and attractive for $\boldsymbol{r > 0.8}$ fm; nuclear force is short-ranged and charge independent.

← *Back to Q122*

Q123

Plot a graph showing the variation of number of undecayed nuclei in a radioisotope as a function of time. Hence define half life of a nucleus and obtain its relation with

average life.

[2019 • Set 55-4-3]

SOLUTION

The graph plots the number of undecayed nuclei N (y-axis) against time t (x-axis).

Shape: a smooth **exponential decay curve** following $N=N_0\,e^{-\lambda t}$. It starts at N_0 at $t=0$ and falls, approaching zero asymptotically as $t\to\infty$. Mark N_0 on the y-axis, the half value $\frac{N_0}{2}$ at $t=T_{1/2}$, and $\frac{N_0}{4}$ at $t=2T_{1/2}$.

Definition of half-life: the **half-life $T_{1/2}$** is the time in which the number of undecayed nuclei of a radioactive sample reduces to half its initial value.

Using $N=N_0\,e^{-\lambda t}$ and putting $N=\frac{N_0}{2}$ at $t=T_{1/2}$:

$$\frac{N_0}{2}=N_0\,e^{-\lambda T_{1/2}}$$

$$\frac{1}{2}=e^{-\lambda T_{1/2}}$$

Taking natural log on both sides:

$$\lambda T_{1/2}=\ln 2=0.693$$

$$T_{1/2}=\frac{0.693}{\lambda}$$

Relation with average (mean) life τ: the mean life is $\tau=\frac{1}{\lambda}$, so $\lambda=\frac{1}{\tau}$.

Substituting:

$$T_{1/2}=0.693\,\tau$$

Thus the **half-life $T_{1/2}=0.693\,\tau$**, or equivalently $\tau=1.44\,T_{1/2}$, so the mean life is slightly longer than the half-life.

Final answer. Half-life $T_{1/2}=\frac{0.693}{\lambda}=0.693\,\tau$, where τ is the average life.

← *Back to Q123*

Q124

(a) Distinguish between isotopes and isobars, giving one example for each.
(b) Why is the mass of a nucleus always less than the sum of the masses of its constituents? Write one example to justify your answer.

OR

(a) Classify the following six nuclides into (i) isotones, (ii) isotopes, and (iii) isobars: ${}^{12}_{6}C$, ${}^{3}_{2}He$, ${}^{198}_{80}Hg$, ${}^{3}_{1}H$, ${}^{197}_{79}Au$, ${}^{14}_{6}C$.
(b) How does the size of a nucleus depend on its mass number? Hence explain why the density of nuclear matter should be independent of the size of the nucleus.

[2019 • Set 55-5-1]

SOLUTION

(a) Distinction between isotopes and isobars:

Isotopes are nuclei of the **same element** having the same atomic number Z but different mass number A (same number of protons, different number of neutrons). Example: ${}^{1}_{1}H$ and ${}^{2}_{1}H$.

Isobars are nuclei having the same mass number A but different atomic number Z (different elements). Example: ${}^{3}_{1}H$ and ${}^{3}_{2}He$.

(b) Mass of nucleus is always less than the sum of the masses of its constituent nucleons because, when free nucleons bind together, energy equal to the **binding energy** BE is released. By mass-energy equivalence $\boldsymbol{E=mc^2}$, this released energy comes from a loss of mass called the **mass defect** $\Delta m=\frac{BE}{c^2}$, so the nuclear mass is reduced.

Example for ${}^{4}_{2}He$:

$$\Delta m = (2m_p+2m_n) - m(\text{He})$$

$$\Delta m = 2(1.007825)+2(1.008665)-4.002603$$

$$\Delta m = 0.030377\ u$$

This gives a binding energy $\mathbf{BE=0.030377\times 931.5\approx 28.3\ MeV}$, confirming the mass loss.

OR (a) Classification of the six nuclides:

(i) **Isotones** (same number of neutrons $N=A-Z$): $^{198}_{80}\text{Hg}$ ($N=118$) and $^{197}_{79}\text{Au}$ ($N=118$).

(ii) **Isotopes** (same Z): $^{12}_{6}\text{C}$ and $^{14}_{6}\text{C}$.

(iii) **Isobars** (same A): $^{3}_{2}\text{He}$ and $^{3}_{1}\text{H}$.

OR (b) The nuclear radius depends on mass number as:

$$R=\text{R}_0\,\text{A}^{1/3}$$

where $\mathbf{R_0=1.2\ fm}$. The volume is:

$$V=\frac{4}{3}\pi\,\text{R}^3=\frac{4}{3}\pi\,\text{R}_0{}^3\,A$$

so $V\propto A$. The mass of the nucleus $M\propto A$. Hence the density is:

$$\rho=\frac{M}{V}\propto\frac{A}{A}=\text{constant}$$

Since A cancels out, the **density of nuclear matter is independent of mass number (size)** and is the same for all nuclei.

> **TIP**
>
> Use $\mathbf{1\ u=931.5\ MeV/c^2}$ to convert any mass defect straight into binding energy.

Final answer. Isotopes share Z, isobars share A; nuclear mass is less by the mass defect, and as $R\propto \text{A}^{1/3}$ the nuclear density is constant.

← *Back to Q124*

Q125

(a) Explain the processes of nuclear fission and nuclear fusion by using the plot of binding energy per nucleon (BE/A) *versus the mass number A.*
(b) A radioactive isotope has a half-life of 10 *years. How long will it take for the activity to reduce to* 3.125% *?*

[2018]

SOLUTION

(a) The plot of binding energy per nucleon $\frac{\text{BE}}{A}$ versus mass number A rises steeply for light nuclei, reaches a maximum of about 8.75 MeV near $\boldsymbol{A=56}$, and then falls slowly for heavy nuclei.

Nuclear fission: a heavy nucleus ($\boldsymbol{A\approx 240}$, with $\frac{\text{BE}}{A}\approx \mathbf{7.6}$ MeV) splits into two medium-mass nuclei ($\boldsymbol{A\approx 120}$, with $\frac{\text{BE}}{A}\approx \mathbf{8.5}$ MeV). The increase of about 0.9 MeV per nucleon over roughly 240 nucleons releases about 200 MeV per fission. The products are more tightly bound, so energy is released.

Nuclear fusion: two very light nuclei (very low $\frac{\text{BE}}{A}\approx \mathbf{1}$ MeV) combine to form a heavier nucleus of higher $\frac{\text{BE}}{A}$. This large gain in binding energy per nucleon releases energy, for example deuterium-tritium fusion liberates about 17.6 MeV.

In both cases the products lie nearer the peak of the curve, so they are more stable and energy is released.

(b) Given half-life $\mathbf{T_{1/2}=10}$ years, find time for activity to fall to 3.125%.

Express 3.125% as a fraction:

$$\frac{3.125}{100}=\frac{1}{32}=\left(\frac{1}{2}\right)^5$$

Since activity falls by a factor $\frac{1}{2}$ in each half-life, the number of half-lives is $\boldsymbol{n=5}$.

Total time:

$$t=n\times T_{1/2}=5\times 10$$

$$t=50 \text{ years}$$

Final answer. The activity reduces to **3.125%** after **5** half-lives, i.e. $\boldsymbol{t=50}$ years.

← *Back to Q125*

Q126

(i) A radioactive nucleus 'A' undergoes a series of decays as given below:

$A\ \alpha\ A_1\ \beta^-\ A_2\ \alpha\ A_3\ \gamma\ A_4$

The mass number and atomic number of A_2 *are 176 and 71 respectively. Determine the mass and atomic numbers of* A_4 *and A.*

(ii) Write the basic nuclear processes underlying β^+ *and* β^- *decays.*

[2017]

SOLUTION

(i) Apply the changes in A and Z for each emission, where for α decay $\Delta A=-4, \Delta Z=-2$; for β^- decay $\Delta A=0, \Delta Z=+1$; and for γ emission there is no change in A or Z.

The decay chain is $A\alpha A_1\beta^-A_2\alpha A_3\gamma A_4$, and we are given A_2 has mass number 176 and atomic number 71.

Finding A (work backwards from A_2). $A_1\beta^-A_2$ means A_1 had the same mass number and one less atomic number:

$$A_1: A=176,\ Z=71-1=70$$

$A\alpha A_1$ means A had 4 more in mass number and 2 more in atomic number:

$$A: A=176+4=180,\ Z=70+2=72$$

Finding A_4 (work forwards from A_2). $A_2\alpha A_3$:

$$A_3: A=176-4=172,\ Z=71-2=69$$

$A_3\gamma A_4$ leaves A and Z unchanged:

$$A_4: A=172,\ Z=69$$

So nucleus A has mass number 180, atomic number 72, and nucleus A_4 has mass number 172, atomic number 69.

(ii) Basic nuclear processes:

In β^- decay a neutron converts into a proton: $n \rightarrow p+e^-+\bar{\nu}_e$.

In β^+ decay a proton converts into a neutron: $p \rightarrow n+e^++\nu_e$.

TIP

When data is given for a middle nucleus, run the conservation rules backward and forward separately; gamma emission never changes A or Z.

Final answer. A is **(180, 72)** and A_4 is **(172, 69)**; β^-: $n \to p + e^- + \bar{\nu}_e$ and β^+: $p \to n + e^+ + \nu_e$.

← *Back to Q126*

Q127

(a) State the law of radioactive decay. Write the SI unit of 'activity'.
(b) There are $4\sqrt{2} \times 10^6$ *radioactive nuclei in a given radioactive sample. If the half life of the sample is* 20 *s, how many nuclei will decay in* 10 *s?*

[2017]

SOLUTION

(a) Law of radioactive decay: the **rate of disintegration** of a radioactive sample at any instant is **directly proportional to the number of undecayed nuclei** N present at that instant.

$$-\frac{dN}{dt} = \lambda N$$

where λ is the decay constant. Integrating gives $N = N_0 e^{-\lambda t}$.

SI unit of activity: **becquerel (Bq)**, where **1 Bq=1** disintegration per second.

(b) Given $N_0 = 4\sqrt{2} \times 10^6$, half-life $T_{1/2} = 20\ s$, time $t = 10\ s$.

Number of nuclei left after time t using $N = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$:

$$N = 4\sqrt{2} \times 10^6 \left(\frac{1}{2}\right)^{10/20}$$

$$N = 4\sqrt{2} \times 10^6 \left(\frac{1}{2}\right)^{1/2}$$

$$N = 4\sqrt{2} \times 10^6 \times \frac{1}{\sqrt{2}}$$

$$N = 4\times 10^6$$

Number of nuclei decayed in 10 s:

$$N_0 - N = 4\sqrt{2}\times 10^6 - 4\times 10^6$$

$$N_0 - N = 4(\sqrt{2}-1)\times 10^6$$

$$N_0 - N \approx 4(0.414)\times 10^6 \approx 1.66\times 10^6$$

Final answer. About **1.66×10^6** nuclei decay in **10** s; activity is measured in becquerel (Bq).

← Back to Q127

Q128

(a) Write the process of β-decay. How can radioactive nuclei emit β-particles even though they do not contain them? Why do all electrons emitted during β-decay not have the same energy?
(b) A heavy nucleus splits into two lighter nuclei. Which one of the two -- parent nucleus or the daughter nuclei has more binding energy per nucleon?

[2017]

SOLUTION

(a) Process of β-decay (β^-): a neutron inside the nucleus converts into a proton, emitting an electron and an antineutrino.

$$n \to p + e^- + \bar{\nu}_e$$

In nuclear-symbol form:

$${}^{A}_{Z}X \to {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e$$

How can a nucleus emit electrons though it does not contain them: the nucleus contains only protons and neutrons, not electrons. The emitted electron is **created at the instant of decay** when a neutron transforms into a proton; it is not stored in the nucleus beforehand.

Why emitted electrons do not all have the same energy: in β-decay the available energy (Q-value) is **shared between the emitted electron and the antineutrino**. Since this sharing varies from one decay to another, the electron carries a **continuous range of energies** from zero up to a maximum value Q, giving a continuous energy spectrum.

(b) When a heavy nucleus splits into two lighter nuclei, the lighter **daughter nuclei have more binding energy per nucleon** than the parent. This is because on the binding energy per nucleon curve, medium-mass nuclei lie higher (more tightly bound) than heavy nuclei. The gain in binding energy per nucleon is why energy is released in fission.

Final answer. The β-electron is created during $n \to p + e^- + \bar{\nu}_e$ and shares energy with the antineutrino; the daughter nuclei have more binding energy per nucleon.

← *Back to Q128*

Q129

(a) Derive the relation between the decay constant and half life of a radioactive substance.
(b) A radioactive element reduces to 25% of its initial mass in 1000 years. Find its half life.

[2017]

SOLUTION

(a) The number of undecayed nuclei at time t follows the decay law:

$$N = N_0 e^{-\lambda t}$$

where N_0 is the initial number of nuclei and λ is the decay constant.

At $t = T_{1/2}$, half the nuclei remain, so $N = \frac{N_0}{2}$:

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

$$\frac{1}{2}=e^{-\lambda T_{1/2}}$$

Taking natural log of both sides:

$$\ln\left(\frac{1}{2}\right)=-\lambda T_{1/2}$$

$$\lambda T_{1/2}=\ln 2=0.693$$

$$T_{1/2}=\frac{0.693}{\lambda}$$

(b) Given the element reduces to 25% of its initial mass in 1000 years.

The remaining fraction is:

$$\frac{N}{N_0}=\frac{25}{100}=\frac{1}{4}=\left(\frac{1}{2}\right)^2$$

Since $\frac{N}{N_0}=\left(\frac{1}{2}\right)^{t/T_{1/2}}$, the number of half-lives is 2:

$$\frac{1000}{T_{1/2}}=2$$

$$T_{1/2}=\frac{1000}{2}=500 \text{ years}$$

Final answer. The decay constant relates to half-life as $\mathbf{T_{1/2}=\frac{0.693}{\lambda}}$, and the half-life of the element is **500** years.

← *Back to Q129*

Q130

(a) Write the basic nuclear process involved in the emission of β^+ *in a symbolic form, by a radioactive nucleus.*
(b) In the reactions given below:
(i) $\mathbf{{}^{11}_{6}C \rightarrow {}^{y}_{z}B + x + \nu}$

(ii) $^{12}_{6}\mathrm{C} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{20}_{a}\mathrm{Ne} + {}^{b}_{c}\mathrm{He}$

Find the values of x, y, and z and a, b and c.

[2016]

SOLUTION

(a) In β^+ decay, a proton inside the nucleus converts into a neutron, emitting a positron and a neutrino.

$$p \rightarrow n + e^+ + \nu_e$$

In nuclear-symbol form:

$$^{A}_{Z}\mathrm{X} \rightarrow {}^{A}_{Z-1}\mathrm{Y} + e^+ + \nu_e$$

(b)(i) For $^{11}_{6}\mathrm{C} \rightarrow {}^{y}_{z}\mathrm{B} + x + \nu$, the atomic number falls from 6 to that of boron (5), so this is a β^+ decay and x is a **positron** e^+ (i.e. $^{0}_{+1}e$).

Conserving mass number: $11 = y + 0 \Rightarrow y = 11$.

Conserving atomic number: $6 = z + 1 \Rightarrow z = 5$.

So $x = e^+$, $y = 11$, $z = 5$.

(b)(ii) For $^{12}_{6}\mathrm{C} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{20}_{a}\mathrm{Ne} + {}^{b}_{c}\mathrm{He}$.

Conserving mass number: $12 + 12 = 20 + b$

$$24 = 20 + b \Rightarrow b = 4$$

Conserving atomic number: $6 + 6 = a + c$

$$12 = a + c$$

For neon, $a = 10$, and then $c = 12 - 10 = 2$ for helium.

So $a = 10$, $b = 4$, $c = 2$.

Final answer. β^+: $p \rightarrow n + e^+ + \nu_e$; reaction (i) gives $x = e^+$, $y = 11$, $z = 5$ and reaction (ii) gives $a = 10$, $b = 4$, $c = 2$.

← *Back to Q130*

Q131

Half life of $^{238}_{92}\mathrm{U}$ *against* α*-decay is* 4.5×10^{9} *years. Calculate the activity of* 1 *g sample of* $^{238}_{92}\mathrm{U}$. *Given Avogadro's number* $= \mathbf{6\times 10^{26}}$ *atoms/kmol.*

[2016]

SOLUTION

The activity of a radioactive sample is $\boldsymbol{R=\lambda N}$, where λ is the decay constant and N is the number of undecayed nuclei present.

First find the decay constant from the half life using $\lambda=\dfrac{\mathbf{0.693}}{\mathbf{T_{1/2}}}$.

Convert the half life into seconds. With $\mathbf{1\ year=3.16\times 10^{7}\ s}$:

$$T_{1/2}=4.5\times 10^{9}\times 3.16\times 10^{7}=1.42\times 10^{17}\ s$$

$$\lambda=\frac{0.693}{1.42\times 10^{17}}=4.88\times 10^{-18}\ \mathrm{s^{-1}}$$

Now find the number of atoms N in the 1 g sample. The molar mass of $^{238}_{92}\mathrm{U}$ is 238 g/mol, and Avogadro's number $\mathbf{=6\times 10^{26}\ atoms/kmol=6\times 10^{23}\ atoms/mol}$.

$$N=\frac{\text{mass}}{\text{molar mass}}\times N_A=\frac{1}{238}\times 6\times 10^{23}$$

$$N=2.52\times 10^{21}\ \text{atoms}$$

The activity is therefore

$$R=\lambda N=(4.88\times 10^{-18})(2.52\times 10^{21})$$

$$R=1.23\times 10^{4}\ \mathrm{Bq}$$

Expressed in curie, using $\mathbf{1\ Ci=3.7\times 10^{10}\ Bq}$, this is about 0.33 μCi. The activity of the 1 g sample is $\boldsymbol{R\approx 1.23\times 10^{4}\ Bq}$ **(about 0.33 μCi)**.

COMMON MISTAKE

Always convert the half life to seconds before finding λ*, otherwise the activity comes out per year instead of per second (becquerel).*

Final answer. The activity of the 1 g sample is about $\mathbf{1.23\times 10^4}$ Bq (nearly **0.33 μCi**).

← *Back to Q131*

Q132

(a) Derive the mathematical expression for law of radioactive decay for a sample of a radioactive nucleus.
(b) How is the mean life of a given radioactive nucleus related to the decay constant?

[2016]

SOLUTION

(a) The law of radioactive decay states that the rate of disintegration of a radioactive sample at any instant is directly proportional to the number of undecayed nuclei present at that instant.

If N is the number of undecayed nuclei at time t, then the rate of decay is

$$-\frac{dN}{dt} \propto N$$

$$-\frac{dN}{dt} = \lambda N$$

where λ is the decay constant. The negative sign shows that N decreases with time.

Separating the variables,

$$\frac{dN}{N} = -\lambda\, dt$$

Integrating both sides, with $N=N_0$ at $t=0$ and $N=N$ at time t,

$$\int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_0^t dt$$

$$\ln N - \ln N_0 = -\lambda t$$

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t$$

Taking the exponential of both sides,

$$N = N_0\, e^{-\lambda t}$$

This is the **law of radioactive decay**, showing that the number of undecayed nuclei falls off exponentially with time.

(b) The mean life τ of a radioactive nucleus is the reciprocal of the decay constant.

$$\tau = \frac{1}{\lambda}$$

It is the average lifetime of all the nuclei of the sample, found by taking the time-weighted average over all decays, $\tau = \frac{\int_0^\infty t\,\lambda\, e^{-\lambda t}\, dt}{\int_0^\infty \lambda\, e^{-\lambda t}\, dt} = \frac{1}{\lambda}$. It is also related to the half life by $T_{1/2} = \tau \ln 2 = 0.693\,\tau$.

Final answer. $N = N_0\, e^{-\lambda t}$, and the mean life is $\tau = \frac{1}{\lambda}$.

← *Back to Q132*

Q133

Plot a graph showing the variation of binding energy per nucleon as a function of mass number. Which property of nuclear force explains the approximate constancy of binding energy in the range **30? How does one explain the release of energy in both the processes of nuclear fission and fusion from the graph?**

[2016]

SOLUTION

The required graph plots binding energy per nucleon ($\frac{B}{A}$) on the y-axis (in MeV) against the mass number A on the x-axis.

Description of the curve: starting from the origin region it rises steeply for light nuclei (with sharp peaks at ^{4}He, ^{12}C, ^{16}O), reaches a maximum value of about 8.8 MeV near $A = 56$ (iron), remains nearly flat at about 8 MeV over the broad range $30 < A < 170$, and then falls gently to about 7.6 MeV for heavy nuclei like

^{238}U. The curve is roughly flat-topped, peaking at iron and tailing down on both sides.

The property of nuclear force that explains the near constancy of $\frac{B}{A}$ in the range **30 < A < 170** is the **saturation of nuclear forces**.

Because the nuclear force is short-ranged, a given nucleon attracts only its few immediate neighbours and not all the other nucleons in the nucleus. So as more nucleons are added, each one still binds with about the same (limited) number of neighbours. The total binding energy B therefore increases nearly in proportion to A, which keeps $\frac{B}{A}$ almost constant in this range.

Energy release in fission and fusion: nuclei tend to move toward the peak of the curve, where binding energy per nucleon is highest and the nucleus is most stable.

In **fission**, a heavy nucleus ($A>200$, $\frac{B}{A} \approx 7.6$ **MeV**) splits into two medium-mass fragments with higher $\frac{B}{A} \approx 8.5$ **MeV**. The increase in binding energy per nucleon means energy is released.

In **fusion**, two very light nuclei (low $\frac{B}{A}$) combine into a heavier nucleus with higher $\frac{B}{A}$. Again the total binding energy increases, so energy is released.

Final answer. The flat region is explained by the saturation (short range) of nuclear forces; both fission and fusion release energy by moving products to higher binding energy per nucleon.

← *Back to Q133*

Q134

Obtain the relation between the decay constant and half life of a radioactive sample. The half life of a certain radioactive material against α*-decay is 100 days. After how much time, will the undecayed fraction of the material be* 6.25%*?*

[2015]

SOLUTION

Starting from the law of radioactive decay, the number of undecayed nuclei at time t is

$$N = N_0 e^{-\lambda t}$$

After one half life ($t = T_{1/2}$), by definition half the nuclei remain, so $N = \frac{N_0}{2}$.

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

$$\frac{1}{2} = e^{-\lambda T_{1/2}}$$

Taking natural logarithm of both sides,

$$-\lambda T_{1/2} = \ln\left(\frac{1}{2}\right) = -\ln 2$$

$$\lambda T_{1/2} = \ln 2$$

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

This is the required relation between half life and decay constant.

Numerical part: here $T_{1/2} = 100$ **days** and the undecayed fraction is

$$\frac{N}{N_0} = 6.25\% = \frac{6.25}{100} = \frac{1}{16}$$

Express this as a power of one-half, since after each half life the amount halves.

$$\frac{N}{N_0} = \frac{1}{16} = \left(\frac{1}{2}\right)^4$$

So 4 half lives must elapse, i.e. $\frac{t}{T_{1/2}} = 4$.

$$t = 4 \times T_{1/2} = 4 \times 100$$

$$t = 400 \text{ days}$$

The undecayed fraction falls to 6.25% after **400 days**.

TIP

When the fraction is a neat power of 12 (like $\mathbf{116=(12)^4}$), just count half lives instead of using logarithms.

Final answer. $\mathbf{T_{1/2}=\frac{0.693}{\lambda}}$, and the undecayed fraction reaches 6.25% after 400 days.

← *Back to Q134*

Q135

Distinguish between nuclear fission and fusion. Show how in both these processes energy is released.
Calculate the energy release in MeV in the deuterium-tritium fusion reaction:
$\mathbf{{}_1^2H + {}_1^3H \rightarrow {}_2^4He + \mathit{n}}$
Using the data:
$\mathbf{\mathit{m}({}_1^2H) = 2.014102\ \mathit{u}}$
$\mathbf{\mathit{m}({}_1^3H) = 3.016049\ \mathit{u}}$
$\mathbf{\mathit{m}({}_2^4He) = 4.002603\ \mathit{u}}$
$\mathbf{m_n = 1.008665\ \mathit{u}}$
$\mathbf{1u = 931.5\ \mathit{MeV}/c^2}$

[2015]

SOLUTION

Nuclear fission is the process in which a heavy nucleus splits into two lighter nuclei of nearly equal mass (along with a few neutrons), for example ${}^{235}_{92}U$ breaking up after capturing a neutron. Nuclear fusion is the process in which two light nuclei combine to form a single heavier nucleus, for example two hydrogen isotopes joining to form helium.

Key distinctions: fission is usually triggered by neutron capture on a heavy nucleus and releases about 200 MeV per event, while fusion needs extremely high temperatures (about 10^7 to 10^8 K) to overcome the Coulomb repulsion between the light nuclei and releases of the order of a few MeV per nucleon.

Energy release in both: in each process the product nucleus or nuclei have a higher binding energy per nucleon than the reactants, i.e. they sit higher on the $\frac{B}{A}$ curve. The total binding energy increases, the total rest mass decreases by Δm, and this lost mass appears as released energy through $\boldsymbol{E=\Delta m\, c^2}$.

Numerical for the D-T reaction $\boldsymbol{^2_1H+{}^3_1H\rightarrow {}^4_2He+n}$:

The mass defect is

$$\Delta m=[m(^2_1\mathrm{H})+m(^3_1\mathrm{H})]-[m(^4_2\mathrm{He})+\mathrm{m}_n]$$

$$\Delta m=[2.014102+3.016049]-[4.002603+1.008665]$$

$$\Delta m=5.030151-5.011268$$

$$\Delta m=0.018883\ u$$

The energy released is

$$Q=\Delta m\times 931.5\ \mathrm{MeV}/u$$

$$Q=0.018883\times 931.5$$

$$Q=17.59\ \mathrm{MeV}$$

The deuterium-tritium fusion reaction releases $\boldsymbol{Q\approx 17.59\ \mathrm{MeV}}$.

COMMON MISTAKE

Add the masses of all reactants and all products first, then subtract; mixing the order of subtraction gives a negative Δm and a wrong (negative) Q.

Final answer. Fission splits a heavy nucleus, fusion joins light nuclei, both release energy via $\boldsymbol{E=\Delta m\, c^2}$; the D-T reaction releases about 17.59 MeV.

← *Back to Q135*

Q136

(a) Deduce the expression, $\boldsymbol{N=N_0\, e^{-\lambda t}}$, for the law of radioactive decay.
(b) (i) Write symbolically the process expressing the β^+ decay of $^{22}_{11}\mathrm{Na}$. Also write the

basic nuclear process underlying this decay.
(ii) Is the nucleus formed in the decay of the nucleus $^{22}_{11}\text{Na}$*, an isotope or isobar ?*

[2014]

SOLUTION

(a) The law of radioactive decay states that the rate of disintegration at any instant is proportional to the number of undecayed nuclei N present at that instant. The probability of decay per unit time is a constant λ, the decay constant.

$$-\frac{dN}{dt}=\lambda N$$

Separating the variables,

$$\frac{dN}{N}=-\lambda\, dt$$

Integrating, with $N=N_0$ at $t=0$,

$$\int_{N_0}^{N}\frac{dN}{N}=-\lambda\int_0^t dt$$

$$\ln\left(\frac{N}{N_0}\right)=-\lambda t$$

$$N=N_0 e^{-\lambda t}$$

(b)(i) The β^+ decay of $^{22}_{11}\text{Na}$ is written as

$$^{22}_{11}\text{Na}\rightarrow {}^{22}_{10}\text{Ne}+e^{+}+\nu_e$$

Here a positron (e^+) and a neutrino (ν_e) are emitted. The basic nuclear process underlying this decay is the conversion of a proton inside the nucleus into a neutron:

$$p\rightarrow n+e^{+}+\nu_e$$

(b)(ii) Comparing parent and daughter, $^{22}_{11}\text{Na}\rightarrow {}^{22}_{10}\text{Ne}$, the mass number stays the same ($A=22$) while the atomic number changes (Z goes from 11 to 10). Nuclei with the same A but different Z are **isobars**, so the nucleus formed is an isobar of sodium.

Final answer. $N=N_0\, e^{-\lambda t}$; the β^+ decay is ${}^{22}_{11}Na \rightarrow {}^{22}_{10}Ne + e^+ + \nu_e$ (from $p \rightarrow n + e^+ + \nu_e$), and Ne-22 is an isobar of Na-22.

← Back to Q136

Q137

For the past some time, Aarti had been observing some erratic body movement, unsteadiness and lack of coordination in the activities of her sister Radha, who also used to complain of severe headache occasionally. Aarti suggested to her parents to get a medical check-up of Radha. The doctor thoroughly examined Radha and diagnosed that she has a brain tumour.
(a) What, according to you, are the values displayed by Aarti ?
(b) How can radioisotopes help a doctor to diagnose brain tumour ?

[2014]

SOLUTION

(a) The values displayed by Aarti are: keen power of **observation** (she carefully noticed the unusual symptoms in her sister), genuine **care and concern** for her sister's health, **awareness and presence of mind** in connecting the symptoms to a possible illness, and a sense of **responsibility** in promptly advising her parents to arrange a medical check-up. These reflect a caring, alert and responsible attitude.

(b) Radioisotopes help a doctor diagnose a brain tumour by acting as radioactive tracers in medical imaging.

A small, safe dose of a short-half-life gamma-emitting radioisotope, such as technetium-99m (^{99m}Tc, half life about 6 hours), is tagged to a suitable chemical and introduced into the patient's body. The radioisotope tends to accumulate differently in tumour tissue than in normal tissue.

The gamma rays emitted from inside the body are detected outside by a gamma camera, which builds up an image of how the radioisotope is distributed. Abnormal regions of uptake then show up as bright or dark spots, revealing the location and size of the tumour.

A short half life is chosen so that the activity inside the body falls quickly and the radiation dose to the patient stays small, while a gamma emitter is used because gamma rays are penetrating enough to escape the body and be detected. This non-invasive method can locate a tumour early and accurately.

Final answer. Aarti shows observation, care, awareness and responsibility; radioisotopes act as tracers (e.g. ^{99m}Tc) whose gamma emission is imaged to locate the tumour.

← *Back to Q137*

Q138

(a) In a typical nuclear reaction, e.g.

$$_1{}^2\mathrm{H} + {}_1{}^2\mathrm{H} \longrightarrow {}_2{}^3\mathrm{He} + n + 3.27\ \mathrm{MeV},$$

although number of nucleons is conserved, yet energy is released. How? Explain.
(b) Show that nuclear density in a given nucleus is independent of mass number A.

[2013]

SOLUTION

(a) In the reaction $^2_1\mathbf{H}+{}^2_1\mathbf{H}\rightarrow {}^3_2\mathbf{He}+n+\mathbf{3.27\ MeV}$, there are 4 nucleons on each side, so the number of nucleons is conserved. Yet energy is released because the way these nucleons are bound changes.

The two deuterons are loosely bound (low binding energy per nucleon), whereas the product $^3_2$He is more tightly bound (higher binding energy per nucleon). So the total binding energy of the products is greater than that of the reactants.

A greater binding energy means the products have a smaller total rest mass than the reactants. This difference in mass, the mass defect Δm, is converted into energy according to Einstein's relation

$$E=\Delta m\, c^2$$

This energy (3.27 MeV) appears mainly as the kinetic energy of the products. Hence energy is released even though nucleon number is conserved, because the released energy comes from the **mass defect**, not from a change in the number of nucleons.

(b) The radius of a nucleus of mass number A is given by

$$R = R_0 A^{1/3}$$

Treating the nucleus as a sphere, its volume is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0{}^3 A$$

Its mass, taking each nucleon to have mass m (about 1 u), is

$$M = A\, m$$

Therefore the nuclear density is

$$\rho = \frac{M}{V} = \frac{A\, m}{\frac{4}{3}\pi R_0{}^3 A}$$

$$\rho = \frac{m}{\frac{4}{3}\pi R_0{}^3}$$

The mass number A cancels out, so ρ does not depend on A. Putting in $\mathbf{R_0 \approx 1.2 \times 10^{-15}\ m}$ and $\mathbf{m \approx 1.66 \times 10^{-27}\ kg}$ gives $\mathbf{\rho \approx 2.3 \times 10^{17}\ kg/m^3}$**, the same for all nuclei**.

Final answer. Energy is released because the product is more tightly bound, so mass defect appears as energy via $\boldsymbol{E = \Delta m\, c^2}$; and since $\boldsymbol{A}$ cancels, nuclear density $\approx \mathbf{2.3 \times 10^{17}}$ kg/m^3 is independent of $\boldsymbol{A}$.

← *Back to Q138*

Q139

(a) Write symbolically the β^- decay process of $^{32}_{15}P$.
(b) Derive an expression for the average life of a radionuclide. Give its relationship

with the half-life.

[2012]

SOLUTION

(a) In β^- decay the mass number A stays the same and the atomic number Z increases by one, since a neutron inside the nucleus turns into a proton with emission of an electron and an antineutrino.

$$^{32}_{15}\mathrm{P} \rightarrow {}^{32}_{16}\mathrm{S} + \mathrm{e}^- + \bar{\nu}_e$$

The underlying basic nuclear process is

$$n \rightarrow p + \mathrm{e}^- + \bar{\nu}_e$$

(b) The mean (average) life τ is the average of the lifetimes of all the nuclei of the sample.

The number of nuclei that decay between time t and t+dt is

$$|\mathrm{dN}| = \lambda N \, \mathrm{dt} = \lambda \, \mathrm{N_0} \, \mathrm{e}^{-\lambda t} \, \mathrm{dt}$$

Each such nucleus has lived for a time t. The mean life is the total lifetime of all nuclei divided by the total number of nuclei:

$$\tau = \frac{1}{\mathrm{N_0}} \int_0^\infty t \, |\mathrm{dN}|$$

$$\tau = \frac{1}{\mathrm{N_0}} \int_0^\infty t \, \lambda \, \mathrm{N_0} \, \mathrm{e}^{-\lambda t} \, \mathrm{dt}$$

$$\tau = \lambda \int_0^\infty t \, \mathrm{e}^{-\lambda t} \, \mathrm{dt}$$

Using the standard integral $\int_0^\infty t \, \mathrm{e}^{-\lambda t} \, \mathrm{dt} = \frac{1}{\lambda^2}$,

$$\tau = \lambda \times \frac{1}{\lambda^2}$$

$$\tau = \frac{1}{\lambda}$$

Relation with half life: since $\mathrm{T_{1/2}} = \frac{\ln 2}{\lambda} = \ln 2 \tau$, we get

$$\mathrm{T_{1/2}} = 0.693 \, \tau \tau = 1.44 \, \mathrm{T_{1/2}}$$

So the **average life is the reciprocal of the decay constant**, and it is about 1.44 times the half life.

CONCEPT

The mean life ($\tau=1/\lambda$) is always longer than the half life ($T_{1/2}=0.693\,\tau$), because a few long-lived nuclei pull the average up.

Final answer. ${}^{32}_{15}P \rightarrow {}^{32}_{16}S + e^- + \nu_e$; the average life is $\tau=\frac{1}{\lambda}=1.44\,T_{1/2}$.

← *Back to Q139*

Q140

Draw a plot of potential energy of a pair of nucleons as a function of their separations. Mark the regions where the nuclear force is (i) attractive and (ii) repulsive. Write any two characteristic features of nuclear forces.

[2012]

SOLUTION

The required graph plots the potential energy U of a pair of nucleons (in MeV) on the y-axis against their separation r (in fm) on the x-axis.

Description of the curve: starting from large positive values at very small r, the curve falls steeply, crosses zero near $r=r_0\approx 0.8$ fm, dips to a minimum of about −100 MeV at roughly $r\approx 1$ fm, and then rises back, approaching zero from below as r increases beyond about 3 fm. The minimum (potential well) corresponds to the equilibrium separation, and the curve becomes flat (force negligible) at large r.

Marking the regions (the force is $F=-\frac{dU}{dr}$):

(i) **Attractive region**: for $r>r_0\approx 0.8$ fm (to the right of the potential minimum the slope is such that the force pulls the nucleons together). This region binds the nucleons.

(ii) **Repulsive region**: for $r0\approx 0.8$ fm, where the potential energy rises steeply and the force is strongly repulsive. This hard core prevents the nucleus from

collapsing.

Two characteristic features of nuclear forces:

1. They are **short-range** forces, effective only up to about 2 to 3 fm, and become negligible beyond, unlike the Coulomb force which has infinite range.

2. They are **charge-independent**, that is, the force between a proton-proton, neutron-neutron, or proton-neutron pair is essentially the same; the nuclear force does not depend on the charge of the nucleons. (They are also much stronger than the Coulomb force and show saturation.)

Final answer. $U(r)$ is repulsive for $r<0.8$ fm and attractive for $r>0.8$ fm; nuclear forces are short-range and charge-independent.

← *Back to Q140*

Q141

(i) What characteristic property of nuclear force explains the constancy of binding energy per nucleon (BE/A) *in the range of mass number 'A' lying* **30 < A < 170***?*
(ii) Show that the density of nucleus over a wide range of nuclei is constant - independent of mass number A.

[2012]

SOLUTION

(i) The constancy of the binding energy per nucleon (BE/A) for **30 < A < 170** is explained by the saturation (short-range) property of the nuclear force.

The nuclear force is strongly attractive but acts only over a very short range, about 2 to 3 fm. So a given nucleon is influenced only by the few nucleons in its immediate neighbourhood, not by all the other nucleons in the nucleus.

Because each nucleon interacts with a fixed limited number of nearest neighbours, the force is said to saturate. The total binding energy then grows in direct proportion to the number of nucleons:

$$\text{BE} \propto A$$

$$\Rightarrow \frac{\text{BE}}{A} \approx \text{constant} \approx 8.5\ \text{MeV/nucleon}$$

If instead every nucleon attracted every other nucleon, BE would grow as $A(A-1)$ and BE/A would rise steeply with A, which is not observed. Hence saturation is the key property.

(ii) The radius of a nucleus of mass number A is given by

$$R = R_0 A^{1/3},\ R_0 \approx 1.2\ \text{fm}$$

The volume of the nucleus (assumed spherical) is

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^{\,3} A$$

So $\boldsymbol{V \propto A}$.

The mass of the nucleus is approximately

$$M \approx A\,u$$

where u is one atomic mass unit, so $\boldsymbol{M \propto A}$.

The nuclear density is therefore

$$\rho = \frac{M}{V} = \frac{A\,u}{\frac{4}{3}\pi R_0^{\,3} A}$$

$$\rho = \frac{u}{\frac{4}{3}\pi R_0^{\,3}}$$

The mass number A cancels, so ρ is independent of A. Putting $\boldsymbol{u = 1.66\times 10^{-27}\ \text{kg}}$ and $\boldsymbol{R_0 = 1.2\times 10^{-15}\ m}$ gives $\boldsymbol{\rho \approx 2.3\times 10^{17}\ \text{kg/m}^3}$, the same enormous value for all nuclei. Hence the density of nuclear matter is **constant and independent of the mass number A**.

TIP

Always remember $\boldsymbol{R=R_0A^{1/3}}$ gives $\boldsymbol{V \propto A}$; this single relation is what makes density independent of A.

Final answer. Saturation of the short-range nuclear force keeps **BE/*A*** constant, and since both nuclear mass and volume are proportional to ***A***, the nuclear density stays constant at about $\mathbf{2.310^{17}\ kg/m^3}$.

← *Back to Q141*

Q142

State the law of radioactive decay.
Plot a graph showing the number (N) of undecayed nuclei as a function of time (t) for a given radioactive sample having half life $T_{1/2}$.
Depict in the plot the number of undecayed nuclei at (i) $\boldsymbol{t = 3T_{1/2}}$ *and (ii)* $\boldsymbol{t = 5T_{1/2}}$.

[2011 • Set 55-1-1]

SOLUTION

Law of radioactive decay: The rate of disintegration of a radioactive sample at any instant is directly proportional to the number N of undecayed (parent) nuclei present at that instant.

$$\frac{dN}{dt} \propto N$$

$$\frac{dN}{dt} = -\lambda N$$

where λ is the decay constant (the probability of decay per nucleus per unit time) and the negative sign shows that N decreases with time.

Separating the variables and integrating from $\boldsymbol{N = N_0}$ at $\boldsymbol{t = 0}$ to N at time t:

$$\int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_0^t dt$$

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t$$

$$N = N_0 e^{-\lambda t}$$

This is the exponential law of radioactive decay, showing that the number of undecayed nuclei falls off exponentially with time.

Graph: Plot N (vertical axis) against time t (horizontal axis). The curve starts at N_0 on the N-axis when $\boldsymbol{t = 0}$ and falls as a smooth exponentially decaying curve that bends towards the time axis, approaching it asymptotically (it never quite reaches zero).

Mark these points on the curve. At $\boldsymbol{t = T_{1/2}}$ the value has dropped to $N_0/2$, at $\boldsymbol{t = 2T_{1/2}}$ to $N_0/4$, and so on, halving every half-life.

(i) At $\boldsymbol{t = 3T_{1/2}}$:

$$N = N_0\left(\frac{1}{2}\right)^3 = \frac{N_0}{8}$$

(ii) At $\boldsymbol{t = 5T_{1/2}}$:

$$N = N_0\left(\frac{1}{2}\right)^5 = \frac{N_0}{32}$$

On the graph, drop a vertical line from $\boldsymbol{t = 3T_{1/2}}$ up to the curve and across to the N-axis to mark $N_0/8$, and similarly from $\boldsymbol{t = 5T_{1/2}}$ to mark $N_0/32$. Thus $\boldsymbol{N(3T_{1/2}) = N_0/8}$ **and** $\boldsymbol{N(5T_{1/2}) = N_0/32}$.

Final answer. $\boldsymbol{N = N_0\, e^{-\lambda t}}$; the undecayed nuclei are $\boldsymbol{N_0/8}$ at $\boldsymbol{t = 3T_{1/2}}$ and $\boldsymbol{N_0/32}$ at $\boldsymbol{t = 5T_{1/2}}$.

← *Back to Q142*

Q143

(i) Define 'activity' of a radioactive material and write its S.I. unit.
(ii) Plot a graph showing variation of activity of a given radioactive sample with time.
(iii) The sequence of stepwise decay of a radioactive nucleus is
$D\ \alpha\ D_1\ \beta^-\ D_2$
If the atomic number and mass number of D_2 *are 71 and 176 respectively, what are their corresponding values for D?*

[2010]

SOLUTION

(i) Activity: The activity of a radioactive sample is the rate at which its nuclei disintegrate, that is the number of decays per unit time.

$$R = -\frac{dN}{dt} = \lambda N$$

The SI unit of activity is the becquerel (Bq), where **1 Bq = 1 disintegration per second**. (The older unit is the curie, **1 Ci = 3.710^{10} Bq**.)

(ii) Graph: Since $N = N_0 e^{-\lambda t}$, multiplying by λ gives the activity

$$R = R_0 e^{-\lambda t},\ R_0 = \lambda N_0$$

Plot activity R (vertical axis) against time t (horizontal axis). The curve starts at R_0 at $t = 0$ and falls as a smooth exponentially decaying curve, dropping to $R_0/2$ at $t = T_{1/2}$, to $R_0/4$ at $t = 2T_{1/2}$, and approaching the time axis asymptotically without ever reaching zero.

(iii) In the decay sequence D α D_1 β^- D_2, work backwards from D_2 which has $Z = 71$, $A = 176$.

In β^- decay a neutron changes to a proton, so the atomic number rises by 1 while the mass number is unchanged:

$$D_1 \to Z + 1 = 71 \Rightarrow Z(D_1) = 70$$

$$A(D_1) = 176 \text{ (unchanged in } \beta^- \text{ decay)}$$

In α decay the nucleus emits a $^4_2$He particle, so the atomic number falls by 2 and the mass number falls by 4:

$$Z(D) - 2 = 70 \Rightarrow Z(D) = 72$$

$$A(D) - 4 = 176 \Rightarrow A(D) = 180$$

Therefore the parent nucleus D has **atomic number 72 and mass number 180** (it is $^{180}_{72}$Hf).

Final answer. Activity $R = \lambda N$ in becquerel (Bq); the parent D has atomic number **72** and mass number **180**.

← *Back to Q143*

Q144

Draw a plot showing the variation of binding energy per nucleon versus the mass number A. Explain with the help of this plot the release of energy in the processes of nuclear fission and fusion.

[2009]

SOLUTION

Graph: Plot the binding energy per nucleon BE/A (in MeV, vertical axis) against the mass number A (horizontal axis). The curve rises steeply for light nuclei (with peaks at ^{4}He, ^{12}C, ^{16}O), reaches a flat maximum of about 8.8 MeV/nucleon near $\boldsymbol{A = 56}$ (iron region), stays nearly constant in the range $\mathbf{30 < A < 170}$, and then decreases slowly to about 7.6 MeV/nucleon for the heaviest nuclei such as uranium.

The key idea is that the higher the value of BE/A, the more tightly bound and the more stable the nucleus. Nuclei in the middle of the curve (iron region) are the most stable.

Energy release in nuclear fusion (light nuclei, the rising part of the curve): Two very light nuclei combine to form a heavier nucleus. The product lies higher on the BE/A curve than the reactants, so it is more tightly bound. The total binding energy after fusion is greater than before, and this gain in binding energy is released as kinetic energy of the products and as gamma radiation.

For example,

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + \text{energy}$$

Energy release in nuclear fission (heavy nuclei, the falling part of the curve): A heavy nucleus such as ^{235}U splits into two medium-mass nuclei. These fragments lie higher on the BE/A curve (around 8.5 MeV/nucleon) than the original heavy nucleus (about 7.6 MeV/nucleon), so they are more tightly bound. The increase in total binding energy, about 200 MeV per fission, is released as energy.

In both processes the products are more tightly bound than the reactants, so **energy is released whenever nucleons rearrange into a configuration of higher binding energy per nucleon**.

Final answer. Both fission of heavy nuclei and fusion of light nuclei move towards the peak of the **BE/*A*** curve, raising the total binding energy, and the gain is released as energy.

← *Back to Q144*

Q145

Distinguish between isotopes and isobars. Give one example for each species. A radioactive isotope has a half-life of 5 *years. How long will it take the activity to reduce to* 3.125%*?*

[2008]

SOLUTION

Isotopes: Nuclei (atoms of the same element) that have the same atomic number Z but different mass number A. They have the same number of protons but a different number of neutrons, so they occupy the same place in the periodic table and have identical chemical properties.

Example of isotopes: ${}^{12}_{6}\mathrm{C}$ and ${}^{14}_{6}\mathrm{C}$ (both have $\boldsymbol{Z = 6}$).

Isobars: Nuclei that have the same mass number A but different atomic number Z. They belong to different elements and have different chemical properties.

Example of isobars: ${}^{40}_{18}\mathrm{Ar}$ and ${}^{40}_{20}\mathrm{Ca}$ (both have $\boldsymbol{A = 40}$).

Numerical: Activity is proportional to the number of undecayed nuclei, so the activity ratio equals the fraction of nuclei left.

$$\frac{R}{\mathrm{R_0}} = 3.125\% = \frac{3.125}{100} = \frac{1}{32}$$

Express this fraction as a power of one-half, using $N = \mathrm{N_0}(\frac{1}{2})^n$ where n is the number of half-lives:

$$\frac{1}{32} = (\frac{1}{2})^5$$

$$\Rightarrow n = 5 \text{ half–lives}$$

The total time is the number of half-lives times the half-life:

$$t = n \times T_{1/2} = 5 \times 5 \text{ years}$$

$$t = 25 \text{ years}$$

Hence the activity reduces to 3.125% of its initial value in **25 years**.

TIP

3.125% = 1/32 = $(1/2)^5$, so it is exactly 5 half-lives; spotting the power of 2 avoids using logarithms.

Final answer. Isotopes: same Z, different A (e.g. $^{12}_{6}C$, $^{14}_{6}C$); isobars: same A, different Z (e.g. $^{40}_{18}Ar$, $^{40}_{20}Ca$); the activity falls to **3.125%** in **25** years.

← *Back to Q145*

Q146

Calculate the energy released during the α*-decay of* $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$.
Given:
1. atomic mass of $^{238}_{92}U$ = *238.05079 u*
2. atomic mass of $^{234}_{90}Th$ = *234.04363 u*
3. atomic mass of $^{4}_{2}He$ = *4.00260 u*
1 *u* = 931.5 MeV/c^2.
Is this decay spontaneous? Give reason.

[2007]

SOLUTION

The energy released in a nuclear decay equals the mass defect times c^2, where the mass defect is the difference between the mass of the parent and the total mass of the products.

The decay is

$$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

Mass defect:

$$\Delta m = m(^{238}\text{U}) - [\, m(^{234}\text{Th}) + m(^{4}\text{He}) \,]$$

$$\Delta m = 238.05079 - [\, 234.04363 + 4.00260 \,]\ u$$

$$\Delta m = 238.05079 - 238.04623\ u$$

$$\Delta m = 0.00456\ u$$

(Note: since atomic masses are used, the electron masses cancel on both sides, so this is valid.)

Energy released, using $\mathbf{1\ u = 931.5\ MeV/c^2}$:

$$Q = \Delta m \times 931.5\ \text{MeV}/u$$

$$Q = 0.00456 \times 931.5\ \text{MeV}$$

$$Q \approx 4.25\ \text{MeV}$$

Spontaneity: The value of Q comes out positive, which means the rest mass of the products is less than that of the parent and energy is given out. A decay with $Q > 0$ is energetically allowed, so this α-decay is **spontaneous**. The released energy appears mainly as the kinetic energy of the emitted α-particle and the recoiling ^{234}Th nucleus.

Final answer. $Q \approx 4.25$ **MeV**; since $Q > 0$, the **α**-decay is spontaneous.

← *Back to Q146*

Q147

Draw the graph to show variation of binding energy per nucleon with mass number of different atomic nuclei. Calculate binding energy/nucleon of $^{40}_{20}$Ca *nucleus.*
Given:
mass of $^{40}_{20}$Ca = *39.962589 u*
mass of proton = 1.007825 u
mass of neutron = 1.008665 u
and $\mathbf{1\ u = 931\ MeV/c^2}$

[2007]

SOLUTION

Graph: Plot the binding energy per nucleon BE/A (in MeV, vertical axis) against the mass number A (horizontal axis). The curve rises steeply for light nuclei, reaches a broad maximum of about 8.75 MeV/nucleon near $\mathbf{A = 56}$ (the iron peak), remains almost flat in the range $\mathbf{30 < A < 170}$, and then falls gradually for heavier nuclei. Higher BE/A means greater stability, so iron-region nuclei are the most stable.

Numerical for $^{40}_{20}$Ca: Here $\boldsymbol{Z = 20}$ protons, $\boldsymbol{N = A - Z = 40 - 20 = 20}$ neutrons.

The mass defect is the difference between the total mass of the free nucleons and the actual nuclear (atomic) mass.

Mass of protons:

$$Z\,m_p = 20 \times 1.007825 = 20.156500\ u$$

Mass of neutrons:

$$N\,m_n = 20 \times 1.008665 = 20.173300\ u$$

Total mass of constituents:

$$Z\,m_p + N\,m_n = 20.156500 + 20.173300 = 40.329800\ u$$

Mass defect:

$$\Delta m = 40.329800 - 39.962589 = 0.367211\ u$$

Total binding energy, using $\mathbf{1\ u = 931\ MeV/c^2}$:

$$\text{BE} = \Delta m \times 931\ \text{MeV}/u$$

$$\text{BE} = 0.367211 \times 931\ \text{MeV}$$

$$\text{BE} \approx 341.9\ \text{MeV}$$

Binding energy per nucleon:

$$\frac{\text{BE}}{A} = \frac{341.9}{40}$$

$$\frac{\text{BE}}{A} \approx 8.55\ \text{MeV/nucleon}$$

Hence the binding energy per nucleon of $^{40}_{20}\mathrm{Ca}$ is about **8.55 MeV/nucleon**, consistent with its position near the top of the BE/A curve.

Final answer. The binding energy per nucleon of $^{40}_{20}\mathbf{Ca}$ is about **8.55 MeV/nucleon**.

← *Back to Q147*

Q148

Explain, with the help of a nuclear reaction in each of the following cases, how the neutron to proton ratio changes during (i) alpha-decay (ii) beta-decay.

[2006]

SOLUTION

(i) Alpha-decay: In α-decay the nucleus emits a $^{4}_{2}\mathrm{He}$ nucleus, so it loses 2 protons and 2 neutrons. Consider

$$^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

For the parent uranium, $Z = 92$ and $N = A - Z = 238 - 92 = 146$:

$$\left(\frac{N}{Z}\right)_{\text{before}} = \frac{146}{92} = 1.587$$

For the daughter thorium, $Z = 90$ and $N = 234 - 90 = 144$:

$$\left(\frac{N}{Z}\right)_{\text{after}} = \frac{144}{90} = 1.600$$

Although equal numbers of neutrons and protons are removed, the starting nucleus is heavy with $N/Z > 1$, so removing equal amounts from numerator and denominator raises the ratio. Hence the neutron-to-proton ratio increases slightly in α-decay.

(ii) Beta-minus decay: In β^- decay a neutron inside the nucleus converts into a proton, emitting an electron and an antineutrino, so N falls by 1 and Z rises by 1. Consider

$$^{234}_{90}\mathrm{Th} \rightarrow {}^{234}_{91}\mathrm{Pa} + {}^{0}_{-1}\mathrm{e} + \bar{\nu}$$

For the parent thorium, $Z = 90$, $N = 144$:

$$\left(\frac{N}{Z}\right)_{\text{before}} = \frac{144}{90} = 1.600$$

For the daughter protactinium, $Z = 91$, $N = 234 - 91 = 143$:

$$\left(\frac{N}{Z}\right)_{\text{after}} = \frac{143}{91} = 1.571$$

Since a neutron is replaced by a proton, the neutron-to-proton ratio decreases in β^- decay.

Thus an unstable nucleus adjusts its N/Z ratio towards the stability line: a neutron-rich nucleus undergoes β^- decay to lower N/Z, while α-decay slightly raises it. **In α-decay the N/Z ratio increases; in β^--decay it decreases.**

Final answer. In **α**-decay the N/Z ratio increases (e.g. **146/92 → 144/90**); in **β^-**-decay it decreases (e.g. **144/90 → 143/91**).

← Back to Q148

Q149

Why is the mass of a nucleus always less than the sum of the masses of its constituents, neutrons and protons?
If the total number of neutrons and protons in a nuclear reaction is conserved, how then is the energy absorbed or evolved in the reaction? Explain.

[2006]

SOLUTION

Why the nuclear mass is less than the sum of its constituents: When free protons and neutrons combine to form a nucleus, they become bound by the strong attractive nuclear force and settle into a state of lower (negative) energy. To pull the nucleons apart again, energy equal to the binding energy must be supplied from outside.

By Einstein's mass-energy equivalence, $E = mc^2$, this energy given out during formation appears as a loss of mass. So the nucleus weighs less than its separated constituents by an amount called the mass defect Δm:

$$\Delta m = [\, Z\, m_p + (A - Z) m_n \,] - M_{\text{nucleus}}$$

$$M_{\text{nucleus}} = [\, Z\, m_p + (A - Z) m_n \,] - \Delta m$$

The binding energy is the energy equivalent of this mass defect:

$$\text{BE} = \Delta m\, c^2$$

This is why the nuclear mass is always smaller than the total mass of the free protons and neutrons.

Energy in a reaction even though nucleon number is conserved: In any nuclear reaction the total number of protons plus neutrons (nucleons) is conserved. However, the way these nucleons are distributed among the nuclei changes, and different nuclei have different binding energies per nucleon, hence different total masses.

The energy released or absorbed is the Q-value, given by the change in rest mass:

$$Q = [\Sigma\, m_{\text{reactants}} - \Sigma\, m_{\text{products}}]c^2$$

If the products are more tightly bound than the reactants, their total mass is less, so $Q > 0$ and energy is released (exoergic reaction). If the products are less tightly bound, their total mass is greater, so $Q < 0$ and energy must be absorbed (endoergic reaction).

Thus it is the change in total binding energy, that is the change in rest mass, and not the nucleon count, that decides the energy evolved or absorbed. **Energy comes from the conversion of mass, since nucleon number is conserved but mass is not.**

> **CONCEPT**
>
> Nucleon number is conserved but total rest mass is not; the missing mass is what becomes the released energy.

Final answer. The nuclear mass is less by the mass defect Δm ($\text{BE} = \Delta m\, c^2$); reaction energy arises because rearranging the conserved nucleons changes the total binding energy and hence the rest mass, with $Q = (\Sigma\, m_{\text{reactants}} - \Sigma\, m_{\text{products}})c^2$.

← Back to Q149

Q150

Draw a graph showing the variation of binding energy per nucleon with mass number for different nuclei. Explain, with the help of this graph, the release of energy by the process of nuclear fusion.

[2006]

SOLUTION

Graph: Plot the binding energy per nucleon BE/A (in MeV, vertical axis) against the mass number A (horizontal axis). The curve rises steeply for light nuclei (with sharp peaks at ^{4}He, ^{12}C, ^{16}O), reaches a broad maximum of about 8.8 MeV/nucleon near $\boldsymbol{A = 56}$ (iron region), stays nearly flat in the range $\mathbf{30 < A < 170}$, and then decreases slowly to about 7.6 MeV/nucleon for the heaviest nuclei. A higher BE/A means a more tightly bound, more stable nucleus.

Release of energy in nuclear fusion: Fusion involves the very light nuclei that lie on the steeply rising left-hand part of the curve, where BE/A is small. When two such light nuclei combine, they form a heavier nucleus that lies higher up the curve, with a larger binding energy per nucleon.

Because the product is more tightly bound than the reactants, the total binding energy after fusion is greater than before. By mass-energy equivalence the gain in binding energy appears as a loss of rest mass, and this energy is released as kinetic energy of the products and as gamma radiation.

A typical fusion reaction is

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

Here the light reactants (BE/A of a few MeV) form $^{4}_{2}$He, which has a high BE/A of about 7 MeV/nucleon, so a large amount of energy is liberated per nucleon. Such reactions power the Sun and stars.

Thus in fusion the products move up the BE/A curve compared with the reactants, and the **increase in total binding energy is released as energy**.

Final answer. Light nuclei lie low on the rising part of the **BE/*A*** curve; fusing them gives a more tightly bound product higher on the curve, and the gain in binding energy is released as energy (e.g. about **17.6 MeV** in the D-T reaction).

← *Back to Q150*

Q151

(a) *Draw a graph showing the variation of potential energy of a pair of nucleons as a function of their separation. Indicate the regions in which nuclear force is* ***(i)*** *attractive, and* ***(ii)*** *repulsive.*
(b) *Write two characteristic features of nuclear force which distinguish it from the coulomb force.*

[2005]

SOLUTION

(a) The graph is plotted with potential energy U (in MeV) on the vertical axis and the separation r (in fm) between the two nucleons on the horizontal axis.

Starting from large r, the curve lies just below the r-axis (small negative U) and rises to meet the axis ($U \to 0$) as r becomes large, showing the force is short range.

As r decreases, U becomes more negative and reaches a minimum value of about −100 MeV at a separation $\mathbf{r_0 \approx 0.8\ fm}$. This minimum is the position of stable equilibrium.

For r smaller than r_0, the curve turns sharply upward and U becomes large and positive (rises very steeply as $r \to 0$).

(i) For $r > r_0 \approx 0.8$ fm the slope $\frac{dU}{dr} > 0$, so the force $F = -\frac{dU}{dr} < 0$ is directed inward. This is the **attractive region** that binds the nucleons together.

(ii) For $r < r_0 \approx 0.8$ fm the slope $\frac{dU}{dr} < 0$, so the force $F = -\frac{dU}{dr} > 0$ is directed outward. This is the **repulsive region**, which prevents the nucleons from collapsing into one another.

(b) Two characteristic features of the nuclear force that distinguish it from the Coulomb force:

1. The nuclear force is a very **short-range** force, effective only up to about 2 to 3 fm and negligible beyond, whereas the Coulomb force is long range and varies as $\frac{1}{r^2}$ to infinity.

2. The nuclear force is **charge independent**, that is, it is the same between proton-proton, proton-neutron and neutron-neutron pairs, while the Coulomb force acts only between charged particles. (The nuclear force is also about 100 times stronger than the Coulomb force at ~ 1 fm.)

Final answer. The PE curve is repulsive for $r < 0.8$ fm and attractive for $r > 0.8$ fm; the nuclear force is short range and charge independent.

← *Back to Q151*

Q152

(a) *Show that the decay rate 'R' of a sample of a radionuclide is related to the number of radioactive nuclei 'N' at the same instant by the expression* $\boldsymbol{R = \lambda N}$.
(b) *The half-life of* ^{238}U *against* α*-decay is* 1.5×10^{17} *s. What is the activity of a sample of* ^{238}U *having* 25×10^{20} *atoms?*

[2005]

SOLUTION

(a) According to the law of radioactive decay, the rate of disintegration of a radioactive sample at any instant is directly proportional to the number N of undecayed nuclei present at that instant. This follows from the random and independent nature of nuclear decay.

$$-\frac{dN}{dt} \propto N$$

$$-\frac{dN}{dt} = \lambda N$$

where λ is the decay constant, defined as the probability of decay per unit time per nucleus.

The decay rate (activity) is defined as $R = -\frac{dN}{dt}$. Therefore,

$$R = \lambda N$$

(b) Given: half-life $T_{1/2} = 1.5 \times 10^{17}\ s$ and number of atoms $N = 25 \times 10^{20}$.

The decay constant is

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{1.5 \times 10^{17}\ s}$$

$$\lambda = 4.62 \times 10^{-18}\ s^{-1}$$

The activity is

$$R = \lambda N = (4.62 \times 10^{-18}) \times (25 \times 10^{20})$$

$$R = 1.155 \times 10^{4}\ \text{Bq}$$

$$R \approx 1.155 \times 10^{4}\ \text{Bq}$$

CONCEPT

Activity is measured in becquerel (Bq), where **1 Bq = 1** disintegration per second.

Final answer. The activity of the sample is $R = \lambda N \approx 1.155 \times 10^{4}$ **Bq** (disintegrations per second).

← *Back to Q152*

Q153

Give the mass number and atomic number of elements on the right-hand side of the decay process.
The graph shows how the activity of a sample of radon-220 changes with time. Use the graph to determine its half-life. Calculate the value of decay constant of radon-220.



[2004]

SOLUTION

Radon-220 is an α-emitter. In α-decay the nucleus emits a helium nucleus $^{4}_{2}\mathrm{He}$, so the mass number decreases by 4 and the atomic number decreases by 2.

$$^{220}_{86}\mathrm{Rn} \longrightarrow {}^{216}_{84}\mathrm{Po} + {}^{4}_{2}\mathrm{He}$$

On the right-hand side the two elements are:

Polonium: mass number $A = 216$, atomic number $Z = 84$.

Alpha particle (helium nucleus): mass number $A = 4$, atomic number $Z = 2$.

(You can check conservation: mass numbers $216 + 4 = 220$ and atomic numbers $84 + 2 = 86$, matching the parent.)

To read the half-life from the activity-versus-time graph: mark the initial activity R_0 on the vertical axis, then locate the activity $\frac{R_0}{2}$. Drop a line from this point to the curve and read the corresponding time on the horizontal axis. This time is the **half-life**, which for radon-220 comes out as $T_{1/2} \approx 55.6\ s$.

The decay constant is

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{55.6\ s}$$

$$\lambda \approx 0.0125\ \mathrm{s}^{-1}$$

Final answer. Products are $^{216}_{84}\mathbf{Po}$ and $^{4}_{2}\mathbf{He}$; from the graph $T_{1/2} \approx 55.6\ s$, giving $\lambda \approx 0.0125\ \mathrm{s}^{-1}$.

← Back to Q153

Q154

Asha's mother read an article in the newspaper about a disaster that took place at Chernobyl. She could not understand much from the article and asked a few questions from Asha regarding the article. Asha tried to answer her mother's questions based on what she learnt in Class XII Physics.
(a) What was the installation at Chernobyl where the disaster took place? What, according to you, was the cause of this disaster?
(b) Explain the process of release of energy in the installation at Chernobyl.
(c) What, according to you, were the values displayed by Asha and her mother?

[2017]

SOLUTION

(a) The installation at Chernobyl (Ukraine) was a **nuclear fission power reactor** (of the RBMK design) used to generate electricity. The disaster of April 1986 was caused by a flawed safety-test procedure carried out together with serious design defects of the reactor (a positive void coefficient and a slow control-rod response). This produced a sudden uncontrolled power surge that caused a steam explosion, which breached the reactor core and released a large amount of radioactive material into the atmosphere.

(b) The energy in the reactor is released by a **controlled nuclear fission chain reaction** of uranium-235. When a slow (thermal) neutron is captured by a ^{235}U nucleus, the nucleus splits into two lighter fragments, releasing 2 to 3 neutrons and about 200 MeV of energy.

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3\,^{1}_{0}n + \text{energy } (\sim 200 \text{ MeV})$$

Most of this energy appears as the kinetic energy of the fission fragments. The fragments are stopped in the surrounding material, heating the coolant and producing steam, which drives turbines to generate electricity. The freed neutrons are slowed by a moderator and go on to cause further fissions, sustaining the **chain reaction**. In a working reactor this chain is kept just self-sustaining using control rods; at Chernobyl this control was lost.

(c) Values displayed: Asha showed scientific awareness, a helpful and patient attitude, and the ability to apply her classroom knowledge to a real-life situation. Her mother showed awareness of current world events, curiosity, and an open-minded willingness to learn from her daughter.

Final answer. Chernobyl was a fission power reactor; its energy came from a controlled $^{235}\mathrm{U}$ chain reaction, and the disaster arose from a flawed test plus design defects.

← *Back to Q154*

Q155

(a) Derive the law of radioactive decay $N = N_0 e^{-\lambda t}$.
(b) The half life of $^{238}\mathrm{U}$ *undergoing* α*-decay is* 4.510^9 *years. Find its mean life.*
(c) What fraction of the initial mass of a radioactive substance will decay in five half-life periods?
OR
(a) State the postulates of Bohr's model of hydrogen atom and derive the expression for Bohr radius.
(b) Find the ratio of the longest and the shortest wavelengths amongst the spectral lines of Balmer series in the spectrum of hydrogen atom.

[2020 • Set 55-4-1]

SOLUTION

(a) Law of radioactive decay: the rate of disintegration is proportional to the number N of undecayed nuclei present.

$$-\frac{dN}{dt} = \lambda N$$

Separating the variables,

$$\frac{dN}{N} = -\lambda\, dt$$

Integrating, with $N = N_0$ at $t = 0$ and $N = N$ at time t:

$$\int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_0^t dt$$

$$\ln N - \ln N_0 = -\lambda t$$

$$\ln\left(\frac{N}{N_0}\right) = -\lambda t$$

$$N = N_0 e^{-\lambda t}$$

(b) Given $\mathbf{T_{1/2} = 4.5 \times 10^9}$ **years**. The mean life is

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} = \frac{4.5 \times 10^9}{0.693} \text{ years}$$

$$\tau \approx 6.5 \times 10^9 \text{ years}$$

(c) After n half-lives the fraction left is $(\frac{1}{2})^n$. For $\boldsymbol{n = 5}$,

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Fraction decayed = $\mathbf{1 - \frac{1}{32} = \frac{31}{32} \approx 96.875\%}$.

$$\frac{31}{32} \approx 96.9\%$$

OR

(a) Postulates of Bohr's model: (1) The electron revolves around the nucleus only in certain stable circular orbits without radiating energy; these are called stationary states. (2) The angular momentum of the electron in these orbits is quantised: $\boldsymbol{L = m\,v\,r = \frac{n\,h}{2\pi}}$, where $\boldsymbol{n = 1, 2, 3,}$ (3) The atom emits or absorbs energy only when the electron jumps between two orbits, the photon energy being $\mathbf{h\nu = E_2 - E_1}$.

For the electron in a hydrogen atom, the Coulomb force provides the centripetal force:

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} = \frac{m\,v^2}{r}$$

$$\Rightarrow m\,v^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} \quad ...(i)$$

From the quantisation condition, $m\,v\,r = \dfrac{n\,h}{2\pi}$, so

$$v = \frac{n\,h}{2\pi\,m\,r} \quad \text{...(ii)}$$

Substituting (ii) into (i):

$$m\left(\frac{n\,h}{2\pi\,m\,r}\right)^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

Solving for r gives the radius of the nth orbit:

$$r_n = \frac{\varepsilon_0\,n^2\,h^2}{\pi\,m\,e^2}$$

For $n = 1$ this gives the **Bohr radius** $a_0 = 0.529$.

(b) For the Balmer series the lower level is $n = 2$, and

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$$

Longest wavelength: transition $n = 3 \to 2$,

$$\frac{1}{\lambda_{max}} = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$$

Shortest wavelength: transition $n = \infty \to 2$,

$$\frac{1}{\lambda_{min}} = R\left(\frac{1}{4} - 0\right) = \frac{R}{4} = \frac{9R}{36}$$

Therefore the ratio of longest to shortest wavelength is

$$\frac{\lambda_{max}}{\lambda_{min}} = \frac{9R/36}{5R/36} = \frac{9}{5}$$

$$\lambda_{max} : \lambda_{min} = 9 : 5$$

Final answer. $N = N_0\,e^{-\lambda t}$; mean life $\tau \approx 6.5 \times 10^9$ years; fraction decayed in 5 half-lives $= \dfrac{31}{32} \approx 96.9\%$ (OR Bohr radius $r_n = \dfrac{\varepsilon_0\,n^2\,h^2}{\pi\,m\,e^2}$ and Balmer ratio **9:5**).

← *Back to Q155*

Q156

Distinguish between 'nuclear fission' and 'nuclear fusion'. Which one of them is used for release of energy in a nuclear reactor? Name one fuel used in a nuclear reactor.

[2010]

SOLUTION

Nuclear fission and nuclear fusion are distinguished as follows:

1. In **fission**, a heavy nucleus (such as ^{235}U) splits into two lighter nuclei, whereas in **fusion**, two light nuclei combine to form a heavier nucleus.

2. Fission is usually triggered by bombarding the heavy nucleus with a slow neutron and can occur at ordinary temperature, whereas fusion requires extremely high temperature ($\sim 10^7\ K$) and pressure to overcome the Coulomb repulsion between the positively charged nuclei.

3. The energy released per nucleon is smaller in fission (~ 0.9 MeV per nucleon) than in fusion (~ 3.5 MeV per nucleon), so fusion is more efficient per unit mass.

A representative fission reaction:

$$^{235}_{92}U + {}^{1}_{0}n \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + 200\ \text{MeV}$$

A representative fusion reaction:

$$^{2}_{1}H + {}^{3}_{1}H \longrightarrow {}^{4}_{2}He + {}^{1}_{0}n + 17.6\ \text{MeV}$$

The process used for the release of energy in a nuclear reactor is **nuclear fission** (specifically the controlled, slow-neutron-induced fission chain reaction), because fusion is extremely difficult to control for power generation.

One fuel used in a nuclear reactor is **uranium-235 (^{235}U)** (plutonium-239, ^{239}Pu, is also used).

Final answer. Fission splits a heavy nucleus, fusion joins light nuclei; a nuclear reactor uses controlled fission, with ^{235}U as a common fuel.

← *Back to Q156*

Q157

What are radiation hazards? Mention four safety measures which should be taken against radioactive radiation.

[2010]

SOLUTION

Radiation hazards are the harmful biological effects produced in living tissue by **ionizing radiation** (α, β, γ rays and neutrons) emitted by radioactive sources.

The radiation ionizes atoms and molecules inside cells and can:

damage the DNA of cells, which may lead to cancer or to genetic mutations passed to offspring;

cause radiation sickness, skin burns and lowering of the body's immunity;

at very high doses, cause death. Even long-term low-dose exposure increases the risk of cancer.

Four safety measures against radioactive radiation:

1. **Shielding**: keep suitable absorbing material between the source and the worker, for example thick lead or concrete walls and lead aprons for γ-rays.

2. **Distance**: stay as far from the source as possible, since the intensity of radiation falls off as $\frac{1}{r^2}$.

3. **Minimum exposure time**: spend the least possible time near the source and use remote-handling tools such as tongs and manipulators.

4. **Monitoring and protective measures**: wear film or TLD dose badges to record the dose received, use protective clothing, never eat or drink in radiation areas, and dispose of radioactive waste safely in marked, lead-lined containers.

Final answer. Radiation hazards are the harmful effects of ionizing radiation on living tissue; key safety measures are shielding, distance, minimum exposure time, and proper monitoring/protective equipment.

← Back to Q157

Q158

What are radio isotopes? Write their any four uses.

[2010]

SOLUTION

Radioisotopes are the **radioactive isotopes** of an element, that is, atoms having the same atomic number Z (same element) but different mass number A, whose nuclei are unstable and decay spontaneously by emitting α, β or γ radiation.

They occur naturally or are produced artificially in nuclear reactors and cyclotrons. Common examples are ^{60}Co, ^{131}I, ^{14}C and ^{32}P.

Four uses of radioisotopes:

1. **In medicine** (diagnosis and therapy): ^{131}I is used to study and treat the thyroid gland, and ^{60}Co is used as a γ-source for the radiotherapy of cancer.

2. **In radiocarbon dating**: the radioisotope ^{14}C is used to estimate the age of ancient organic remains (wood, bones, fossils).

3. **As tracers in industry and research**: small amounts of a radioisotope are added to detect leaks in underground pipes and to measure the thickness of metal sheets.

4. **In agriculture and sterilization**: ^{32}P is used to trace the uptake of fertilizers by plants, and γ-radiation is used to sterilize medical equipment and to preserve food.

Final answer. Radioisotopes are unstable, radiation-emitting isotopes of an element; they are used in medicine, carbon dating, as industrial tracers, and in agriculture/sterilization.

← Back to Q158